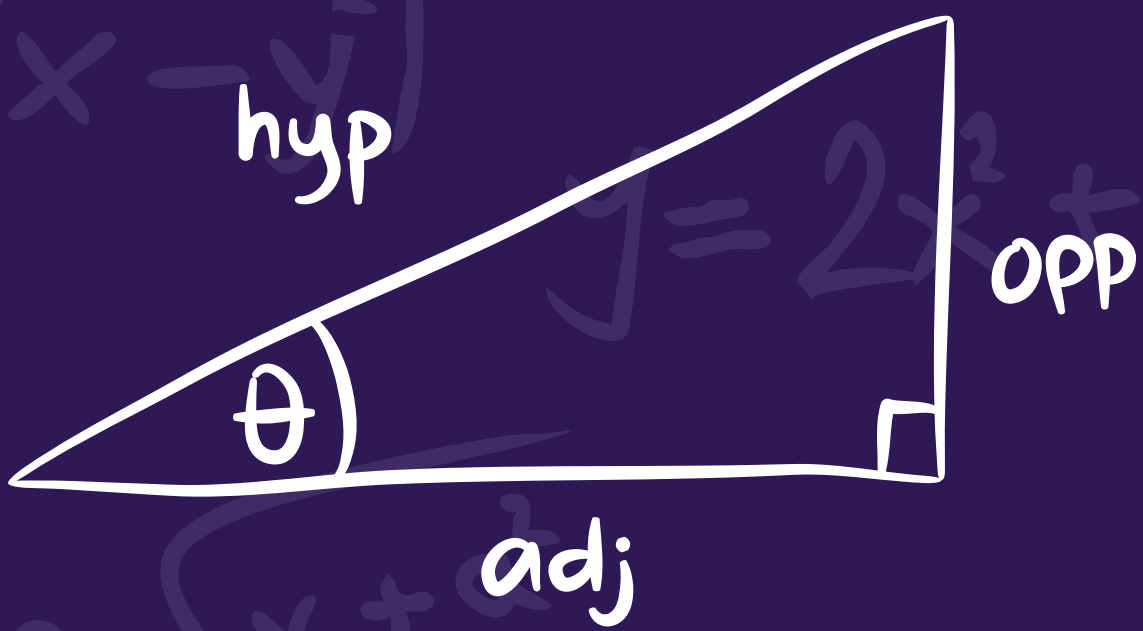




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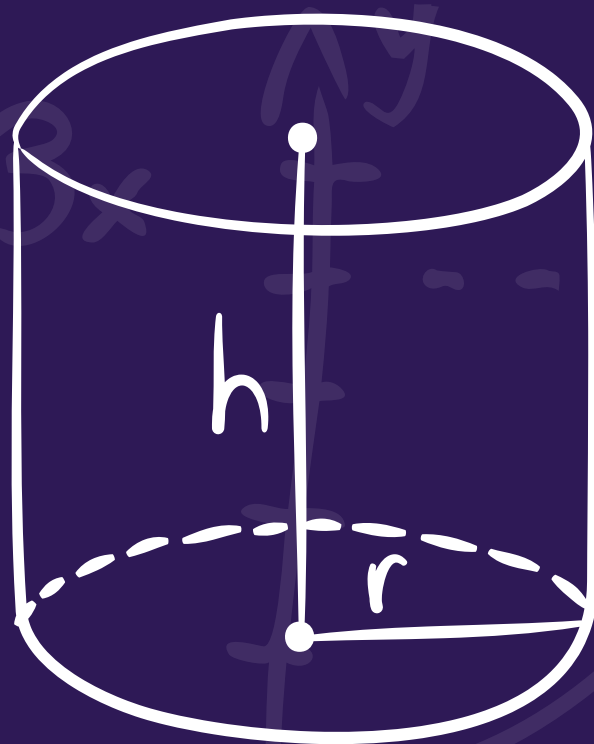
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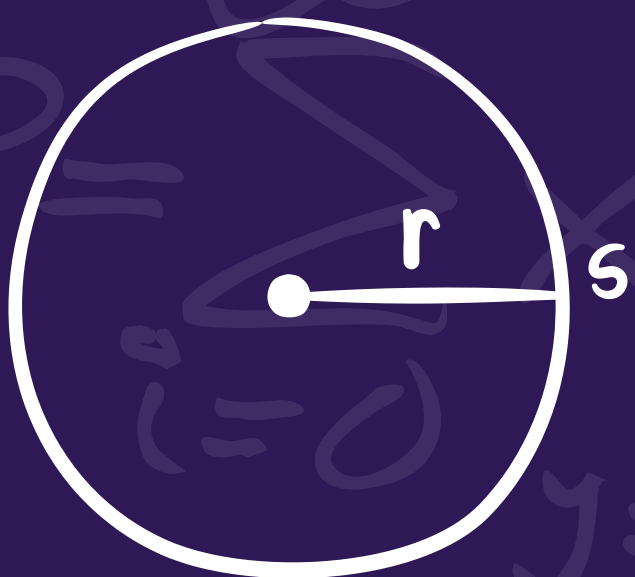


$$\sin(\theta) = \frac{\text{opp}}{\text{hyp}}$$

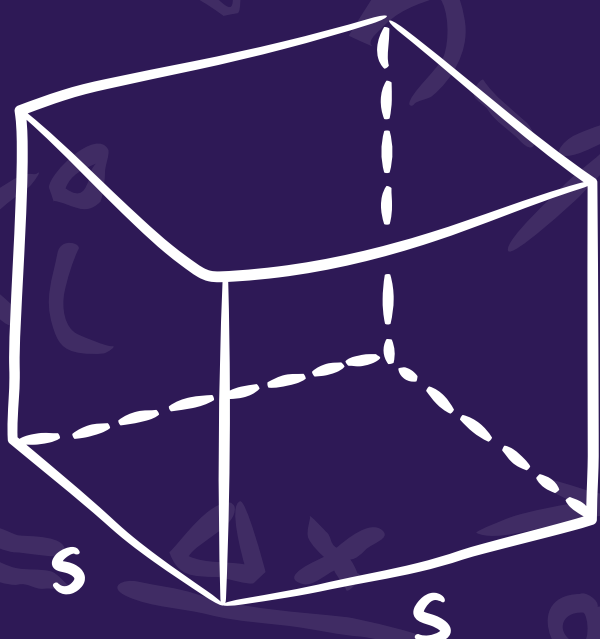
$$a + (b + c) = (a + b) + c$$



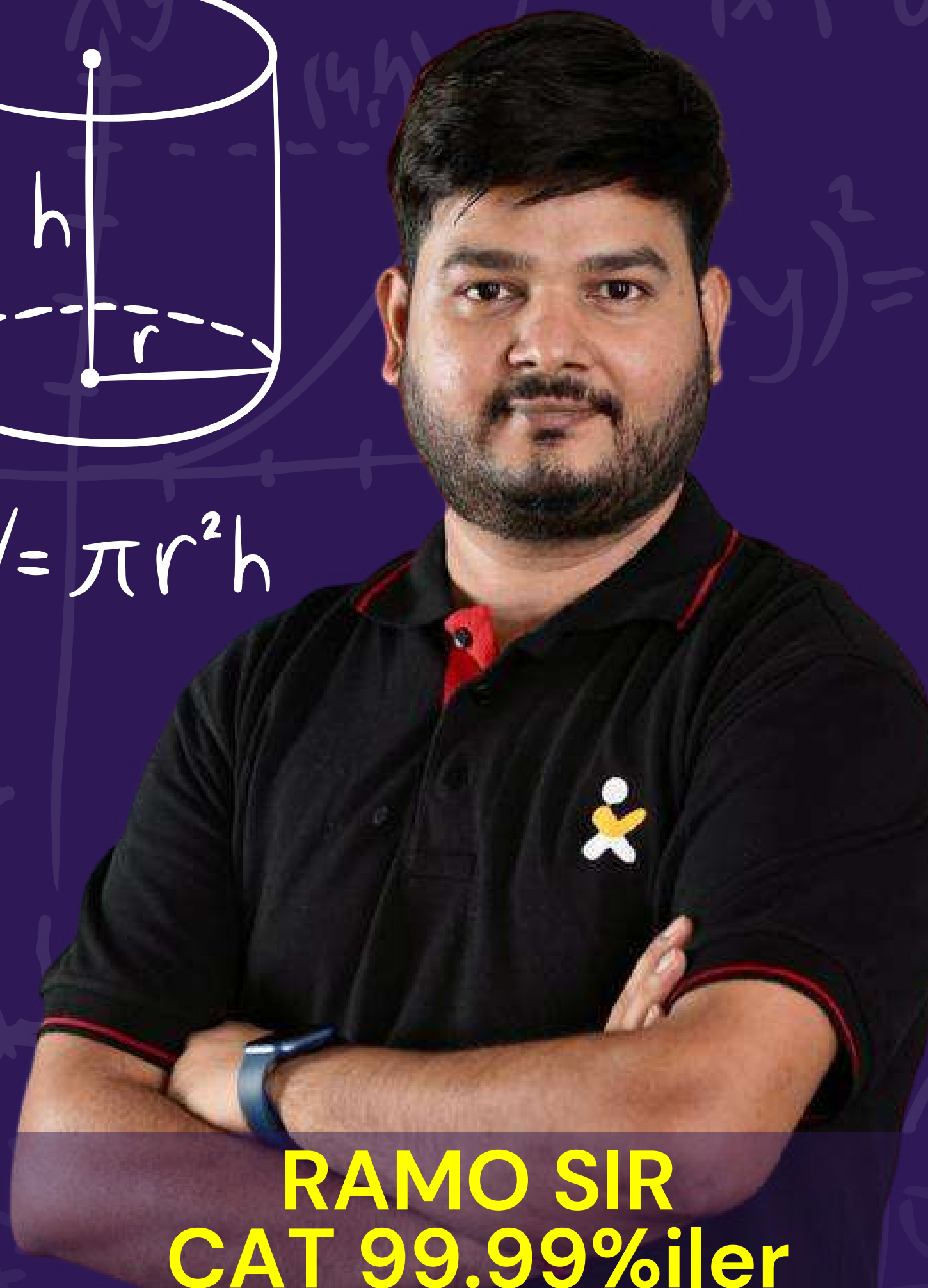
$$V = \pi r^2 h$$



$$A = \pi r^2$$



$$V = s^3$$



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INDEX

Sr. No.	Topic Name
1.	Number system
2.	Algebra
3.	Simplification, Surds and Indices
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5.	Height and distance
6.	Geometry
7.	Co-ordinate Geometry
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Number System



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1.1 IMPORTANT CONCEPTS

1. UNIT DIGIT

Cyclicity = 1

{0, 1, 5, 6}

Cyclicity = 2

$$4^{\text{odd}} = 4$$

$$9^{\text{odd}} = 9$$

$$4^{\text{even}} = 6$$

$$9^{\text{even}} = 1$$

{4, 9}

Cyclicity = 4

{2, 3, 7, 8}

➤ **Step 1** - Divide the number by 4 and find out the remainder (R)

➤ **Step 2** - $\{2, 3, 7, 8\}^R \rightarrow$ unit digit.

Eg. $2^{41} \times 7^{42} \times 8^{43}$. Find unit digit

Sol. $2 \times 7^{42} \times 8^{43}$

Cyclicity = 4 divide power by 4 find R.

Unit digit $\rightarrow 2^1 \times 7^2 \times 8^3 \rightarrow 2 \times 9 \times 2 \rightarrow 6$

Eg. Find unit digit of $1! + 2! + 3! + \dots + 100!$

Sol. $1! + 2! + 3! + 4! + 5! + \dots + 100!$ (from 5! to 100! Every term has 0 in the last)

$$1 + 2 + 6 + 24 \rightarrow 2 + 1 = 3$$

NOTE :

If any power gets divided by 4 completely, then take $R = 4$

2. DIVISIBILITY

• **Divisibility by 2** - Number will be divisible by 2 if last digit be is divisible by 2.

• **Divisibility by 4** - Number will be divisible by 4 if last 2 digit is divisible by 4.

• **Divisibility by 8** - Number will be divisible by 8 if last 3 digits is divisible by 8.

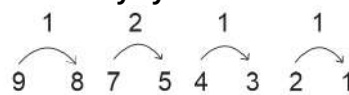
• **Divisibility by 16** - Number will be divisible by 16 if last 4 digits is divisible by 16.

Note - A number will be perfect square if its last digit has (0, 1, 4, 5, 6, 9)

• **Divisibility by 3** - Number will be divisible by 3 if sum of digits of number is divisible by 3.

• **Divisibility by 9** - Number will be divisible by 9 if sum of digits of number is divisible by 9.

• **Divisibility by 11** -



$1 + 2 + 1 + 1 = 5 \Rightarrow$ This sum must be 0, 11 or multiple of 11, so 98754321 Not divisible by 11.

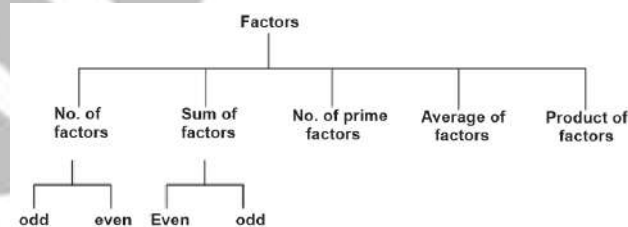
• **Divisibility by 7, 11, 13** - If a number is divisibly by 7, 11, 13, then it will be divisible by 1001. The number will be in form XYZXYZ

Eg. If a number 238abc is divisible by 7, 11, 13, then find $a+b+c$?

Sol. since it is divisible by 7, 11, 13 then $a=2, b=3, c=8$

Therefore $a+b+c = 2+3+8=13$

3. FACTORS



• **Number of factors** - $a^p \times b^q \times c^r$

$$\text{Number of factors} = (p+1)(q+1)(r+1)$$

$$\text{Eg. } 2160 = 2^4 \times 3^3 \times 5^1$$

$$\text{Number of factors} = 5 \times 4 \times 2 = 40$$

• **Number of even factors** - of $2^p b^q c^r = p(q+1)(r+1)$

$$\text{Eg. } 100 = 2^2 \times 5^2$$

$$\begin{bmatrix} 2^0 & 5^0 \\ 2^1 & 5^1 \\ 2^2 & 5^2 \end{bmatrix}$$

$$2 \times 3 \rightarrow 6 \text{ No. of even factors.}$$

$$100 \rightarrow 1, 2, 4, 5, 10, 20, 25, 50, 100 \rightarrow 6$$

• **Number of odd factors** - $2^p b^q c^r = 2^0 (q+1)(r+1)$
 $\Rightarrow (q+1)(r+1)$

$$\text{Eg. } 100 = 2^2 \times 5^2$$



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$$\begin{array}{l} \text{odd} \left[\begin{array}{cc} 2^0 & 5^0 \end{array} \right] \\ \text{even} \left[\begin{array}{cc} 2^1 \times & 5^1 \end{array} \right] \\ \text{even} \left[\begin{array}{cc} 2^2 \times & 5^2 \end{array} \right] \end{array}$$

$$1 \times 3 = 3$$

No. of odd factors of 100 is 3.

• **Sum of factors**

$$a^p b^q c^r = (a^0 + a^1 + \dots + a^p)(b^0 + b^1 + \dots + b^q)(c^0 + c^1 + \dots + c^r)$$

Eg. $100 = 2^2 \times 5^2$

$$2^0 \rightarrow 5^0$$

$$2^1 \rightarrow 5^1$$

$$2^2 \rightarrow 5^2$$

$$(5^0 + 5^1 + 5^2)(2^0 + 2^1 + 2^2)$$

• **Sum of even factors**

$$2^p b^q c^r = (2^1 + 2^2 + \dots + 2^p)(b^0 + b^1 + \dots + b^q)(c^0 + c^1 + \dots + c^r)$$

Eg. $240 = 2^4 \times 3^1 \times 5^1$

Sum of even factors

$$= (2^1 + 2^2 + 2^3 + 2^4)(3^0 + 3^1)(5^0 + 5^1)$$

$$\Rightarrow 30 \times 4 \times 6 = 720$$

• **Sum of odd factors -**

$$2^p b^q c^r \Rightarrow 2^0(b^0 + b^1 + \dots + b^q)(c^0 + c^1 + \dots + c^r)$$

Eg. $240 = 2^4 \times 3^1 \times 5^1$

$$(2^0)(3^0 + 3^1)(5^0 + 5^1) = 1 \times 4 \times 6 = 24.$$

• **Product of factors -** $(N)^{\frac{\text{number of factors}}{2}}$

$$\text{Average of factors} = \frac{\text{Sum of factor}}{\text{No. of factors}}$$

• **Number of Prime factors** - If $a^p b^q c^r$ then No. of prime factors is $p + q + r$.

Eg. No. of prime factors of $2^{13} 3^{15} 5^{16} \rightarrow 13 + 15 + 16 = 44$

4. **NUMBER OF ZEROES**

$$\text{No. of zeros} = \text{No. of } 5 = \left[\frac{n}{5} \right] + \left[\frac{n}{5^2} \right] + \left[\frac{n}{5^3} \right] + \dots$$

Eg. Find No. of zeros in 100! ?

Sol -

$$\begin{array}{r|l} 5 & 100 \\ 5 & 20 \\ & 4 \end{array}$$

$$20 + 4 = 24$$

5. **REMAINDER THEOREM**

• **Dividend** = divisor $\times$ quotient + remainder

Eg.

$$\begin{array}{r} \text{dividend} \\ \text{divisor} \leftarrow 5 \overline{) 17} \leftarrow \text{Quotient} \\ \underline{15} \\ 2 \rightarrow \text{Remainder} \end{array}$$

$$17 = 5 \times 3 + 2$$

NOTE :

$$\begin{array}{ccc} x \overline{) m} & x \overline{) n} & x \overline{) m+n} \\ \hline r_1 & r_2 & r_3 \end{array}$$

$$\text{Divisor} = x = r_1 + r_2 - r_3$$

Eg - When a certain number is divided by a certain divisor leaves remainder 43 and another number is divided by same divisor leaves remainder 37. If sum of both number is divided by same divisor leaves remainder 13. Find divisor?

Sol. Divisor = $43 + 37 - 13 = 67$

$$\frac{x}{D} \rightarrow \text{Rem.} = R$$

$$\frac{2x}{D} \rightarrow \text{Rem.} = \frac{2R}{D}$$

$$\frac{x}{D} \rightarrow \text{Rem.} = R$$

$$\frac{x^2}{D} \rightarrow \text{Rem.} = \frac{R^2}{D}$$

$$\frac{x}{D} \rightarrow \text{Rem.} = \frac{R^3}{D}$$

• **Concept of Remainder**

$$\frac{A+B+C+D}{M} \rightarrow R = \frac{A_R + B_R + C_R + D_R}{M}$$

$$\frac{A \times B \times C \times D}{M} \rightarrow R = \frac{A_R \times B_R \times C_R \times D_R}{M}$$

➤ **Concept Negative Remainder -**

$$\begin{array}{r} 11 \overline{) 97} \text{ (8)} \\ \underline{88} \\ 9 \end{array} \quad \begin{array}{r} 11 \overline{) 97} \text{ (8)} \\ \underline{99} \\ -2+11=9 \end{array}$$

➤ **Remainders of algebraic expressions**

$$\frac{(x+a)^n}{x} \rightarrow R = a^n$$

Or



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$$\frac{(tx+a)^x}{x} \rightarrow R = a^n \quad [t - \text{multiple of } x]$$

6. FERMAT THEOREM

$$\frac{a^{p-1}}{p} = R \rightarrow 1$$

P is a Prime number and a, p both are co-prime

Eg. Find the remainder in $\frac{2^{100}}{101}$?

$$\text{Sol} - \frac{2^{100}}{101} \rightarrow R = 1.$$

7. WILSON THEOREM

$$\frac{(P-1)!}{P} = P - 1 \text{ remainder}$$

P is a prime no.

Eg. Find the remainder in $\frac{28!}{29}$?

Sol - Remainder is 28.

8. LCM & HCF-

- Least Common Multiple (LCM)** - LCM of x, y, z is the smallest number which is exactly divisible by x, y, z.

Eg. Multiple of 12 - 12, 24, 36, 48, 60, 72, 84, 96

16 - 16, 32, 48, 64, 80, 96

Common - 48, 96

Least - 48. $\rightarrow$ LCM

- Important type**

Q. Find the smallest number which is exactly divisible by x, y, z?

Sol. LCM of x, y, z.

Q. Find the smallest number which is divided by x, y, z and leaves remainder r?

Sol. LCM of (x, y, z) + r

Q. Find the smallest number k which is divided by x, y, z and leaves remainder a, b, c respectively.

Sol. LCM of (x, y, z) - k, where

$$k = x - a = y - b = z - c$$

9. HIGHEST COMMON FACTOR (HCF)

HCF of x, y, z is the largest number which can divide x, y, z exactly

Eg. Factors of 12 $\rightarrow$ 1, 2, 3, 4, 5, 6, 12

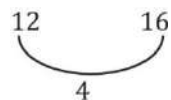
Factor of 16 $\rightarrow$ 1, 2, 4, 8, 16

Common $\rightarrow$ 1, 2, 4

Highest $\rightarrow$ 4 $\rightarrow$ HCF

NOTE:

HCF can never be greater than the difference of two numbers. HCF may be difference or factor of difference.



NOTE:

If two number are divided by their difference or factor of difference then it leaves same remainder.

- Important Type-**

Q. Find the largest number which can divide x, y, z exactly.

Sol. HCF of x, y, z

Q. Find the largest number which can divided x, y, z and leaves remainder 'r' in each case.

Sol. HCF of x-r, y-r, z-r

or

$$\text{HCF of } |x-y|, |y-z|, |z-x|$$

Q. Find the largest number which can divided x, y, z and leaves remainder a, b, c.

Sol. HCF of (x-a), (y-b), (z-c)

10. Relation between HCF and LCM

- Product of two number = Product of their HCF and LCM

NOTE:

If HCF of two number is H then number can be Hx and Hy where x and y are relatively prime numbers.

11. LCM and HCF of fraction

$$\text{LCM of fraction} = \frac{\text{LCM of numerator}}{\text{HCF of denominator}}$$

$$\text{LCM of } \frac{a}{b}, \frac{c}{d}, \frac{e}{f} = \frac{\text{LCM of } a, c, e}{\text{HCF of } b, d, f}$$

$$\text{HCF of fraction} = \frac{\text{HCF of numerator}}{\text{LCM of denominator}}$$

$$\text{HCF of } \frac{a}{b}, \frac{c}{d}, \frac{e}{f} = \frac{\text{HCF of } a, c, e}{\text{LCM of } b, d, f}$$

12. SEQUENCE AND SERIES

- Arithmetic Progression (AP)**

AP series

$$\rightarrow a, a + d, a + 2d, a + 3d, \dots$$

Where a = 1st term, d = Common difference

- n^{th} term, $a_n = a + (n - 1)d$

- Sum of n terms, $S_n = \frac{n}{2} [2a + (n - 1)d] = \frac{n}{2} [a + l]$, where l is the last term.

NOTE:

- Sum of 1st 'n' natural numbers = $\frac{n(n+1)}{2}$
- Sum of 1st 'n' odd numbers = $n^2 \rightarrow \left[\text{where } n = \frac{\text{last term} + 1}{2} \right]$



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- Sum of 1st 'n' even numbers
 $= n(n+1) \rightarrow \left[n = \frac{\text{Last term}}{2} \right]$
- Sum of squares of 1st 'n' natural numbers
 $= \frac{n(n+1)(2n+1)}{6}$
- Sum cubes of 1st n natural numbers $= \left[\frac{n(n+1)}{2} \right]^2$
- Sum of squares of 1st n even numbers $= \frac{2n(n+1)(2n+1)}{3}$
- **Geometric Progression (GP)**

- nth term $= a_n = ar^{n-1}$
 where a is the first term, r is the common ratio, and n is the number of terms.
- Sum of n terms of a geometric progression
 $S_n = \frac{a(1-r^n)}{1-r}$ when $r \neq 1$
 $S_n = na$ when $r = 1$
 Where n = number of terms, a = first term, and d = common difference
- Sum of infinite GP $= \frac{a}{1-r}$, where $-1 < r < 1$.

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IMPORTANT CONCEPTS

1. SQUARE FORMULA

- $(a + b)^2 = a^2 + b^2 + 2ab$
- $(a - b)^2 = a^2 + b^2 - 2ab$
- $(a + b)^2 = (a - b)^2 + 4ab$
- $(a - b)^2 = (a + b)^2 - 4ab$
- $(a^2 - ab + b^2)(a^2 + ab + b^2) = a^4 + a^2b^2 + b^4$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $a^2 - b^2 = (a + b)(a - b)$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

2. CUBE FORMULA

- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

3. SPECIAL CASE 1

- If $a^2 - ab + b^2 = 0$ then $a^3 + b^3 = 0$
- If $b = 1$, then $a^2 - a + 1 = 0$, then $a^3 + 1 = 0$ or $a^3 = -1$

4. SPECIAL CASE 2

- If $a^2 + a + 1 = 0$ then $a^3 - 1 = 0$ or $a^3 = 1$

5. SPECIAL CASE 3

- If $\frac{a}{b} + \frac{b}{a} = 1$ then $a^3 + b^3 = 0$
- If $\frac{1}{a} - \frac{1}{b} = \frac{1}{a-b}$ then $a^3 + b^3 = 0$

6. SPECIAL CASE 4

- If $\frac{a}{b} + \frac{b}{a} = -1$ then $a^3 - b^3 = 0$
- If $\frac{a}{b} + \frac{b}{a} = \frac{1}{a+b}$ then $a^3 - b^3 = 0$

7. SPECIAL CASE 5

- If $ab(a + b) = 1$ then $\frac{1}{a^3b^3} - a^3 - b^3 = 3$
- $a^3 + b^3 + c^3 - 3abc =$
 $= \frac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2]$
 $= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
 $= (a+b+c)[(a+b+c)^2 - 3(ab+bc+ca)]$
 - If $(a+b+c) = 0$
Then $a^3 + b^3 + c^3 - 3abc = 0$
 $a^3 + b^3 + c^3 - 3abc = 0$
 - If $a^3 + b^3 + c^3 - 3abc = 0$
 a, b and c are distinct no
then, $a + b + c = 0$
 - $a^3 + b^3 + c^3 - 3abc = 0$
 a, b and c all are +ve integer no then, $a = b = c$.
 - $a^2 + b^2 + c^2 - ab - bc - ca = 0$
 $a^2 + b^2 + c^2 = ab + bc + ca$
then $a = b = c$
 - If a, b and c are in A.P. then
 $a^3 + b^3 + c^3 - 3abc = 9bd^2$ (d =common difference)

8. TYPE BASE QUESTIONS

- If $x - \frac{1}{x} = a$ then $x^2 + \frac{1}{x^2} = a^2 + 2$
and $x^4 + \frac{1}{x^4} = (a^2 + 2)^2 - 2$
- If $x + \frac{1}{x} = a$ then $x^3 + \frac{1}{x^3} = a^3 - 3a$
If $x - \frac{1}{x} = a$ then $x^3 - \frac{1}{x^3} = a^3 + 3a$



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- If $x + \frac{1}{x} = a$
Then $x^5 + \frac{1}{x^5} = (a^2 - 2)(a^3 - 3a) - a$
- If $\left(x - \frac{1}{x}\right) = a$, then
 $x^5 - \frac{1}{x^5} = (a^2 + 2)(a^3 + 3a) - a$
- $x + \frac{1}{x} = a$, then $x^6 + \frac{1}{x^6} = (a^3 - 3a)^2 - 2$
- If $x + \frac{1}{x} = \sqrt{2}$ then $x^2 + \frac{1}{x^2} = 0$ or $x^4 + 1 = 0$ or $x^4 = -1$
- If $x + \frac{1}{x} = \sqrt{3}$ then $x^3 + \frac{1}{x^3} = 0$ or $x^6 = -1$
- If $x + \frac{1}{x} = 2$ then $x = 1$
If $x + \frac{1}{x} = -2$ then $x = -1$
- If $x^n + \frac{1}{x^n} = a$ then $x^n - \frac{1}{x^n} = \pm \sqrt{a^2 - 4}$
If $x^n - \frac{1}{x^n} = b$ then $x^n + \frac{1}{x^n} = \pm \sqrt{b^2 + 4}$
- When $a + \frac{1}{a} = 1$,
Or,
 $a^2 - a + 1 = 0$
Then $a^3 = -1$
Or,
 $a^3 + 1 = 0$
 $\Rightarrow$ When $a + \frac{1}{a} = -1$
Or,
 $a^2 + a + 1 = 0$
then,
 $a^3 = 1$
- Rationalizing factor of the surd $x = \sqrt{a} \pm \sqrt{b}$ and
 $\frac{1}{x} = \sqrt{a} \mp \sqrt{b}$

Note:

If $xy = 1$ then $\frac{1}{1+x} + \frac{1}{1+y} = 1$

9. MISCELLANEOUS

- $1 + A + B + AB = (1+A)(1+B)$
- $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$
If a, b , and c are in A.P. then,
 - $a^2 + b^2 + c^2 - ab - bc - ca = 3d^2$ (d = common difference)
- If $a + \frac{1}{b} = x, b + \frac{1}{c} = y, c + \frac{1}{a} = z$, then
 $abc + \frac{1}{abc} = xyz - (x+y+z)$

10. COMPONENDO & DIVIDENDO

If $\frac{a}{b} = \frac{x}{y}$ then C & D $\Rightarrow \frac{a+b}{a-b} = \frac{x+y}{x-y}$

11. SPECIAL CASE

If $x = \frac{2ab}{a+b}$ then $\frac{x+na}{x-na} + \frac{x+nb}{x-nb} = 2$

12. QUADRATIC EQUATION

Quadratic equation $ax^2 + bx + c = 0$

- sum of roots $= \alpha + \beta = -\frac{b}{a}$
- product of roots $= \alpha\beta = \frac{c}{a}$
- roots of equation are
 $\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$
- An equation whose roots are α and β is given by $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
- Discriminant (D) $= b^2 - 4ac$



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14.2 IMPORTANT CONCEPTS

1. LAWS OF INDICES

- $a^m \times a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $a^{m^1} = a^{m \times m \times \dots \times m \text{ times}} \neq a^{m^n}$

$$d. (ab)^n = a^n b^n, \left(\frac{a}{b}\right)^n \rightarrow \frac{a^n}{b^n}$$

$$e. (-a)^n = \begin{cases} a^n \rightarrow n = \text{even} \\ -a^n \rightarrow n = \text{odd} \end{cases}$$

$$f. \frac{1}{a^n} = a^{-n}$$

$$g. a^m = a^n, \text{ then } m = n \text{ or } a^m = b^m \text{ then } a = b$$

2. LAWS OF SURDS

- $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}, \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$
- $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a} = \sqrt[n]{\sqrt[m]{a}}$
- $(\sqrt[n]{a})^m = (a)^{\frac{m}{n}}$
- $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

3. IMPORTANT RESULTS

$$2. \text{ If } x = \sqrt{a + \sqrt{a + \sqrt{a + \sqrt{a + \dots}}}} = \frac{\sqrt{4a+1} + 1}{2}$$

$$3. \text{ If } y = \sqrt{a - \sqrt{a - \sqrt{a - \sqrt{a - \dots}}}} = \frac{\sqrt{4a+1} - 1}{2}$$

$$\text{Therefore } x - y = 1$$

$$2. \text{ If } z = \sqrt{a \sqrt{a \sqrt{a \sqrt{a \dots n \text{ times}}}}} = a^{\frac{1}{2^n}}$$

$$3. \text{ If } p = \sqrt{a \sqrt{a \sqrt{a \sqrt{a \dots}}}} = a$$

$$4. \text{ If } q = \sqrt{a + \sqrt{a - \sqrt{a + \sqrt{a - \dots}}}} = \frac{\sqrt{4a-3} + 1}{2}$$

$$5. \text{ If } r = \sqrt{a - \sqrt{a + \sqrt{a - \sqrt{a - \dots}}}} = \frac{\sqrt{4a-3} - 1}{2}$$

4. SQUARE FORMULA

- $(a + b)^2 = a^2 + b^2 + 2ab$
- $(a - b)^2 = a^2 + b^2 - 2ab$
- $(a + b)^2 = (a - b)^2 + 4ab$
- $(a - b)^2 = (a + b)^2 - 4ab$
- $(a^2 - ab + b^2)(a^2 + ab + b^2) = a^4 + a^2b^2 + b^4$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $a^2 - b^2 = (a + b)(a - b)$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

5. CUBE FORMULA

- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- $a^3 + b^3 + c^3 - 3abc =$

$$= \frac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= (a+b+c)[(a+b+c)^2 - 3(ab + bc + ca)]$$



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Trigonometry



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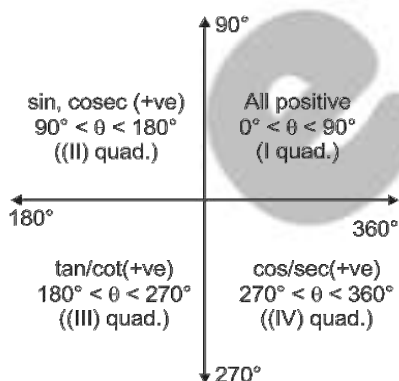
IMPORTANT CONCEPTS

➤ Trigonometry Table

	0°	30°	45°	60°	90°
sinθ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cosθ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tanθ	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND
cotθ	ND	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
secθ	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	ND
cosecθ	ND	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

*ND → Not defined

➤ SIGNS OF TRIGONOMETRIC FUNCTIONS IN DIFFERENT



➤ Change of Functions in Different Quadrants:

- $\sin(90^\circ - \theta) = \cos \theta$; $\tan(90^\circ + \theta) = -\cot \theta$
- $\sin(180^\circ - \theta) = \sin \theta$; $\tan(180^\circ + \theta) = \tan \theta$
- $\sin(180^\circ + \theta) = -\sin \theta$; $\tan(360^\circ - \theta) = -\tan \theta$
- $\sin(360^\circ + \theta) = \sin \theta$; $\csc(360^\circ + \theta) = \csc \theta$

➤ $\sin(360^\circ - \theta) = -\sin \theta$; $\cos(180^\circ - \theta) = -\cos \theta$

➤ IMPORTANT TRIGONOMETRIC IDENTITIES

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\sec^2 \theta - \tan^2 \theta = 1$
- If $(\sec \theta + \tan \theta) = k$, then $(\sec \theta - \tan \theta) = 1/k$
- $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

▪ If $(\operatorname{cosec} \theta - \cot \theta) = p$, then $(\operatorname{cosec} \theta + \cot \theta) = \frac{1}{p}$

➤ $\sin(A + B) = \sin A \cos B + \cos A \sin B$

➤ $\sin(A - B) = \sin A \cos B - \cos A \sin B$

➤ $\cos(A + B) = \cos A \cos B - \sin A \sin B$

➤ $\cos(A - B) = \cos A \cos B + \sin A \sin B$

➤ $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

➤ $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

➤ $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$

➤ $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$

➤ $\sin 2A = 2 \sin A \cos A$

▪ $\sin 2A = \left(\frac{2 \tan A}{1 + \tan^2 A} \right)$

➤ $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$

▪ $\cos 2A = \left(\frac{1 - \tan^2 A}{1 + \tan^2 A} \right)$

➤ $\tan 2A = \left(\frac{2 \tan A}{1 - \tan^2 A} \right)$

➤ $\cot 2A = \left(\frac{\cot^2 A - 1}{2 \cot A} \right)$

➤ $\sin 3A = 3 \sin A - 4 \sin^3 A$

➤ $\cos 3A = 4 \cos^3 A - 3 \cos A$

➤ $\tan 3A = \left(\frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} \right)$

➤ $\cot 3A = \left(\frac{\cot^3 A - 3 \cot A}{3 \cot^2 A - 1} \right)$

➤ $\sin(A + B) \cdot \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

➤ $\cos(A + B) \cdot \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

➤ $\tan(A + B + C)$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - (\tan A \tan B + \tan B \tan C + \tan C \tan A)}$$

▪ If $A + B + C = 180^\circ = \pi$,

$\tan A + \tan B + \tan C = \tan A \tan B \tan C$

and $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$



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- $2\sin A \cos B = \sin(A+B) + \sin(A-B)$
- $2\cos A \sin B = \sin(A+B) - \sin(A-B)$
- $2\cos A \cos B = \cos(A+B) + \cos(A-B)$
- $2\sin A \sin B = \cos(A-B) - \cos(A+B)$
- $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right)$
- $\cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right) \sin\left(\frac{D-C}{2}\right)$
- $\sin(60^\circ - A) \sin A \sin(60^\circ + A) = \frac{1}{4} \sin 3A$
- $\cos(60^\circ - A) \cos A \cos(60^\circ + A) = \frac{1}{4} \cos 3A$
- $\tan(60^\circ - A) \tan A \tan(60^\circ + A) = \tan 3A \{A \neq 30^\circ\}$
- If $A \neq 0^\circ, A \neq 60^\circ$, then
 $\operatorname{cosec} A \operatorname{cosec}(60^\circ - A) \operatorname{cosec}(60^\circ + A) = 4 \operatorname{cosec} 3A$
- If $A \neq 90^\circ, A \neq 30^\circ$, then
 $\sec A \sec(60^\circ - A) \sec(60^\circ + A) = 4 \sec 3A$
- If $A \neq 0^\circ, A \neq 60^\circ$, then
 $\cot A \cot(60^\circ - A) \cot(60^\circ + A) = \cot 3A$
- $\tan \theta + \tan(60^\circ + \theta) + \tan(120^\circ + \theta) = 3 \tan 3\theta$
- $\cot \theta + \cot(60^\circ + \theta) - \cot(60^\circ - \theta) = 3 \cot 3\theta$
- $\cos^3 \alpha + \cos^3(120^\circ + \alpha) + \cos^3(240^\circ + \alpha) = \frac{3}{4} \cos 3\alpha$
- $\sin^3 \theta + \sin^3(120^\circ + \theta) + \sin^3(240^\circ + \theta) = \frac{-3}{4} \sin 3\theta$
- **ADVANCE IDENTITIES TO REMEMBER**
 - $\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta$
 - $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$
 - $\cos 6\theta = 32 \cos^6 \theta - 48 \cos^4 \theta + 18 \cos^2 \theta - 1$
 - $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta) = 2$
 - $(1 + \cot \theta + \sec \theta)(1 + \tan \theta - \operatorname{cosec} \theta) = 2$
 - $(\sin \theta + \cos \theta + 1)(\sin \theta + \cos \theta - 1) = 2\sin \theta \cos \theta$
 - $\frac{\sec \theta + \tan \theta - 1}{\tan \theta - \sec \theta + 1} = \sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 - \sin \theta}$
 - $\tan \theta (1 + \sec 2\theta)(1 + \sec 4\theta)(1 + \sec 8\theta) = \tan 8\theta$
 - $\sec^4 \theta - \tan^4 \theta = \sec^2 \theta + \tan^2 \theta = 1 + 2\tan^2 \theta = 2\sec^2 \theta - 1$
 - $\operatorname{cosec}^4 \theta - \cot^4 \theta = \operatorname{cosec}^2 \theta + \cot^2 \theta = 1 + 2\cot^2 \theta = 2\operatorname{cosec}^2 \theta - 1$
 - $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{1}{\sec \theta - \tan \theta} = \sec \theta + \tan \theta$
 - $\sin x = \sin y \Rightarrow x = n\pi + (-1)^n y; n \in \mathbb{Z} \text{ (integers/पूर्णांक)}$
 - $\cos x = \cos y \Rightarrow x = 2n\pi \pm y; n \in \mathbb{Z}$
 - $\tan x = \tan y \Rightarrow x = n\pi + y; n \in \mathbb{Z}$

- $\sin(x+y+z) = \sin x \cos y \cos z + \cos x \sin y \cos z + \cos x \cos y \sin z - \sin x \sin y \sin z$
 $\sin(x+y+z) = \cos x \cos y \cos z [\tan x + \tan y + \tan z - \tan x \tan y \tan z]$
- $\cos(x+y+z) = \cos x \cos y \cos z - \sin x \sin y \cos z - \sin x \cos y \sin z - \cos x \sin y \sin z$
 $\cos(x+y+z) = \cos x \cos y \cos z [1 - \tan x \tan y - \tan y \tan z - \tan z \tan x]$
- $\tan(x+y+z) = \frac{\tan x + \tan y + \tan z - \tan x \tan y \tan z}{1 - \tan x \tan y - \tan y \tan z - \tan z \tan x}$

• **RESULTS TO FIND CONSTANT VALUES**

- If $a \sin \theta + b \cos \theta = m$ and $a \cos \theta - b \sin \theta = n$, then

$$a^2 + b^2 = m^2 + n^2$$

- If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, then
 $\Rightarrow \cos \theta - \sin \theta = \pm \sqrt{2} \sin \theta$

- If $a \sec \theta + b \tan \theta = c$
 $b \sec \theta + a \tan \theta = d$, then

$$a^2 - b^2 = c^2 - d^2$$

- If $a \sec \theta - b \tan \theta = c$
 $b \sec \theta - a \tan \theta = d$
or $a \tan \theta - b \sec \theta = d$, then

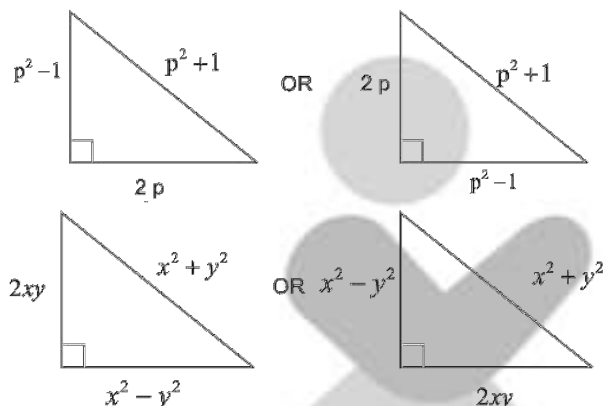
$$a^2 - b^2 = c^2 - d^2$$

• **WHEN SUM OR DIFFERENCE BETWEEN THE ANGLES ARE GIVEN**

- If $A + B = 90^\circ$, then
 $\tan A \tan B = 1$
 $\sin A \sec B = 1$
 $\cos A \operatorname{cosec} B = 1$
 $\cot A \cot B = 1$
 $\sin^2 A + \sin^2 B = 1$
 $\cos^2 A + \cos^2 B = 1$
- If $A + B = 45^\circ$, then
 $(1 + \tan A)(1 + \tan B) = 2$
 $(1 - \cot A)(1 - \cot B) = 2$
- If $A - B = 45^\circ$, then
 $(1 + \tan A)(1 - \tan B) = 2$
- If $A + B = 135^\circ$, then
 $(1 - \tan A)(1 - \tan B) = 2$
 $(1 + \cot A)(1 + \cot B) = 2$
- If $A + B + C = 180^\circ$, then
 $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
 $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$
- If $A + B + C = 90^\circ$, then
 $\cot A + \cot B + \cot C = \cot A \cot B \cot C$
 $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$
- If $A + B = 90^\circ$, then
 $\tan A - \tan B = 2 \tan(A-B)$



• Important Figures for quick results



• Maximum and minimum values of trigonometric identities

➤ Type 1-

$a \sin \theta \pm b \cos \theta$, $a \sin \theta \pm b \sin \theta$, $a \cos \theta \pm b \cos \theta$

- Maximum value = $\sqrt{a^2 + b^2}$
- Minimum value = $-\sqrt{a^2 + b^2}$
- Type 2- $(\sin \theta \cos \theta)^n$
- Minimum value = $\left(\frac{1}{2}\right)^n$
- Maximum value can go up to infinity
- Type 3-
 $a \cos^2 \theta + b \sec^2 \theta$, $a \sin^2 \theta + b \operatorname{cosec}^2 \theta$, $a \tan^2 \theta + b \cot^2 \theta$
- Minimum value = $2\sqrt{ab}$ $b \leq a$
- Type 4-
 $a \sec^2 \theta + b \operatorname{cosec}^2 \theta$
- Minimum value = $(\sqrt{a} + \sqrt{b})^2$



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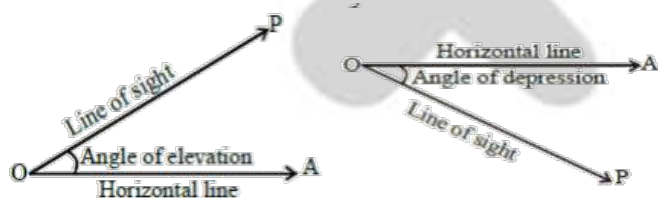
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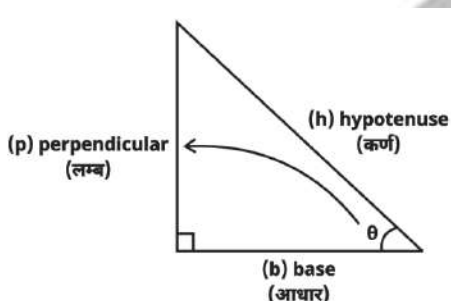


ANGLE OF ELEVATION AND ANGLE OF DEPRESSION

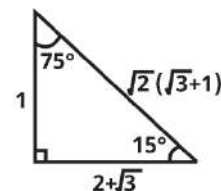
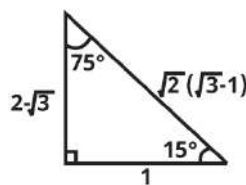
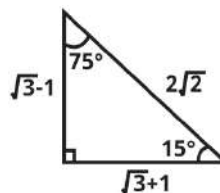
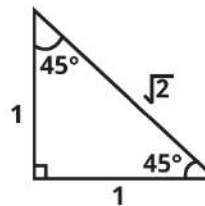
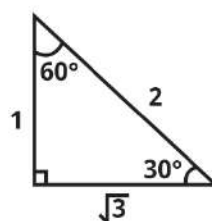
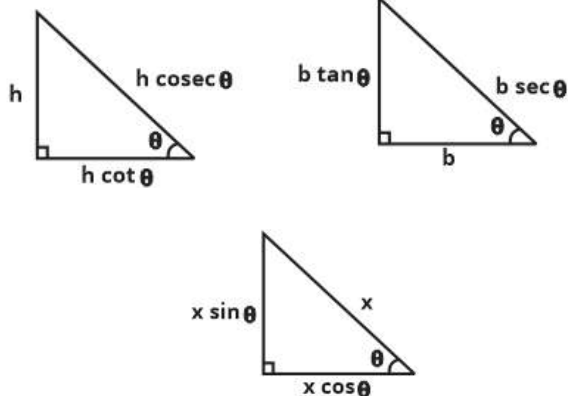


17.2 IMPORTANT CONCEPTS

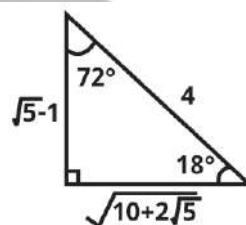
- $\sin \theta = \frac{p}{h}$
- $\operatorname{cosec} \theta = \frac{h}{p}$
- $\cos \theta = \frac{b}{h}$
- $\sec \theta = \frac{h}{b}$
- $\tan \theta = \frac{p}{b}$
- $\cot \theta = \frac{b}{p}$



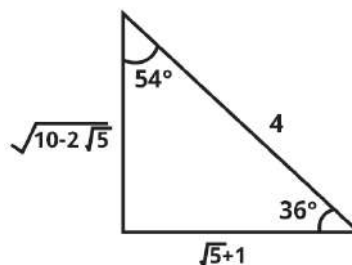
1. DISTANCE RATIOS IN DIFFERENT ANGLES



- $\tan 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3} = \cot 75^\circ$
- $\tan 75^\circ = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2 + \sqrt{3} = \cot 15^\circ$



- $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$
- $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$



$$\cos 36^\circ = \frac{\sqrt{5}+1}{4}, \sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

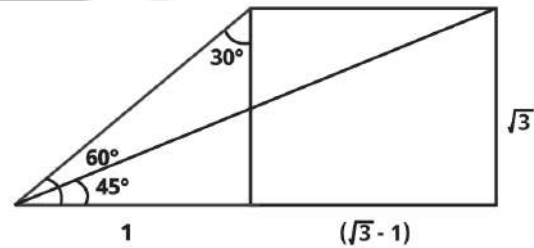
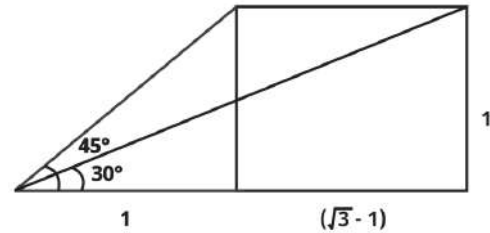
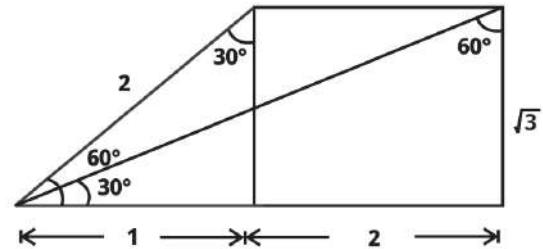
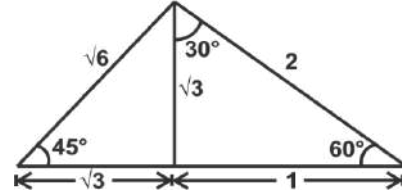
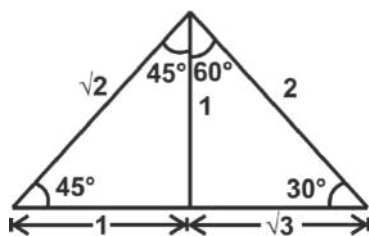
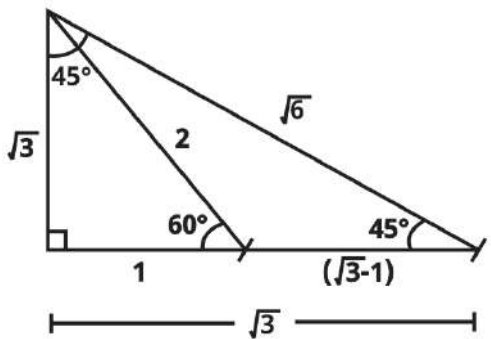
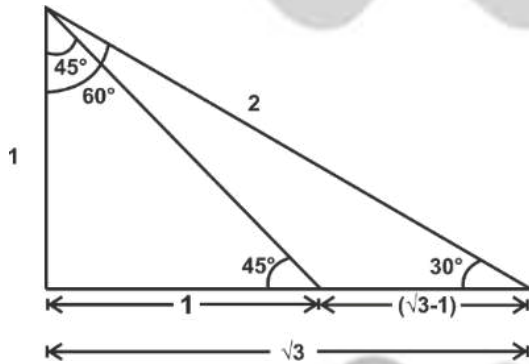
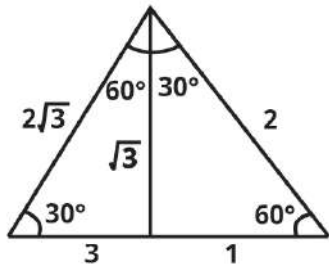
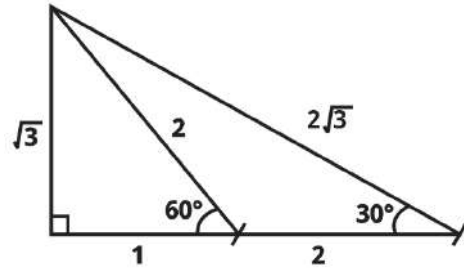
2. Distance Ratio with Angle of Elevations from Two different points



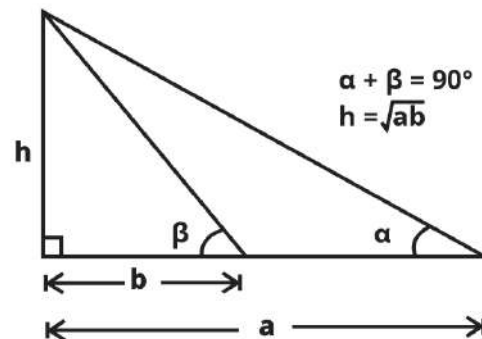
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3. When the angles of elevation are complementary angles



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Exam Oriented Points (Quick Revision)

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1 LINES & ANGLES:

1.1 Angle

Every angle has a measure. The unit of angle measure is a standard angle called a **"degree"**.

1 degree = $1^\circ = 60$ minutes
i.e., 60'

1' = 60 seconds i.e., 60"

1.2 TYPES OF ANGLES:

- An angle whose measure is 90° is called a **right angle**. i.e., $\theta = 90^\circ$
- An angle whose measure is less than 90° is called an **acute angle**.
i.e., $0^\circ < \theta < 90^\circ$
- An angle whose measure is more than 90° but less than 180° is called an **obtuse angle**.
i.e., $90^\circ < \theta < 180^\circ$
- An angle whose measure is 180° is called a **straight angle**.



- An angle whose measure is more than 180° is called a **reflex angle**.
i.e., $180^\circ < \theta < 360^\circ$
- Complementary Angle:**
Two angles, whose sum is 90° , are called complementary angles.
If, $\angle x + \angle y = 90^\circ$
 $\angle x$ & $\angle y$ are called complementary angles.
e.g., $(40^\circ, 50^\circ)$, $(45^\circ, 45^\circ)$, $(35^\circ, 55^\circ)$, etc

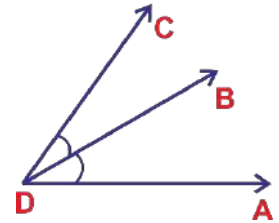
vii) Supplementary Angles:

Two angles, whose sum measures 180° , are called the supplementary angles.
If, $\angle p + \angle q = 180^\circ$
 $\angle p$ & $\angle q$ are called supplementary angles.
e.g., $(120^\circ, 60^\circ)$, $(100^\circ, 80^\circ)$, etc

viii) Adjacent Angles:

Two angles are called adjacent angles if they have:

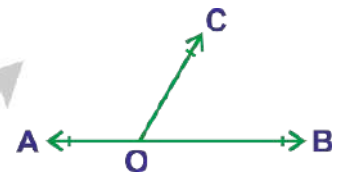
- the same vertex,
- a common arm, and
- Uncommon arms are on either side of the common arm.



e.g., $\angle ADB$ & $\angle CDB$ are adjacent angles.

ix) Linear pair of Angles:

Two adjacent angles are said to form a linear pair of angles, if their non-common arms are two opposite rays.



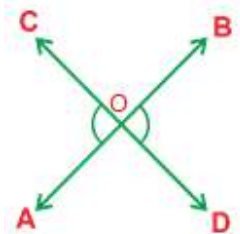
i.e., if $\angle AOC + \angle BOC = 180^\circ$,

then $\angle AOC$ & $\angle BOC$ form a linear pair.

x) Vertically Opposite Angles:

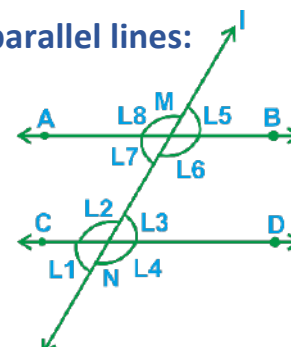
Two angles are called a pair of vertically opposite angles, if their arms form two pairs of opposite rays.

i.e., $\angle AOC$ & $\angle DOB$ are vertically opposite. Also, $\angle AOD$ & $\angle COB$ are vertically opposite.



If two lines intersect, then vertically opposite angles are equal. Here, $\angle AOC = \angle DOB$ & $\angle AOD = \angle COB$

1.3 When a transversal intersects two parallel lines:



Here,

$L7 = L5$
 $L6 = L8$
 $L1 = L3$
 $L2 = L4$

vertically opposite angles

$L3 = L5$
 $L2 = L8$
 $L1 = L7$
 $L4 = L6$

→ Corresponding angles

$L3 = L7$
 $L2 = L6$
 $L1 = L5$
 $L4 = L8$

→ Alternate angles

$L7 + L2 = 180^\circ$
 $L3 + L6 = 180^\circ$

→ Sum of consecutive interior angles is 180°

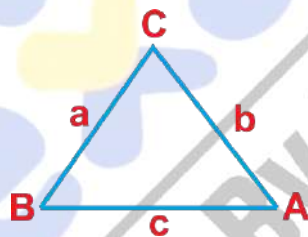
2 TRIANGLES :

2.1 Types of Triangles:

On the basis of sides –

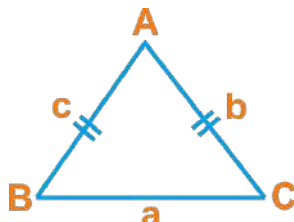
i) Scalene Triangle: -

A triangle with 3 unequal sides is called a scalene triangle.
e.g., In $\triangle ABC$,
for $a \neq b \neq c$,
 $\triangle ABC$ is scalene



ii) Isosceles Triangle:

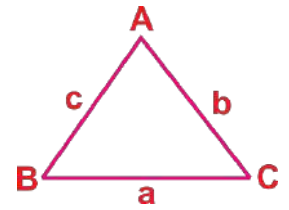
A triangle with (at least) two equal sides is called an isosceles triangle.
e.g., If a, b & c are the sides of $\triangle ABC$ and $a \neq c = b$, then $\triangle ABC$ is isosceles triangle.



iii) Equilateral Triangle:

A triangle in which all three sides are equal is called an equilateral triangle.

i.e., $\triangle ABC$ is equilateral, if $a = b = c$



On the basis of angles:

iv) Acute Triangle: -

A triangle with all three internal angles less than 90° i.e., all three internal

RaMo Mantra :-

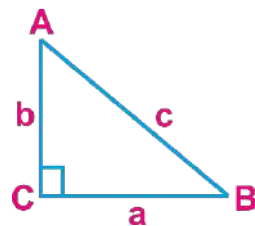
If a, b & c are the sides of a triangle, such that $c > b, c > a$
And, $a^2 + b^2 > c^2$
=> Triangle is an acute triangle

angles are acute

v) Right Triangle: -

A triangle with one angle that measures 90° is called a right triangle.

$\triangle ABC$ is right triangle, if $\angle C = 90^\circ$



RaMo Mantra :

If a, b & c are the sides of a triangle, such that, $c > a, c > b$
and $a^2 + b^2 = c^2$
then, Triangle is called right angled

vi) Obtuse Triangle:

A triangle with one angle greater than 90° i.e., obtuse angle, is known as an obtuse angled triangle.

RaMo Mantra :

If a, b and c are the sides of a triangle such that $c > a, c > b$, and
and $a^2 + b^2 < c^2$
then, Triangle is called obtuse angled

2.2 Condition for formation of a triangle: -

The formation of a triangle is possible if the sum of the lengths of any two sides of a triangle is always greater than the length of third side.

OR

The difference of the lengths of any two side of a triangle is always smaller than the length of the third side.

i.e., if $\triangle ABC$ is a triangle with sides a , b and c , then

$$\begin{aligned} a + b &> c & \text{and} & |a - b| < c \\ b + c &> a & & |b - c| < a \\ c + a &> b & & |c - a| < b \end{aligned}$$

Generalisation: For a $\triangle ABC$ with sides a , b and c

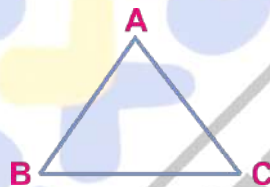
- (1) $|b - c| < a < |b + c|$
- (2) $|a - c| < b < |a + c|$
- (3) $|b - a| < c < |a + b|$

2.3 Properties of a triangle:-

i) Angle sum property: -

The sum of the three internal angles of a triangle is always 180° .

i.e., In $\triangle ABC$, $\angle A + \angle B + \angle C = 180^\circ$



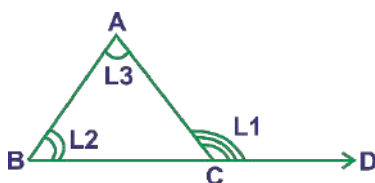
ii) Exterior angle property: -

An exterior angle of a triangle is equal to the sum of the opposite interior angles.

i.e., In $\triangle ABC$,
 $\angle ACD = \angle CAB + \angle CBA$
 $\angle 1 = \angle 2 + \angle 3$

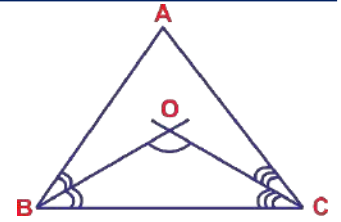
OR

Ext. $\angle C = \angle A + \angle B$



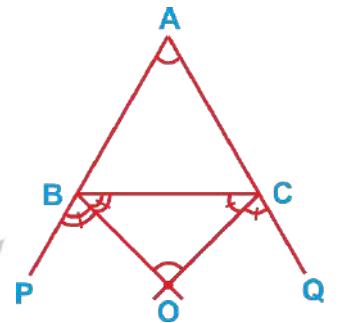
iii) If the bisectors of angles $\angle ABC$ and $\angle ACB$ of a triangle ABC meet at a point O , then,

$$\angle BOC = 90^\circ + \frac{\angle A}{2}$$



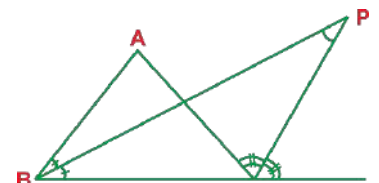
iv) If sides AB and AC of a $\triangle ABC$ are produced to P and Q respectively and if the bisectors of $\angle PBC$ and $\angle QCB$ intersect at O , then:

$$\angle BOC = 90^\circ - \frac{\angle A}{2}$$



v) The angle between internal bisector of one base angle and the external bisector of the other base angle of a triangle is equal to one-half of the vertical angle.

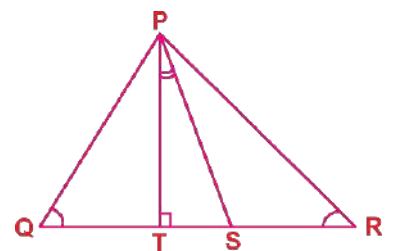
i.e., In adjacent fig., $\angle BPC = \frac{1}{2}\angle BAC$



vi) In $\triangle PQR$, PS is the bisector of $\angle QPR$, & $PT \perp QR$.

Then

$$\angle TPS = \frac{|\angle Q - \angle R|}{2}$$



2.4 Congruent Triangle:

Two triangles are congruent if and only if one of them can be made to superimpose on the other so as to cover it exactly. Corresponding elements of congruent triangles are equal.

If $\triangle ABC \cong \triangle DEF$, then

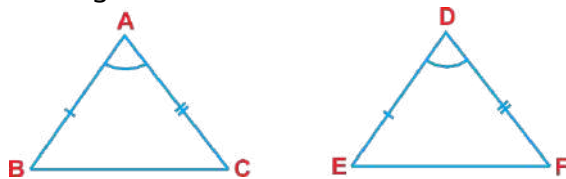
$$\Rightarrow AB = DE; BC = EF; AC = DF$$

$$\Rightarrow \angle A = \angle D; \angle B = \angle E; \angle C = \angle F$$

Sufficient conditions for congruence of triangles:

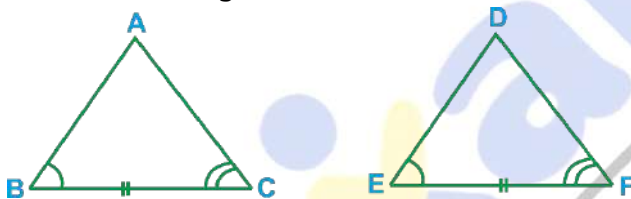
▪ **SAS (Side-Angle-Side):**

Two triangles are congruent if two sides and the included angle of one triangle are equal to the corresponding sides and the included angle of the other triangle.



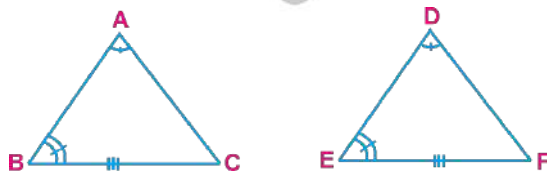
▪ **ASA (Angle-Side-Angle):**

Two triangles are congruent if two angles and the included side of one triangle are equal to the corresponding two angles and the included side of other triangle.



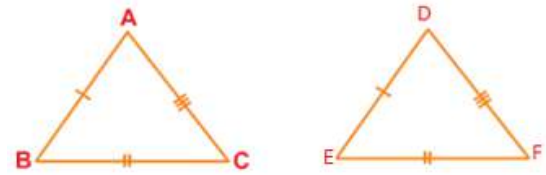
▪ **AAS (Angle-Angle-Side):**

If any two angles and non-included side of one triangle are equal to the corresponding angles and side of another triangle, the two triangles are congruent.



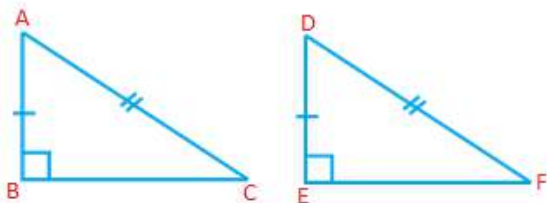
▪ **SSS (Side-Side-Side):**

Two triangles are congruent if the three sides of one triangle are equal to the corresponding three sides of the other triangle.



▪ **(RHS) Right angle - Hypotenuse - Side:**

Two right triangles are congruent if the hypotenuse and one side of one triangle are respectively equal to the hypotenuse and one side of the other triangle.



2.5 Similarity of Triangles:

Two triangles, $\triangle ABC$ and $\triangle PQR$, are similar if and only if corresponding angles have the same measures.

OR

If and only if the lengths of corresponding sides are proportional.

RaMo Mantra :

If $\triangle ABC \sim \triangle PQR$, then

$$\rightarrow \frac{AB}{PQ} = \frac{AC}{PR} = \frac{BC}{QR}$$

This is equivalent to saying that one triangle is an enlargement or diminution of the other.

So, if $\triangle ABC \sim \triangle PQR$, then

$$= \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{h_1}{h_2} = \frac{M_1}{M_2} = \frac{R_1}{R_2} = \frac{r_1}{r_2} = \frac{P_1}{P_2} = \sqrt{\frac{\Delta_1}{\Delta_2}}$$

where h_1 is height, R_1 is circumradius, M_1 is median, r_1 is inradius, P_1 is perimeter and Δ_1 is area of $\triangle ABC$.

and h_2 is height, R_2 is circumradius, M_2 is median, P_2 is perimeter, r_2 is inradius and Δ_2 is area of $\triangle PQR$.

OR

→ Simply, every corresponding ratio will be equal.

- Criteria for similarity of triangles,
i) AAA OR AA ii) SSS
iii) SAS

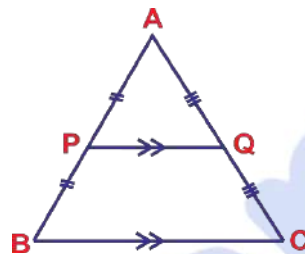
→ Every congruent triangles are similar triangles.

But converse is not always true.

2.6 MPT (Mid-point theorem) :

The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and equal to half of it.

If $PQ \parallel BC$ & P is the midpoint of AB and Q is the midpoint of AC.



Then, $PQ = \frac{1}{2}BC$

Converse is also true:

→ The line drawn through the mid-point of one side of a triangle, parallel to another side, intersects, the third side at its mid-point.

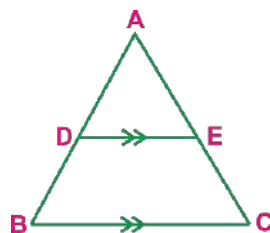
$$\Rightarrow \frac{\text{area}(\Delta PQA)}{\text{area}(\Delta BCA)} = \frac{1}{4}$$

2.7 BPT (Basic Proportionality Theorem) :

If $DE \parallel BC$, then

$$\Delta ABC \sim \Delta ADE,$$

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$



Also,

$$\frac{AD}{DB} = \frac{AE}{EC}$$

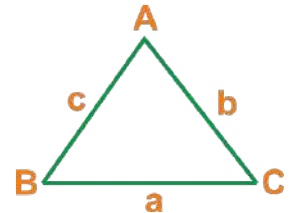
2.8 Inequality Relations in a Triangle:

In ΔABC ,

If $\angle B = \angle C$, then $b = c$

If $\angle B > \angle C$, then $b > c$

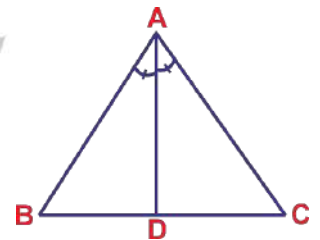
If $\angle B < \angle C$, then $b < c$



2.9 Angle Bisector Theorem:

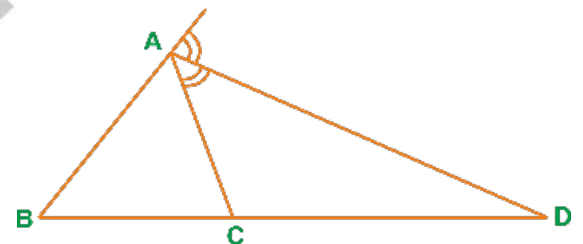
- i) In ΔABC , AD is the angle bisector of $\angle BAC$, then

$$\frac{AB}{BD} = \frac{AC}{DC}$$



- ii) In ΔABC , if AD is an angle bisector of ext. $\angle A$, then

$$\frac{AB}{BD} = \frac{AC}{DC}$$



2.10 Centroid:

- The point in which the three medians of the triangle intersect is known as the centroid of a triangle.
- The median is a line that joins the midpoint of a side and the opposite vertex of the triangle.
- The centroid of the triangle divides the median in the ratio of 2 : 1.

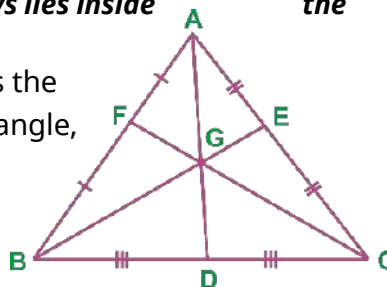
Centroid always lies inside the triangle

In $\triangle ABC$, if G is the centroid of triangle, then

$$AG : GD = 2 : 1$$

$$BG : GE = 2 : 1$$

$$CG : FG = 2 : 1$$



- Median divides the triangle into two parts of equal area. Here,

$$\text{area}(\triangle ABD) = \text{area}(\triangle ADC)$$

$$\text{area}(\triangle BCF) = \text{area}(\triangle ACF)$$

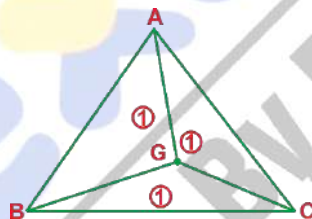
$$\text{area}(\triangle ABE) = \text{area}(\triangle CBE)$$

RaMo Mantra :

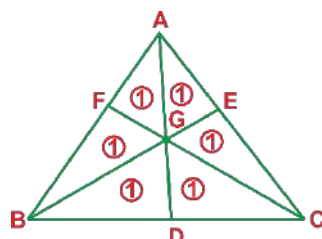
With respect to centroid,

triangle is divided into three parts of equal area i.e., the centroid divides the triangle in three equal parts

i.e. Area (AGB) = Area (AGC) = Area (BGC)



All three medians divide the triangle into 6 equal parts.



- $4\{AD+BE+CF\} > 3\{AB + BC + CA\}$
- $4\{AD^2 + BE^2 + CF^2\} = 3\{AB^2 + BC^2 + CA^2\}$
- $4\{\text{Area of triangle formed by medians (as sides)}\} = 3\{\text{area of } \triangle ABC\}$
i.e. $4\{\text{area of triangle with sides CF, BE \& AD}\}$
 $= 3\{\text{area of } \triangle ABC\}$

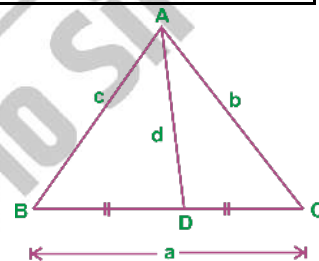
In Simple terms:

- $4 [\sum \text{medians}] > 3 [\sum \text{sides}]$
- $4 [\sum (\text{medians})^2] = 3 [\sum (\text{sides})^2]$
- $4 \{\text{Area of triangle formed by medians}\} = 3 \{\text{Area of } \triangle ABC\}$

v) **Formula for length of Medians :-**

$$\text{Median } AD = \frac{1}{2} \sqrt{2(b^2 + c^2) - a^2}$$

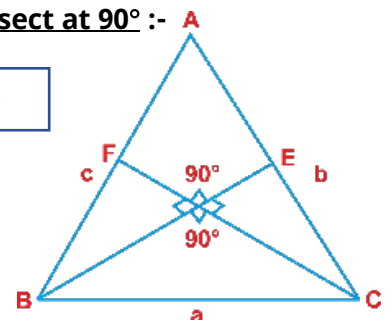
$$AD = \frac{1}{2} \sqrt{2(AB^2 + AC^2) - BC^2}$$



vi) **If medians intersect at 90° :-**

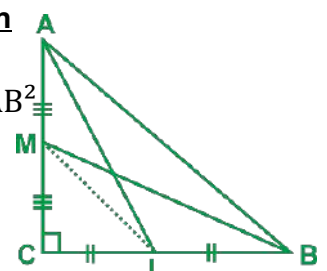
$$\rightarrow 5BC^2 = AB^2 + AC^2$$

$$\rightarrow 5a^2 = b^2 + c^2$$

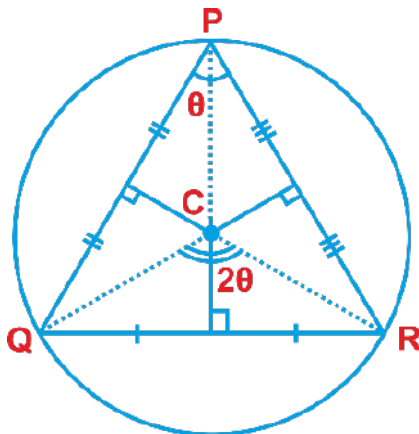


vii) **Relation of Medians in a right triangle:**

$$4(AL^2 + BM^2) = 5AB^2$$



2.11 Circumcenter :-



- i) The circumcenter of a triangle is defined as the point where the perpendicular bisectors of the sides of that particular triangle intersect.

$$PC=QC= RC = \text{circumradius (R)}$$

ii) Here, $\angle QCR = 2(\angle QPR)$

iii) Circumradius(R) = $\frac{abc}{4\Delta}$
where a, b and c are the sides of a triangle and 'Δ' is the area of triangle.

iv) $R = \frac{a}{2 \sin A}$

v) $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

vi) **Location of circumcenter: -**

- The circumcenter of an acute triangle is inside the triangle.
- The circumcenter of an obtuse triangle is outside the triangle.
- The circumcenter of a right-angled triangle is on the hypotenuse.

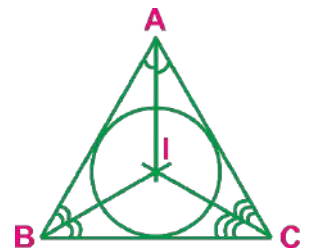
It is the midpoint of hypotenuse.

So, $R = \frac{\text{hypotenuse length}}{2}$ in right triangle

2.12 Incenter:-

- i) Intersection point of all the angle bisectors of a triangle is known as incenter.

It is equidistant from all the sides of the triangle.



- ii) Radius of incircle is known as inradius "r".

$$r = \frac{\Delta}{s}, \text{ where } \Delta \text{ is the area of the triangle \& "s" is semi perimeter.}$$

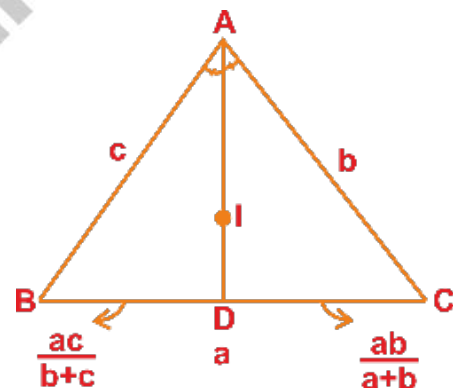
iii) $\angle BIC = 90^\circ + \frac{\angle A}{2}$

- iv) Incenter always lies inside the triangle.

- v) In right angle Δ,

$$r = \frac{p+b-h}{2}$$

p – perpendicular
b – base
h – hypoteneus



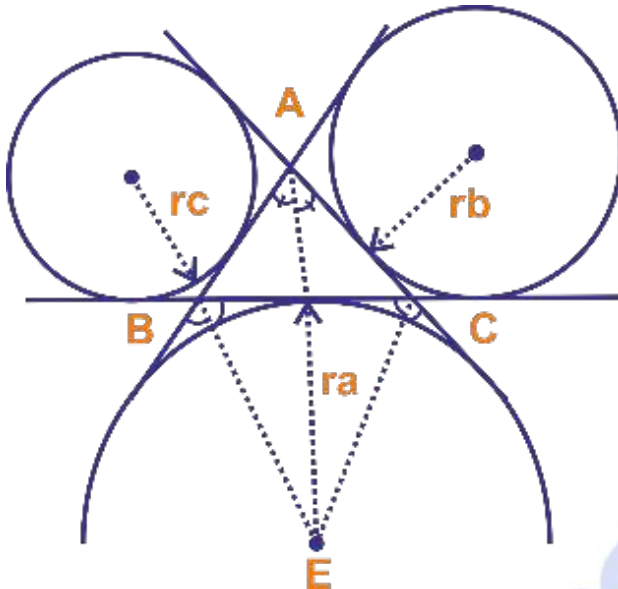
vi) $\frac{AI}{ID} = \frac{b+c}{a}, \frac{BD}{DC} = \frac{AB}{AC}$

- vii) Distance between circumcenter(R) & incenter(r).

$$\text{Distance} = \sqrt{R^2 - 2rR}$$

2.13 Ex-center :-

The intersection point of internal angle bisector of one angle and bisectors of other two opposite exterior angles.



$$i) \angle BEC = 90^\circ - \frac{\angle A}{2}$$

ii) ex-radii :

$$r_a = \frac{\Delta}{s-a}; r_b = \frac{\Delta}{s-b}; r_c = \frac{\Delta}{s-c}$$

$$iii) r_a = \frac{rs}{s-a} = \sqrt{\frac{s(s-b)(s-c)}{s-a}}$$

where, $s = \frac{a+b+c}{2}$

$$iv) \Delta (\text{area}) = \sqrt{r r_a r_b r_c}$$

$$v) r_a + r_b + r_c = 4R + r$$

$$vi) r_a r_b + r_b r_c + r_c r_a = s^2$$

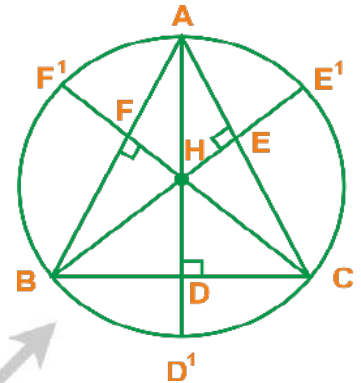
$$vii) r_a^2 + r_b^2 + r_c^2 = (4R + r)^2 - 2s^2$$

2.14 Orthocenter:-

The point where the three altitudes of a triangle intersect is **Orthocenter**.

The orthocenter is not always inside the triangle. If the triangle is obtuse, it will be outside.

Here, "H" is orthocenter.



$$i) \angle BHC = 180^\circ - \angle A$$

$$\angle AHC = 180^\circ - \angle B$$

$$\angle AHB = 180^\circ - \angle C$$

$$ii) \Delta BFH \sim \Delta CEH$$

$$BH \times HE = CH \times HF$$

Also,

$$BH \times HE = CH \times HF = AH \times HD$$

$$iii) FH = FF'; HE = EE'; HD = DD'$$

$$iv) BD \times DC = AD \times DH$$

$$BE \times EH = CE \times EA$$

$$CF \times FH = AF \times FB$$

$$v) (AB + BC + CA) > (AD + CF + BC)$$

$$\Sigma \text{ sides} > \Sigma \text{ Altitudes}$$

vi) **Location of orthocenter:**

- Inside in acute triangle
- Outside if it is an obtuse triangle
- At right vertex if it is a right triangle

RaMo Mantra :

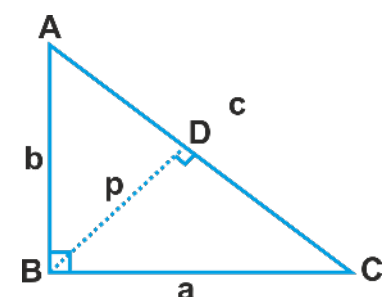
$$\rightarrow BD^2 = CD \times AD$$

$$\rightarrow BD = \frac{AB \times BC}{AC}$$

$$\rightarrow AD = \frac{AB^2}{AC};$$

$$CD = \frac{BC^2}{AC}$$

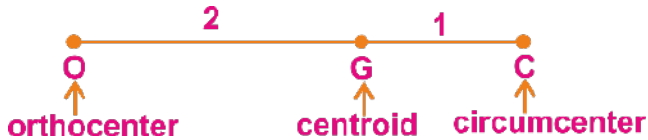
$$\rightarrow AD : CD = AB^2 : BC^2$$



$$\rightarrow \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

Relation between Orthocenter, centroid and circumcenter:

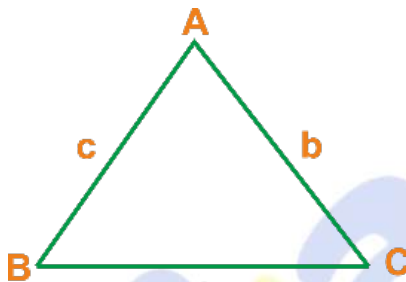
In any triangle, orthocenter, centroid and circumcenter are co-linear and centroid divides the line in 2:1 as in figure.



2.15 Sine Rule :-

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

2.16 Cosine Rule :-



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{b^2 + a^2 - c^2}{2ab}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

2.17 Expression for area :-

Perimeter = a+b+c

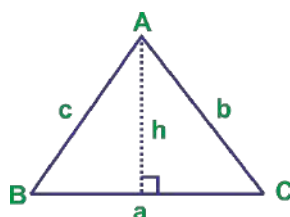
Semi perimeter

$$(s) = \frac{a+b+c}{2}$$

$$\text{Area } (\Delta) = \frac{1}{2} \times$$

height $\times$ base

Generally,

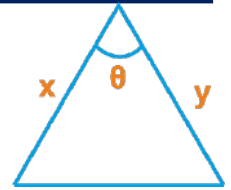


$$\text{Area of triangle} = \frac{1}{2} ac \sin B$$

$$\text{Area of triangle} = \frac{1}{2} ab \sin C$$

$$\text{Area of triangle} = \frac{1}{2} bc \sin A$$

$$\therefore \text{Area of triangle} = \frac{1}{2} \times xy \sin \theta$$



By heron's formula,

$$\text{Area of triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

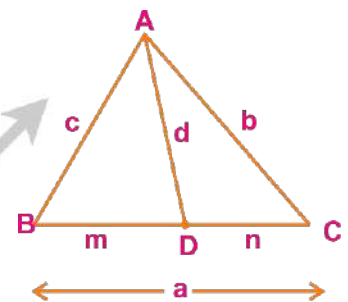
2.18 How to find length of cevian – Stewart's Theorem

i) **Stewart's theorem :-**

Here, AD = d

(known as cevian)

Cevian is any line from the vertex to the base opposite to it.



$$b^2m + c^2n = a(d^2 + mn)$$

If cevian is median :-

$$\text{So, } d = \frac{1}{2} \sqrt{2(b^2 + c^2) - a^2}$$

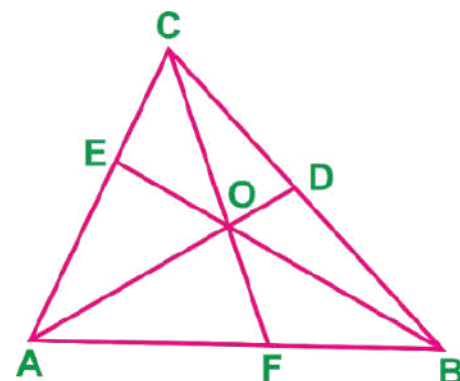
if cevian is angle bisector :-

$$d = 2 \sqrt{\frac{bc}{b+c} s(s-a)}$$

where $s = \frac{a+b+c}{2}$

and side of length a is divided in the proportion b : c

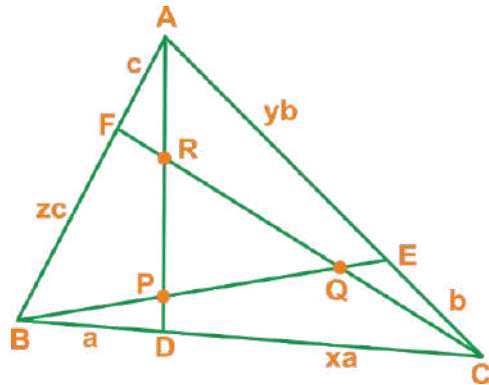
2.19 Ceva's Theorem :-



In $\triangle ABC$, let AD, BE and CF are the cevians (i.e., any line from vertex to opposite base), then

$$\frac{AF}{FB} \times \frac{BD}{DC} \times \frac{CE}{EA} = 1$$

2.20 Routh Theorem:



If in $\triangle ABC$, points D, E and F lie on segments BC, CA and AB, then writing -

$$\frac{CD}{BD} = x; \frac{AE}{CE} = y \text{ and } \frac{BF}{AF} = z$$

The area of the triangle formed by the cevians AD, BE and CF is:

$$= \text{area of } \triangle ABC \times \left\{ \frac{(xyz-1)^2}{(xy+y+1)(yz+z+1)(zx+x+1)} \right\}$$

2.21 Standard Equilateral Triangle

∴

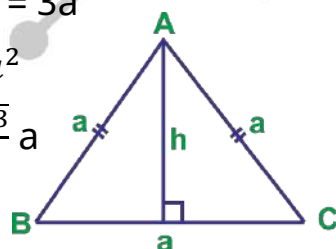
$$\text{Perimeter} = 3a$$

$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$\text{height} = \frac{\sqrt{3}}{2} a$$

$$R = \frac{a}{\sqrt{3}},$$

$$r = \frac{a}{2\sqrt{3}}$$



$$\frac{\text{area(circumcircle)}}{\text{area(incircle)}} = 4 : 1$$

RaMo Mantra :

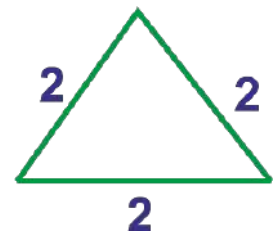
If side = 2

Perimeter = 6

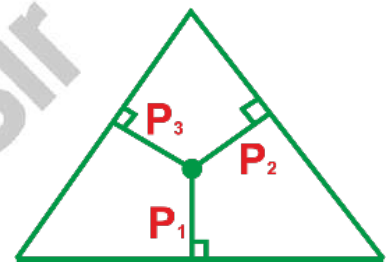
Height = $\sqrt{3}$

$$R = \frac{2}{\sqrt{3}}, r = \frac{1}{\sqrt{3}}$$

Area = $\sqrt{3}$ sq. unit.



- All angle bisectors, all medians, all $\perp$ bisectors & all altitudes are same.
- All 4 types of center are on same point.
- If length of perpendiculars on sides from any point inside the equilateral are P_1, P_2 and P_3 . Then, side is given by :-



$$a = \frac{2}{\sqrt{3}} (P_1 + P_2 + P_3)$$

$$\Rightarrow P_1 + P_2 + P_3 = h$$

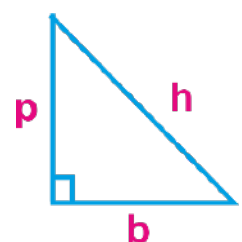
2.22 Right angled Triangle :-

$$\bullet \text{ Area} = \frac{1}{2} \times p \times b$$

$$\bullet r = \frac{p+b-h}{2}$$

$$\bullet R = \frac{h}{2}$$

$$\bullet 2(r + R) = p + b$$

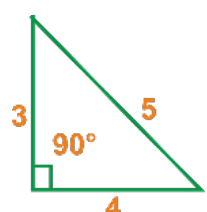


$$\bullet \text{ Area of triangle} = s(s - \text{hypotenuse})$$

$$\text{Area of triangle} = s(s - 2R) \text{ or}$$

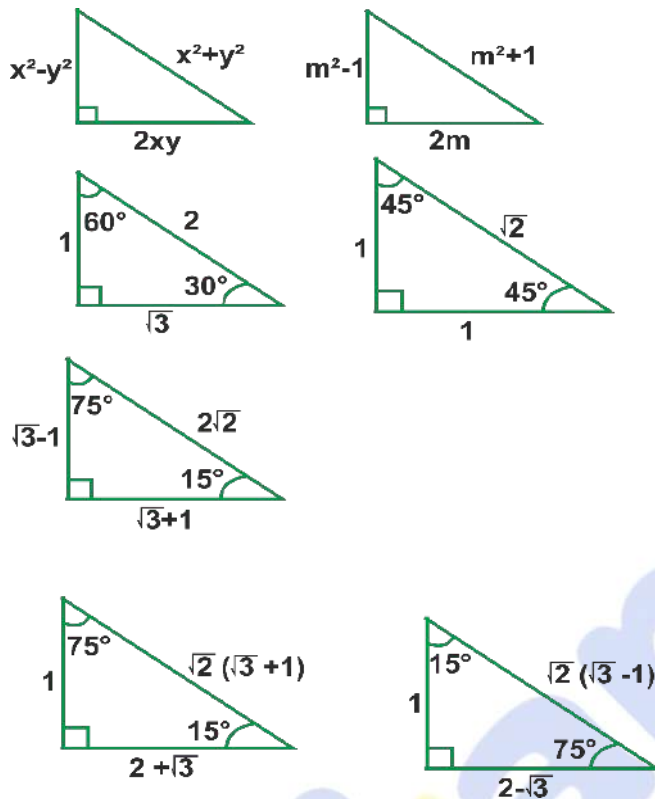
$$\text{Area of triangle} = s(s - 2m) \text{ \{m is the shortest median\}}$$

- If, Area = 6 sq. unit and Perimeter = 12 units, then



$r = 1$ unit, $R = 2.5$ unit
 $R : r = 5 : 2$

• **Special right angled triangle:-**



• **Pythagoras triplets for right angled triangle :-**

We can generate by using these

$2m$, $m^2 + 1$, $m^2 - 1$

$2xy$, $x^2 + y^2$, $x^2 - y^2$

(3, 4, 5)

(6, 8, 10)

(5, 12, 13)

(7, 24, 25)

(8, 15, 17)

(9, 40, 41)

(11, 60, 61)

(12, 35, 37)

(13, 84, 85)

(16, 63, 65)

(20, 21, 29)

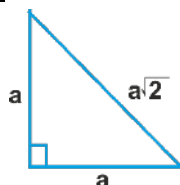
• **Isosceles-right triangle :-**

$P = \text{perimeter} = a(2 + \sqrt{2})$

$\text{area} = \frac{1}{2} a^2$

$= \frac{p^2}{2(6 + 4\sqrt{2})}$

$= \frac{p^2}{4} (3 - 2\sqrt{2})$



• **For isosceles triangle :-**

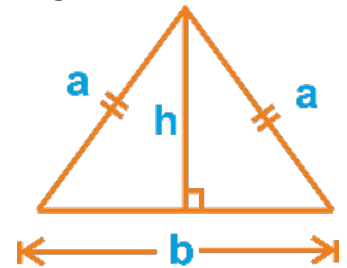
height; $h = \frac{a^2}{2R}$

circumradius;

$R = \frac{a^2}{\sqrt{4a^2 - b^2}}$

$b =$

$\frac{a}{R} \sqrt{(2R + a)(2R - a)}$



2.23 Mass Point Geometry:-

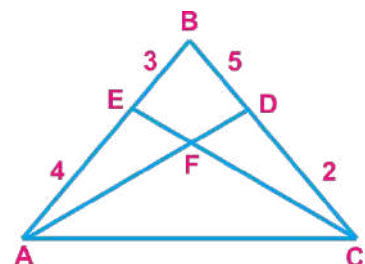


$$m_1 x = m_2 y$$

$$\frac{m_1}{m_2} = \frac{x}{y}$$

Let's take an example to understand -

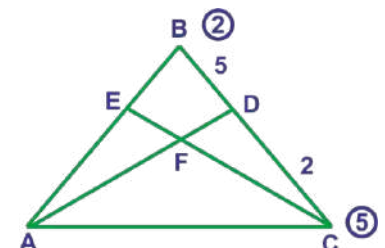
Q. In a $\triangle ABC$, side BC is divided by D in the ratio of 5:2 and BA is divided by E in a ratio 3:4 as shown in the figure. Find the ratios in which F divides the cevians AD & CE i.e., $EF:FC = ?$, $DF:FA = ?$



Sol.

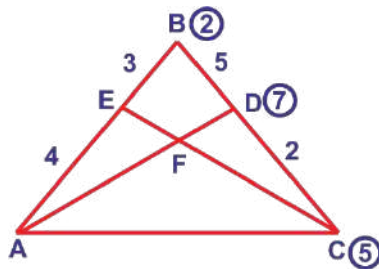
Step-1: In order to have D as the balancing point of BC .

We assign a mass of 2 kg to B and 5 kg to C .



Step-2: Now mass at D -

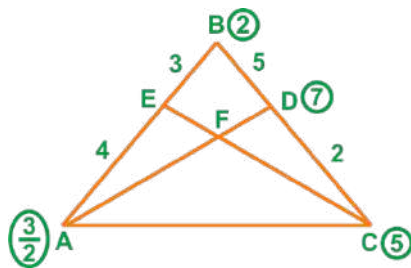
$2\text{kg} + 5\text{kg} = 7\text{kg}$, i.e., 7kg at D.



Step-3: Now on side AB, to have E as a balancing point, we assign weight on A

$$= 2 \times \frac{3}{4} \text{ kg}$$

$$= \frac{3}{2} \text{ kg at A}$$



Step-4: Now mass at E.

$$= \left(2 + \frac{3}{2}\right) \text{ kg}$$

$$= \frac{7}{2} \text{ kg}$$

Step-5: Now the ratio of DF:FA to have F as balancing point

$$DF : FA = \frac{3}{2} : 7 = 3:14$$

$$\text{And, } EF : FC = 5 : \frac{7}{2} = 10:7$$

3 Quadrilateral :-

$$\bullet \angle A + \angle B + \angle C + \angle D = 360^\circ$$

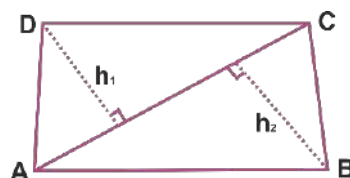
$$\bullet \text{Area}(ABCD) = \frac{1}{2}AC(h_1 + h_2)$$

• If four sides a, b, c and d are given, then

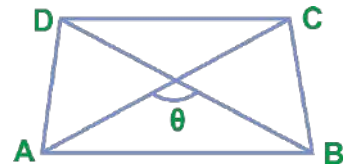
$$\text{area}(ABCD) =$$

$$\sqrt{(s-a)(s-b)(s-c)(s-d)}$$

$$\text{where, } s = \frac{a+b+c+d}{2}$$



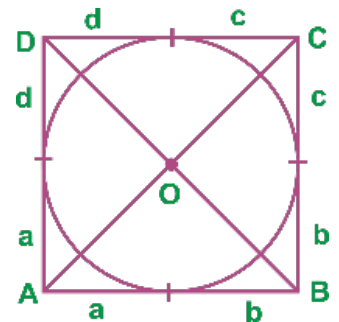
- If d_1, d_2 are the diagonals :
 $\text{area} = \frac{1}{2} d_1 d_2 \sin \theta$
where θ is the angle at which diagonals intersect.



- If diagonals intersect at 90° .

$$AB^2 + CD^2 = BC^2 + AD^2$$

- If a circle touches all the sides of quad. ABCD, then
Sum of opposite sides is equal.
 $\Rightarrow AB + CD = BC + AD$



- O is the center of the circle,
(Not the intersection point of diagonals), then
 $\Rightarrow \angle AOB + \angle COD = \angle DOA + \angle BOC$

• Cyclic quadrilateral :-

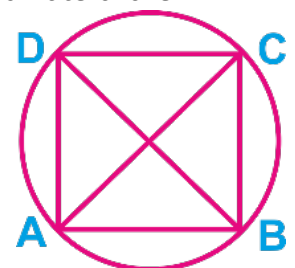
If all the vertex of quadrilateral are on the circle,

Quadrilateral is called cyclic

$$\text{Here, } \angle A + \angle C = 180^\circ$$

$$\angle B + \angle D = 180^\circ$$

$$AC \times BD = AB \times CD + BC \times AD$$

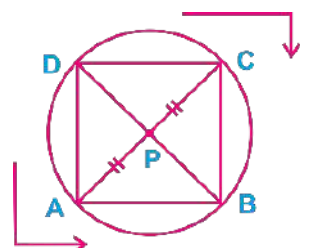


• If one diagonal bisects other:

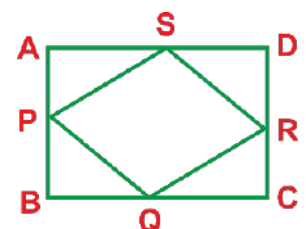
$$\triangle DPC \sim \triangle APB$$

$$\triangle DPA \sim \triangle CPB$$

$$AB \times AD = BC \times CD$$

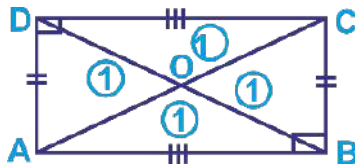


- Quadrilateral formed by joining the mid-points of the pairs of consecutive sides of a quadrilateral is a parallelogram.
PQRS is a || gm.



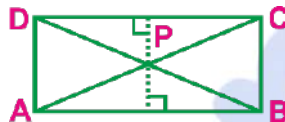
3.1 Rectangle :-

$AB = DC; AD = BC; AC = BD$
 $\angle A = \angle B = \angle C = \angle D = 90^\circ$



- Diagonals AC & BD bisect each other.
- Diagonals divide rectangle into four equal parts.
- $\text{area}(\triangle ABC) = \text{area}(\triangle BCD) = \text{area}(\triangle ABD) = \text{area}(\triangle ACD) = \frac{1}{2}(\text{area of } ABCD)$
- $\text{area}(\triangle OAB) = \text{area}(\triangle OBC) = \text{area}(\triangle OCD) = \text{area}(\triangle OAD) = \frac{1}{4}(\text{area of } ABCD)$

If P is any point inside the rectangle.



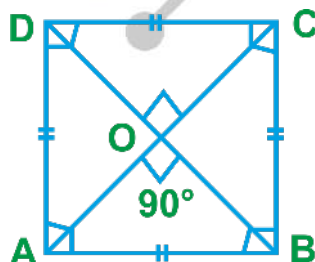
$$AP^2 + CP^2 = BP^2 + DP^2$$

- Quadrilateral formed by joining the mid-points of the consecutive sides of a rectangle is a rhombus.



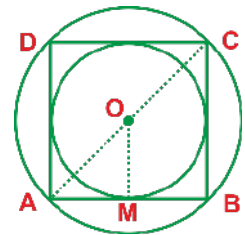
3.2 Square :-

- $AB = BC = CD = DA = a$



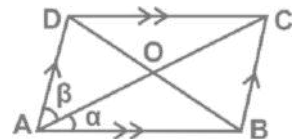
- $\angle A = \angle B = \angle C = \angle D = 90^\circ$
- $AC = BD = a\sqrt{2}$
- $\text{area} = a^2$
- Diagonals bisect each other at 90°
- O is mid point of diagonals.
- Square has all the properties of rectangle.

- $\text{area}(\triangle AOB) = \text{area}(\triangle BOC) = \text{area}(\triangle COD) = \text{area}(\triangle DOA) = \frac{1}{4}a^2$
- $R = OA = OC$ & $OM = r$
 $\Rightarrow R : r = \frac{a\sqrt{2}}{2} : \frac{a}{2}$
 $= \sqrt{2} : 1$



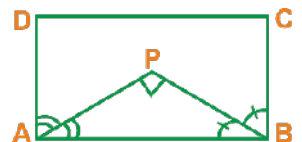
3.3 Parallelogram :-

- $AB \parallel CD; AB = CD$
- $BC \parallel AD; AD = BC$
- $\angle A = \angle C; \angle B = \angle D$
- $\angle A + \angle B = \angle C + \angle D = \angle C + \angle B = \angle A + \angle D = 180^\circ$
- AC & BD, but bisect each other.
- $\text{area of } \triangle ABC = \triangle BCD = \triangle ABD = \triangle ACD = \frac{1}{2}(\text{area of } ABCD)$
- $\text{area}(\triangle AOB) = \text{area}(\triangle BOC) = \text{area}(\triangle COD) = \text{area}(\triangle DOA) = \frac{1}{4} \text{area}(ABCD)$
- $\text{area}(ABCD) = \text{height} \times \text{base}$
- $\text{area}(\text{||gm } ABCD) = AD \times AB \sin \angle DAB$

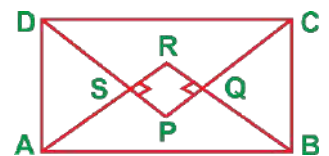


In a ||gm, the bisectors of any two consecutive angles intersect at right angle.

$$\angle APB = 90^\circ$$



- The angle bisectors of a parallelogram form a rectangle. PQRS is a rectangle.



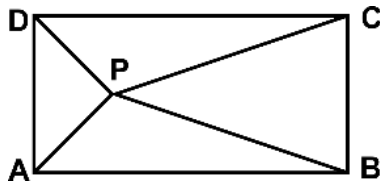
- A quadrilateral is a parallelogram if its opposite sides are equal.
- A quadrilateral is a parallelogram if its opposite angles are equal.
- If diagonals of a quadrilateral bisect each other, then the quadrilateral is a || gm.

vi) A quadrilateral is a parallelogram, if its one pair of opposite sides are equal and parallel.

vii) If the diagonals of a $\parallel$ gm are equal and intersect at 90° , the $\parallel$ gm is a square.

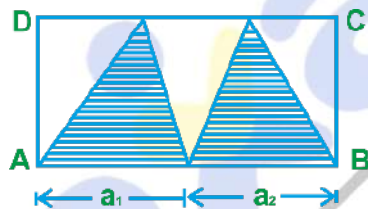
viii) area of $\parallel$ gm = $(s - a)(s - b)$, where s is semiperimeter.

ix) If P is any point inside $\parallel$ gm, then

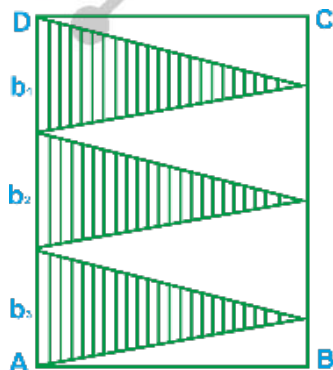


$$\begin{aligned} \text{Area}(\triangle APD) + \text{area}(\triangle BPC) \\ &= \text{area}(\triangle APB) + \text{area}(\triangle PDC) \\ &= \frac{1}{2} \text{area}(\parallel \text{gm}) \end{aligned}$$

x) If ABCD is a $\parallel$ gm, then
Area of shaded region = $\frac{1}{2}(\parallel \text{gm area})$



xi) Area of shaded region = $\frac{1}{2}(\text{area of } \parallel \text{gm})$

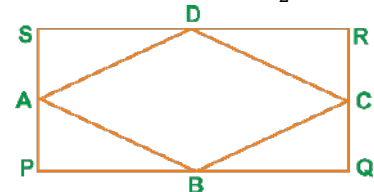


xii) If A, B, C and D are the midpoints of PS, PQ, QR & RS respectively, then

$$\text{Area}(\parallel \text{gm } ABCD) = \frac{1}{4} \text{area}$$

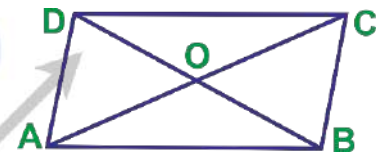
(PQRS)

$$(\text{ABCD}) \text{ Perimeter} = \frac{1}{2}(\text{PQRS})$$



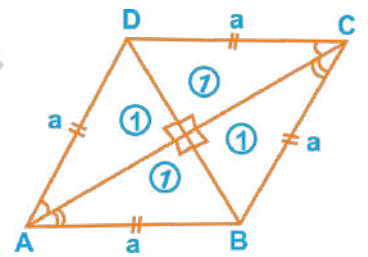
$$\begin{aligned} \text{xiii) } AB^2 + BC^2 + CD^2 + DA^2 &= 2(AB^2 + BC^2) \\ &= 2[2(BO^2 + CO^2)] \\ &= (2BO)^2 + (2CO)^2 \\ &= AC^2 + BD^2 \end{aligned}$$

$$\text{So, } \sum (\text{diagonals})^2 = \sum (\text{sides})^2$$



3.4 Rhombus :-

- $AB = BC = CD = DA$,
- AC is not equal to DB, {but bisect each other at 90° }



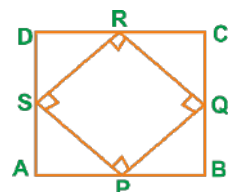
- Diagonals bisect the vertex angle.
- Area = height $\times$ base = $\frac{1}{2} \times d_1 \times d_2$

$$\begin{aligned} 4a^2 &= d_1^2 + d_2^2 \\ a &= \frac{1}{2} \sqrt{d_1^2 + d_2^2} \end{aligned}$$

- A $\parallel$ gm is a rhombus if adjacent sides are equal.

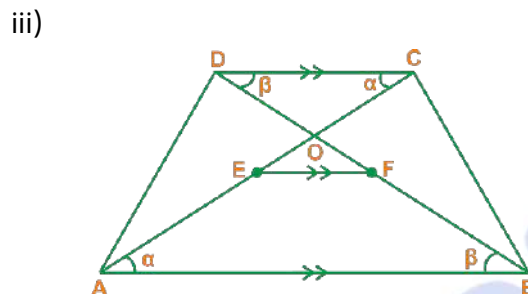
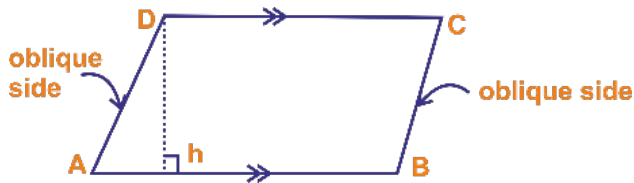
- Diagonal divides the rhombus into 4 congruent right triangles.

- Quadrilateral formed by mid points of sides of a rhombus is a rectangle. Here ABCD is a rhombus. and PQRS is a rectangle.



3.5 Trapezium :-

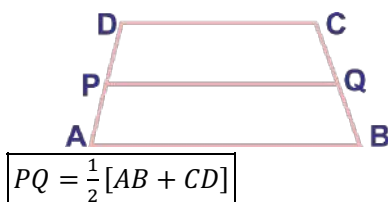
- i) $AB \parallel CD$, $\angle A + \angle D = \angle B + \angle C = 180^\circ$
Area = $\frac{1}{2} h (AB + CD)$



- $\triangle AOB \sim \triangle COD$
- $AO \times DO = CO \times BO$
- $AC^2 + BD^2 = AD^2 + BC^2 + 2AB \cdot CD$
- If E & F are the mid points of diagonals

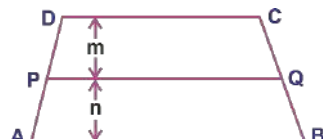
$$EF = \frac{1}{2} [AB - CD]$$

- iv) P is the mid point of AD.
Q is the mid point of BC.



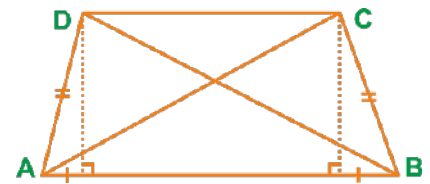
- v) $PQ = \frac{m}{m+n} \times \text{longer side (AB)}$
 $+ \frac{n}{m+n} \times \text{shorter side (DC)}$

Note: m towards shorter side
(Here, it is CD)



vi) Isosceles trapezium :-

- $\angle A + \angle D = \angle B + \angle C = 180^\circ$
- $\angle A + \angle C =$
 $\angle B + \angle D =$
 180°
- Isosceles trapezium is a cyclic quadrilateral
- $AC = BD$
- $\angle A = \angle B$; $\angle C = \angle D$
- $AC^2 + BD^2 = AD^2 + BC^2 + 2AB \cdot CD$

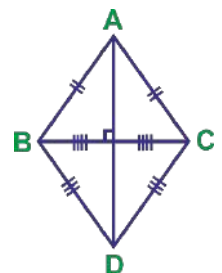


$$AC^2 = AD^2 + AB \cdot CD$$

- Only isosceles trapezium can be inscribed in a circle.

3.6 Kite :-

- Two pair of adjacent sides will be equal.
- Larger diagonal perpendicularly bisect the shorter diagonal.
- Area = $\frac{1}{2} d_1 \times d_2$
- d_1, d_2 are the diagonals.
- Larger diagonal divides kite in 2 congruent $\triangle$ s.
- A convex kite always inscribes a circle.



4 Polygon :-

- A region bounded by many lines.
- Sum of internal angles = $(n - 2) \times 180^\circ$, where n is the number of sides.
- Sum of external angles of polygon = 360°



- Number of diagonals = $\frac{n(n-3)}{2}$
- Each internal angle = $\left(180^\circ - \frac{360^\circ}{n}\right)$
- Each external angle = $\frac{360^\circ}{n}$
- $\frac{\text{Int. angle}}{\text{ext. angle}} = \frac{(n-2)}{2}$

4.1 Regular Polygon :-

No. of sides (n)	Σ Int.	Each int.	Σ Ext.	Each ext. ($\frac{360^\circ}{n}$)	Int. : ext.
3	180°	60°	360°	120°	1 : 2
4	360°	90°	360°	90°	1 : 1
5	540°	108°	360°	72°	3 : 2
6	720°	120°	360°	60°	2 : 1
7	900°	900°/7	360°	360°/7	5 : 2
8	1080°	135°	360°	45°	3 : 1
9	1260°	140°	360°	40°	7 : 2
10	1440°	144°	360°	36°	4 : 1

❖ General formula for a regular polygon :- [n-sided]

• Area = $n \frac{a^2}{4} \cot \left(\frac{\pi}{n} \right)$

At n = 3,

Area of Equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

At n = 4,

Square area = a^2

At n = 6,

Area of regular hexagon = $\frac{3\sqrt{3}}{2} a^2$

❖ Circum radius :-

$R_n = \frac{a}{2} \operatorname{cosec} \left(\frac{\pi}{n} \right)$

❖ Inradius :-

$r_n = \frac{a}{2} \cot \left(\frac{\pi}{n} \right)$

❖ Side :-

$a = 2r \tan \left(\frac{\pi}{n} \right)$

$a = 2R \sin \frac{180^\circ}{n}$

❖ Area :-

$= nr^2 \tan \left(\frac{\pi}{n} \right)$

$= nR^2 \sin \left(\frac{2\pi}{n} \right)$

4.2 Regular Hexagon :-

Number of sides = 6

Σ interior angles =

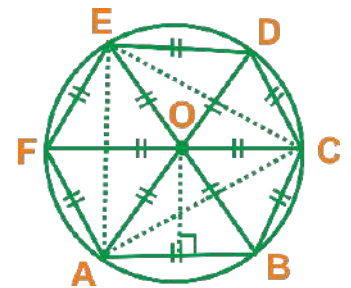
720°;

each interior angle =

120°

Each exterior angle =

60°

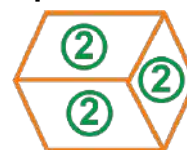


- area = 6 × area of eq. Δ
- area = $6 \left(\frac{\sqrt{3}}{4} a^2 \right) = \frac{3\sqrt{3}}{2} a^2$
- Circumradius = a
- In-radius = $\frac{\sqrt{3}}{2} a$
- $d_1 = AC = EC = AE = FB = BD = DF = a\sqrt{3}$ {smaller diagonals}

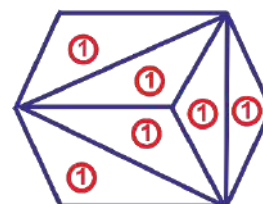
- $d_2 = AD = BE = CF = 2a$ {larger diagonals}
- $d_1 : d_2 = \sqrt{3} : 2$
- For standard value, with side = 2,
Area $6\sqrt{3}$ sq. unit
Perimeter = 12
- Area of hexagon, when (d_1 ; smaller diagonal is given)
 $A = \frac{\sqrt{3}}{2} d_1^2$
- Area of hexagon when (d_2 ; larger diagonal is given)
 $A = \frac{3\sqrt{3}}{8} d_2^2$

❖ Divisions in equal area :-

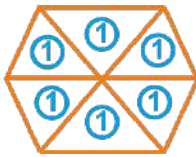
• 3 equal area division :-



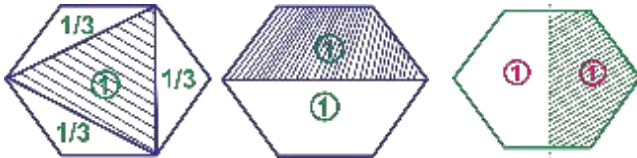
• 6 equal area division :-



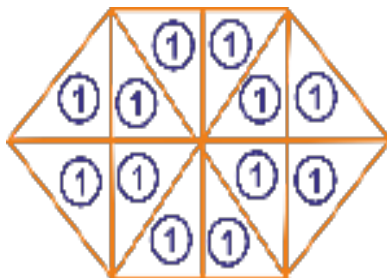
• 6 equal area division :-



- **2 equal area division :**



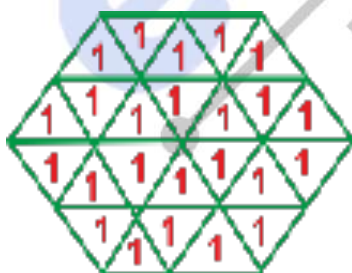
- **12 equal area division :**



- **4 equal area division :**



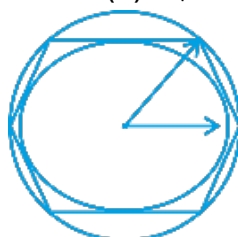
- **24 equal area division :**



Special Case-

- Inradius(r) : Circumradius(R) = $\sqrt{3} : 2$

=> Incircle :
Circumcircle = 3 : 4



=> Circumcircle : area of hexagon :

Incircle

$$= \pi a^2 : \frac{3\sqrt{3}}{2} a^2 : \pi \frac{3a^2}{4}$$

$$= 4\pi : 6\sqrt{3} : 3\pi$$

❖ **Regular Hexagon from equilateral**

Δ :-

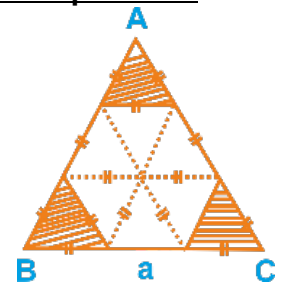
= Area of hexagon :

area of equilateral Δ

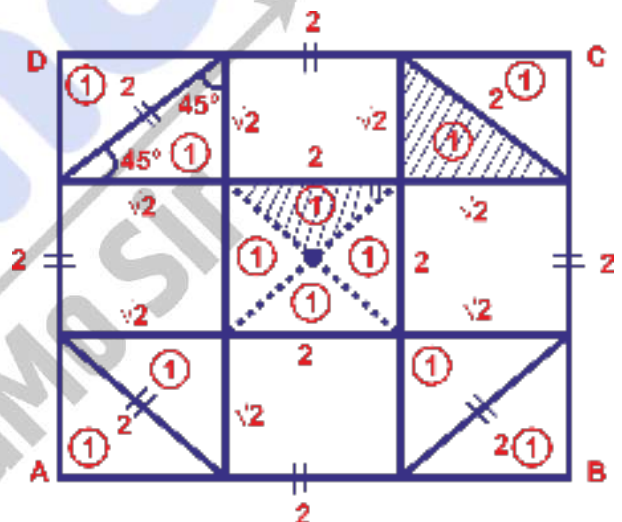
ABC

$$= 6 : 9$$

$$= 2 : 3$$



4.3 Regular Octagon :-



Octagon area = 1 square + 4 rectangles + 4 triangles

$$= 2 \times 2 + 4 \times 2 \times \sqrt{2} + 4 \times \frac{1}{2} \sqrt{2} \times \sqrt{2}$$

$$= 4 + 8\sqrt{2} + 4$$

$$\text{Area} = 8(1+\sqrt{2})$$

$$= 2 \times 2^2(1+\sqrt{2})$$

$$\text{Area} = 2a^2(1+\sqrt{2})$$

- Side of square ABCD

$$x = a(1+\sqrt{2})$$

- Side of octagon

$$a = x(\sqrt{2}-1)$$

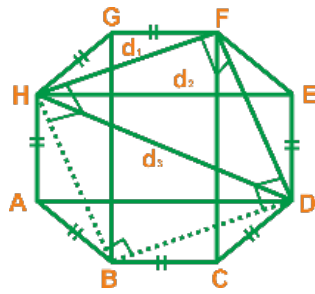
$$\frac{x}{a} = \frac{\text{side(outer square)}}{\text{side(inner square)}} = \frac{1+\sqrt{2}}{1}$$

$$\frac{x^2}{a^2} = 3 + 2\sqrt{2}$$

$$\frac{a^2}{x^2} = 3 - 2\sqrt{2}$$

- Number of diagonals = 20

- a (side of octagon)
 $d_1 = HF = a\sqrt{2 + \sqrt{2}}$
 $d_2 = HE = a(1 + \sqrt{2})$
 $d_3 = HD = a\sqrt{2}(\sqrt{2 + \sqrt{2}})$
 $d_1 : d_2 : d_3 = 2^{\frac{1}{4}} : \sqrt{1 + \sqrt{2}} : 2^{\frac{3}{4}}$



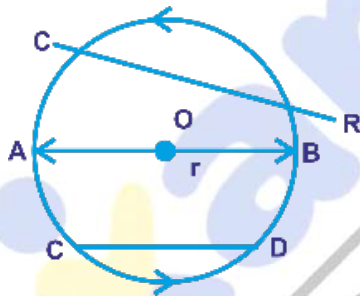
$$d_1 : d_2 : d_3 = 2^{\frac{1}{4}} : (1 + \sqrt{2})^{\frac{1}{2}} : 2^{\frac{3}{4}}$$

5 Circle :-

"A circle is the locus of a point which moves in a plane in such a way that its

distance from a given fixed point in the plane is always constant"

- OB = OA = radius
 - AB is diameter.
 - Diameter = 2 × radius
 - CD is chord.
 - CR is secant line.
 - AB is the largest chord.
- ARC : A continuous piece of a circle is called an arc of the circle. CD is an arc.



Minor arc :-

A minor arc of a circle is the collection of those points of the circle that lie on or inside a central angle.

CD is minor arc.

DC is major arc.

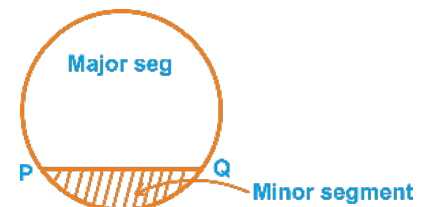
Major arc :-

A major arc of a circle is the collection of points of the circle that lie on or outside a central angle.

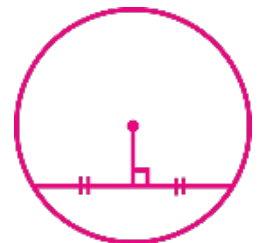
DC is major arc.

5.1 Segment of a circle :-

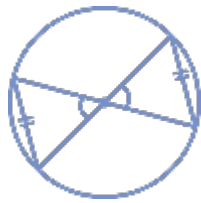
Chord PQ divides the circle into two parts, each part is called a segment of circle.



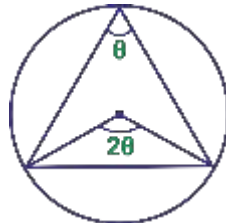
- Segment containing minor arc is called minor segment
- Segment containing major arc is called major segment
- The major and minor segments of a circle are called the alternate segments of each other.
- The perpendicular from the centre of a circle to a chord bisects the chord.
- There is one and only one circle passing through 3 non-collinear points.
- Equal chords of a circle are equidistant from centre.



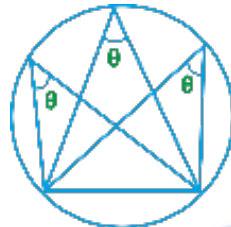
- Equal chords of a circle subtend equal angles at centre.



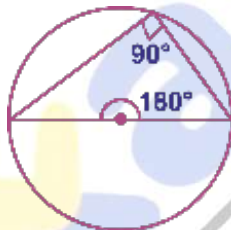
- If a chord subtends 2θ at centre of a circle, it will subtend θ at circumference.



- Any chord subtends equal angle at circumference in the same segment.



- Angle in semicircle is 90°

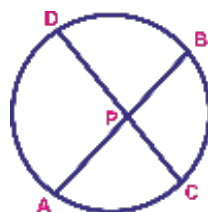


5.2 Alternate segment Theorem :-

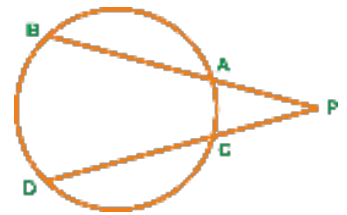
The angle between a chord and a tangent through one of the end points of the chord is equal to the angle in the alternate segment.



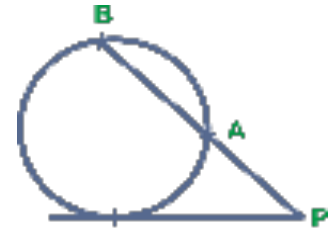
- $PA \times PB = PC \times PD$



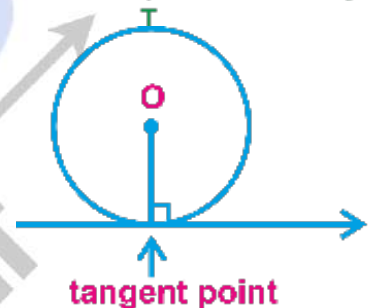
- $PA \times PB = PC \times PD$



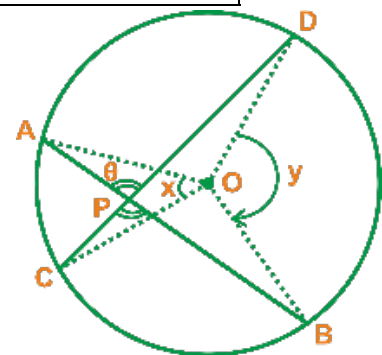
- $PT^2 = PA \times PB$



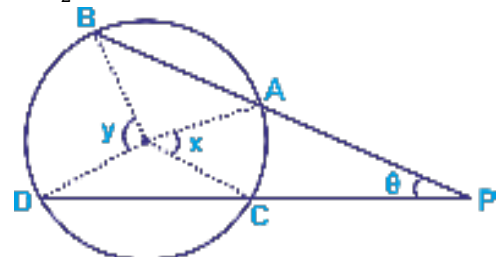
- Tangent point



- $\angle APC = \angle BPD = \frac{1}{2}(x + y)$



- $\angle APC = \frac{1}{2}|y - x|$

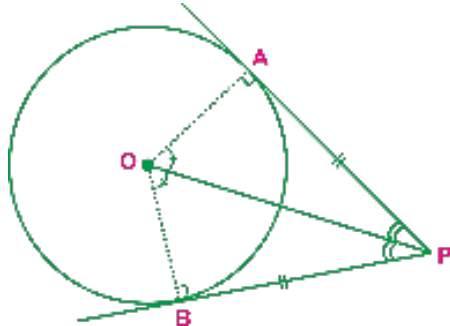


5.3 Tangents from external points

:-

$$PA = PB$$

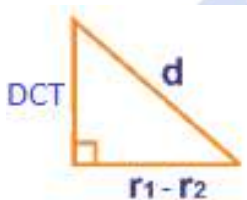
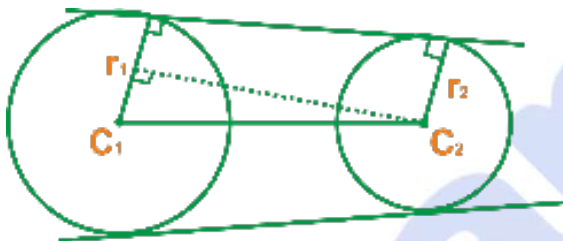
Here, $\triangle OPA \cong \triangle OPB$



5.4 Direct common tangent :-

Here, length of $\overline{C_1C_2} = d$

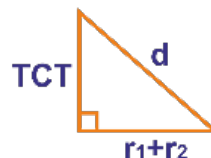
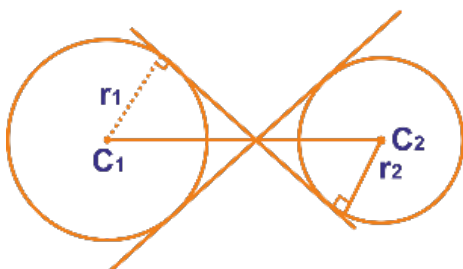
$$DCT = \sqrt{d^2 - (r_1 - r_2)^2}$$



5.5 Transverse common tangent :-

$$TCT = \sqrt{d^2 - (r_1 + r_2)^2}$$

Here, $d > (r_1 + r_2)$

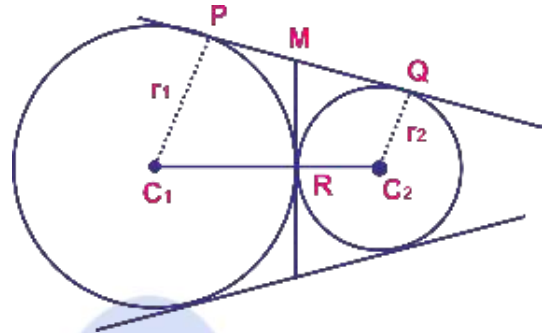


5.6 For Externally touching circles

:-

$$PQ = 2\sqrt{r_1 r_2}$$

$$PM = MQ = MR = \sqrt{r_1 r_2}$$

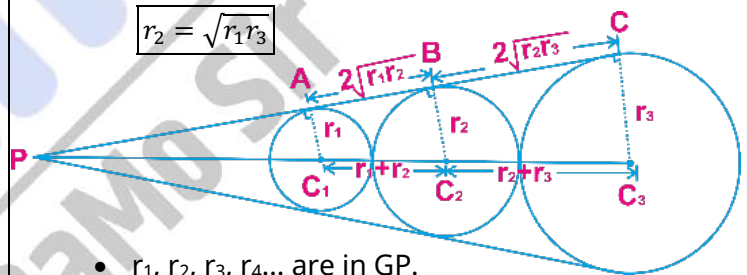


- $\triangle PAC_1 \sim \triangle PBC_2 \sim \triangle PCC_3$

$$\frac{2\sqrt{r_1 r_2}}{2\sqrt{r_2 r_3}} = \frac{r_1 + r_2}{r_2 + r_3}$$

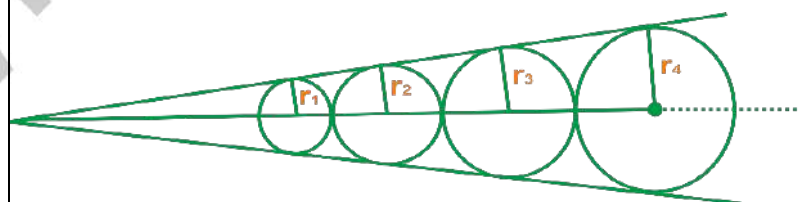
$$r_2^2 = r_1 r_3$$

$$r_2 = \sqrt{r_1 r_3}$$



- $r_1, r_2, r_3, r_4 \dots$ are in GP.

$$r_2 = \sqrt{r_1 r_3}$$



- When two circles are away,
Maximum number of tangents = 4
Minimum number of tangents = 0
- When two circles touch externally,
Maximum number of tangents = 3
Minimum number of tangents = 1
- When two circles cut each other :-
Maximum number of tangents = 2
Minimum number of tangents = 0

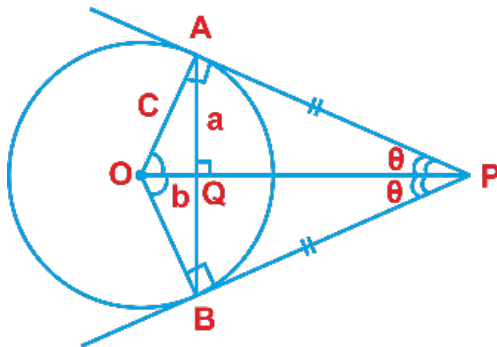
- When two circles touch internally,
Maximum number of tangents = 1
Minimum number of tangents = 1

- $PA = PB, AQ = QB, PQ \perp AB$

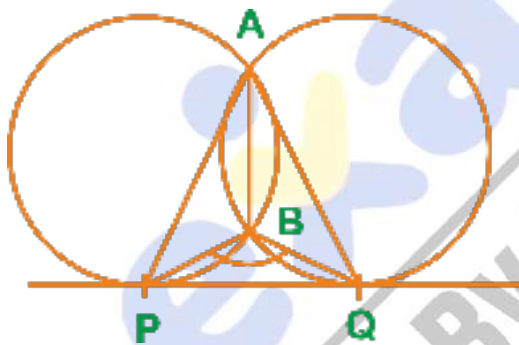
$\triangle PAQ \cong \triangle PBQ$ such that $PA = \frac{a \times c}{b}$

$\triangle PAO \cong \triangle PBO$

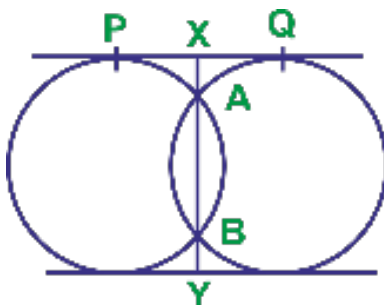
$\angle AOP = \angle BOP$



- $\angle PAQ + \angle PBQ = 180^\circ$

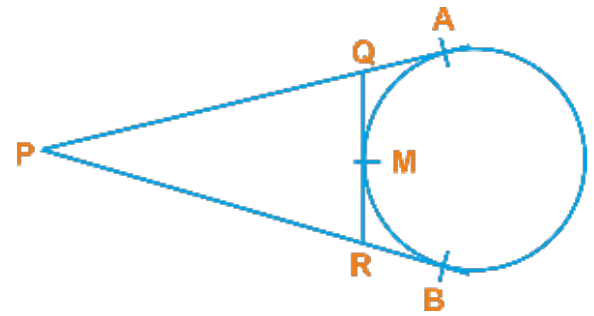


- $PQ^2 = XY^2 - AB^2$



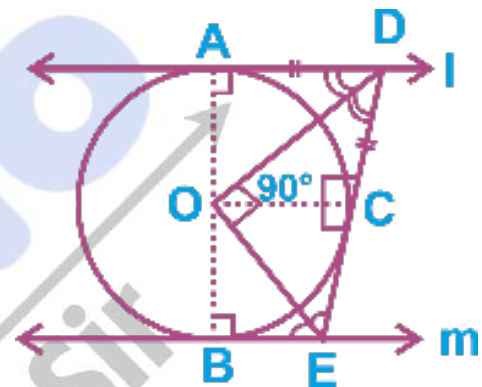
- $PA = PB, QA = QM \text{ \& } RM = RB$
So, $PQ + QA = PR + RB$

$$PQ + QM = PR + RM$$

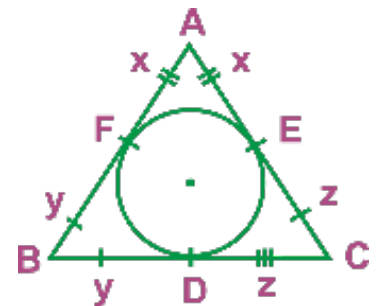


- Perimeter of $\triangle PQR = PA + PB$

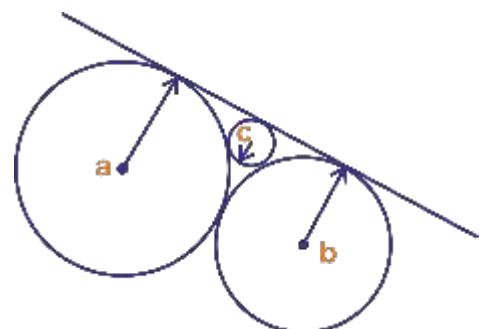
- in fig., if $l \parallel m$, then $\angle DOE = 90^\circ$



- Perimeter of $\triangle ABC = 2(x + y + z)$
- $AB + CD = AC + BD$
- $AF + BD + CE = AE + CD + BF = \frac{1}{2}(\text{Perimeter of } \triangle ABC)$



- $\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$, where a, b, c are radii

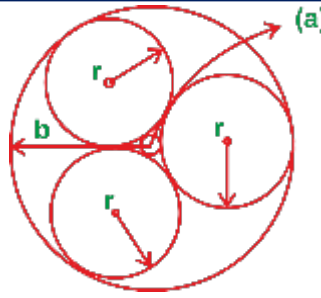


- Radius of the smaller circle :-

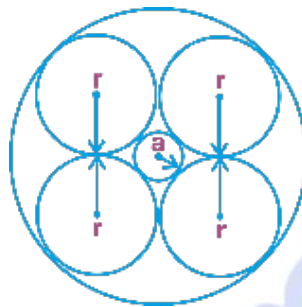
$$a = \left(\frac{2}{\sqrt{3}} - 1\right)r$$

- Radius of larger circle (outer circle)
 $b = \left(\frac{2}{\sqrt{3}} + 1\right)r$

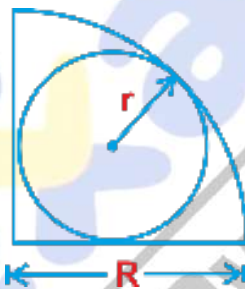
- $b = a + 2r$



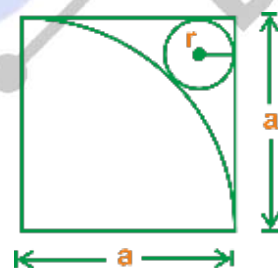
- $a = \sqrt{2} - 1)r$
{smaller circle}
 $b = (\sqrt{2} + 1)r$
{outer circle}



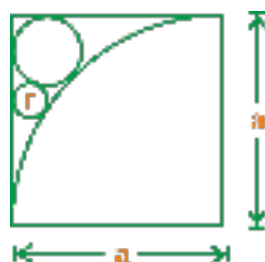
- $r = (\sqrt{2} - 1)R$



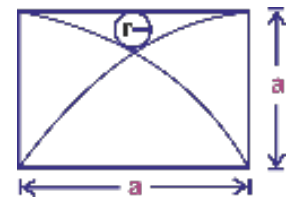
- $r = (\sqrt{2} - 1)^2 a$
 $r = (3 - 2\sqrt{2})a$



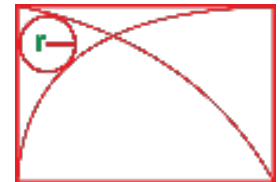
- $r = (\sqrt{2} - 1)^2 \frac{a}{2}$



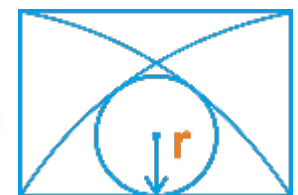
- $r = \frac{a}{16}$



- $r = \frac{a}{6}$



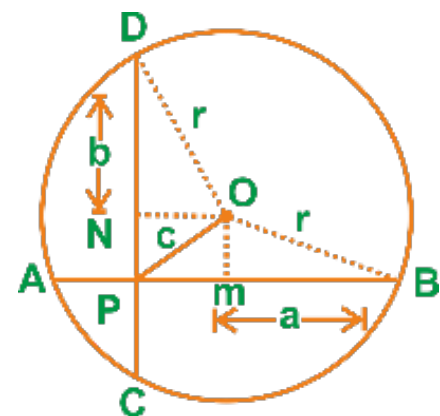
- $r = \frac{3a}{8}$



- If two chords of a circle of lengths $2a$ and $2b$ are mutually perpendicular and the distance of the point at which the chords intersect from the centre of the circle is c .

then radius of circle 'r' is given by,

$$r = \frac{\sqrt{a^2 + b^2 + c^2}}{2}$$



Exam Oriented Points
(Quick Revision)

IMPORTANT CONCEPTS

1. WHEN A TRANSVERSAL LINE INTERSECTS TWO PARALLEL LINES

$L7 = L5$
 $L2 = L4$
 $L6 = L8$
 $L1 = L3$

vertically opposite angles

$L3 = L5$
 $L2 = L8$
 $L1 = L7$
 $L4 = L6$

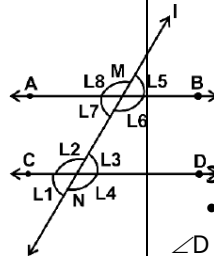
Corresponding angles

$L3 = L7$
 $L2 = L6$
 $L1 = L5$
 $L4 = L8$

Alternate angles

$L7 + L2 = 180^\circ$
 $L3 + L6 = 180^\circ$

Sum of consecutive interior angles is 180°



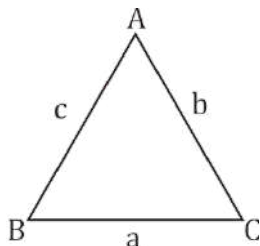
2. TRIANGLE

Right angled triangle	Acute-angled triangle	Obtuse-angled triangle
Largest side, $b^2 = a^2 + c^2$	Largest side, $b^2 < a^2 + c^2$	Largest side, $b^2 > a^2 + c^2$

- Perimeter of triangle (p) = $a + b + c$
- Semi-perimeter of triangle (s) = $\frac{p}{2} = \frac{a + b + c}{2}$

- Area of triangle Δ

- Heron's formula, $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

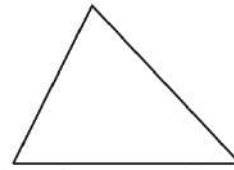


- $\Delta = \frac{1}{2} \times \text{base} \times \text{height}$

- $\Delta = \frac{1}{2} \times ac \times \sin B = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A$

- If coordinate of three vertices are given

$A(x_1, y_1)$



$B(x_2, y_2)$

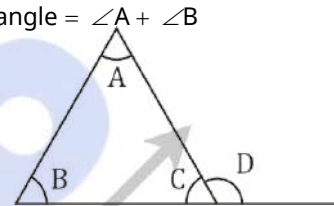
$C(x_3, y_3)$

$$\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

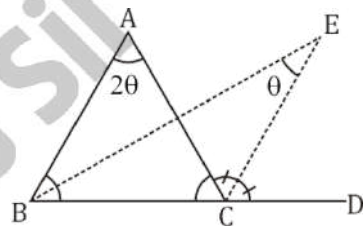
PROPERTIES OF TRIANGLES

- Exterior angle

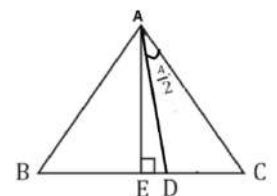
$$\angle D = \text{exterior angle} = \angle A + \angle B$$



- The angle between internal bisector of a base angle and external bisector of the other base angle is half of the remaining vertex angle. i.e. $2\angle BEC = \angle BAC$.



- In ΔABC , $AE \perp BD$ and AD is angle bisector of $\angle A$, then $\angle EAD = \frac{1}{2} |\angle B - \angle C|$



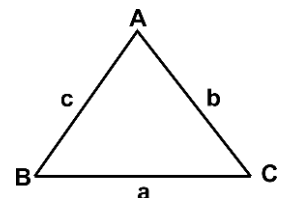
- Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

- Cosine Rule

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$



$$\cos C = \frac{b^2 + a^2 - c^2}{2ab}$$

- Stewart's Theorem:

$$b^2m + c^2n = a(d^2 + mn)$$

4. CENTERS OF TRIANGLE

• Circumcenter:

- The length from all 3 vertices to the circumcenter is equal and is called circumradius.
- $\angle QCR = 2\angle QPR$...
- Circumradius of a triangle where PQ, QR and PR are sides c, a, and b respectively.

$$R = \frac{abc}{4(\text{area of } \Delta)}$$

➤ Location of circumcenter

- The circumcenter of an acute triangle is inside the triangle.
- The circumcenter of an obtuse triangle is outside the triangle.
- The circumcenter of a right-angled triangle is on the hypotenuse.
- It is the midpoint of hypotenuse.

So, $R = \frac{\text{hypotenuse length}}{2}$ in right triangle

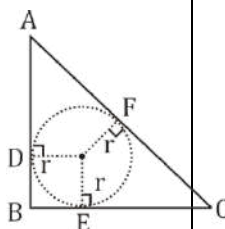
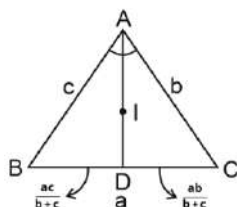
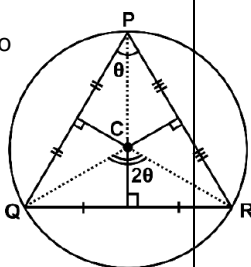
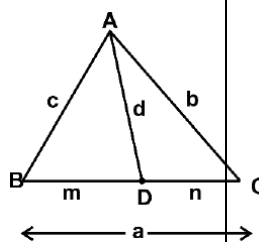
5. INCENTER

- $\angle BIC = 90^\circ + \frac{\angle A}{2}$, $\angle AIC = 90^\circ + \frac{\angle B}{2}$,
 $\angle BIA = 90^\circ + \frac{\angle C}{2}$
- $AI : ID = b+c : a$; $BI : IE = a+c : b$;
 $CI : IF = a+b : c$

$$\frac{AB}{AC} = \frac{BD}{DC}$$

- Inradius of a right angle ΔABC

$$r = \frac{AB + BC - AC}{2}$$



- Area of any triangle is product of inradius and semi perimeter

$$A = r.s$$

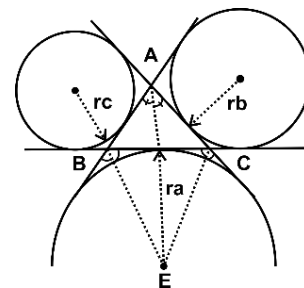
6. EX-CIRCLE

- $\angle BEC = 90^\circ - \frac{A}{2}$

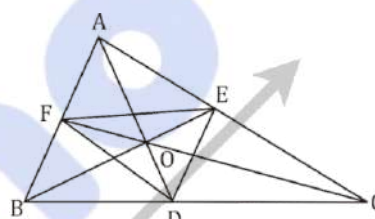
- Ex-radii :

$$r_a = \frac{\Delta}{s-a}; r_b = \frac{\Delta}{s-b};$$

$$r_c = \frac{\Delta}{s-c}$$



7. RELATION BETWEEN R(IN-RADIUS), R (CIRCUM-RADIUS) AND ANGLES OF TRIANGLE



- The distance (d) between the circumcenter (r_c) and incentre (r_i) of a triangle is

$$d = \sqrt{r_c^2 - 2r_c r_i}$$

- $r = 4R(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2})$

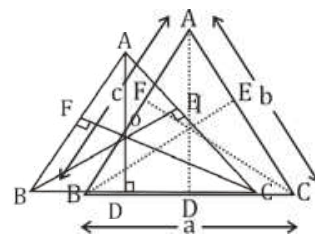
- Area of triangle = $r^2 (\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2})$

8. ORTHOCENTER

- $\angle BOC = 180^\circ - \angle A$, $\angle AOC = 180^\circ - \angle B$, $\angle AOB = 180^\circ - \angle C$

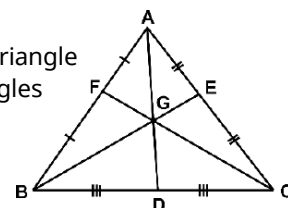
$$h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$$

- $AO \times OD = BO \times OE = CO \times OF$
- Sum of sides > Sum of altitudes
 $AB + BC + CA > AD + BE + CF$

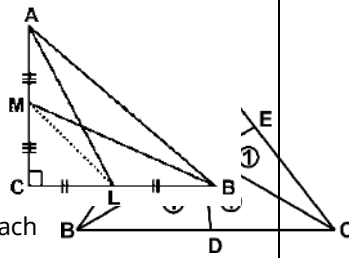


9. CENTROID

- Median divides the area of triangle into two equal area of triangles
Area of $\Delta ABD = \text{Area of } \Delta ACD$



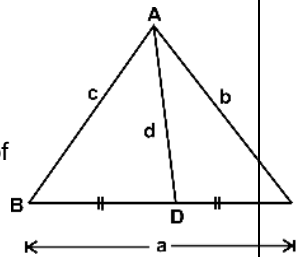
- Area of six smaller triangles formed by 3 medians and 3 sides are equal and is equal to $\frac{1}{6} \times \text{Area} \triangle ABC$.



- Centroid G divides each median in the ratio 2 : 1.
 $AG : GD = BG : GE = CG : GF = 2 : 1$
- Lengths of medians

$$AD = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$$

- In a triangle three times of sum of square of sides equal to four times of sum of square of medians i.e



$$AB^2 + BC^2 + AC^2 = \frac{4}{3} (AD^2 + BE^2 + CF^2)$$

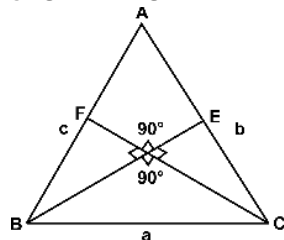
- Area of triangle formed by joining mid-points of two sides and centroid is $\frac{1}{12}$ th of area of triangle.
- $\text{Ar} \triangle OFE = \text{Ar} \triangle OFD = \text{Ar} \triangle OED = \frac{1}{12} \text{Ar} \triangle ABC$
- O is also centroid of $\triangle DEF$

- Let m_1, m_2 & m_3 are three medians, then by Heron's formula.

$$\text{Area of } \triangle = \frac{4}{3} \sqrt{S_m(S_m - m_1)(S_m - m_2)(S_m - m_3)}$$

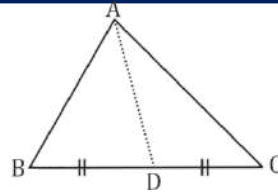
$$\text{Where, } S_m = \text{semi median} = \frac{m_1 + m_2 + m_3}{2}$$

- If the medians intersect at 90° , then $5BC^2 = AB^2 + AC^2$



- Relations of medians in right angled triangle, vii) $4(AL^2 + BM^2) = 5AB^2$

- Apollonius theorem



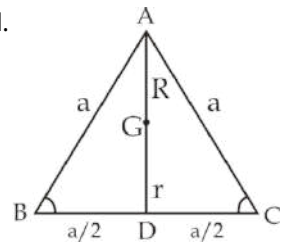
$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$

10. TYPES OF TRIANGLES

- Equilateral Triangle:**

All sides and angles are equal.

- $AB = BC = CA = a$
- $\angle A = \angle B = \angle C = 60^\circ$



- Altitude, $AD, h = \frac{\sqrt{3}}{2} a$

- All centers (orthocenter, circumcenter, incentre & centroid) lie on same point.

- All medians = All altitudes = All perpendicular

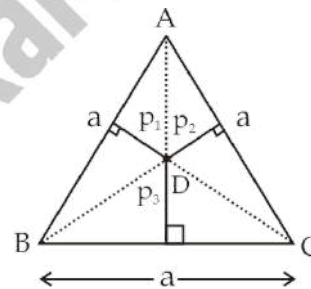
$$\text{Bisector} = \text{All angle bisector} = \frac{\sqrt{3}}{2} a$$

- Circumradius, $R = \frac{a}{\sqrt{3}}$; Inradius, $r = \frac{a}{2\sqrt{3}}$;

$$\frac{R}{r} = \frac{2}{1}$$

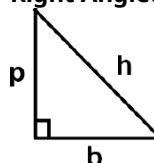
- $\frac{\pi R^2}{\pi r^2} = \frac{\text{area of circumcircle}}{\text{area of incircle}} = \frac{4}{1}$

$$\text{area of } \triangle = \frac{\sqrt{3}}{4} a^2 \left(\because \triangle = \frac{1}{2} a \times a \times \sin 60^\circ \right)$$



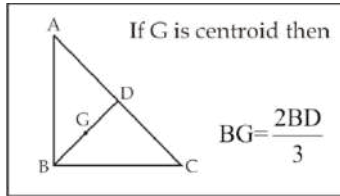
$$a = \frac{2}{\sqrt{3}} [p_1 + p_2 + p_3]$$

- Right Angled Triangle**



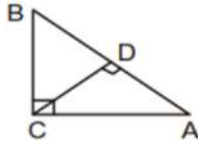
- Area = $\frac{1}{2} pb$

- $R = \frac{h}{2}$
- $r = \frac{p+b-h}{2}$
- $\Delta = s(s-2R) = r^2 + 2Rr$



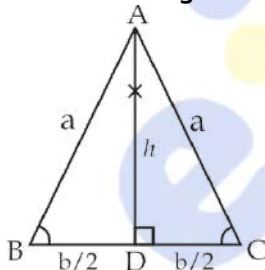
$$BD = \frac{AB \cdot BC}{AC}$$

- Important Results:



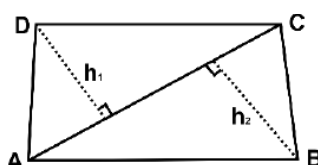
(A)	$CD^2 = BD \times DA$
(B)	$BC \times CA = BA \times CD$
(C)	$BC^2 = BD \times BA$
(D)	$AC^2 = AD \times BA$
(E)	$\frac{BD}{DA} = \frac{BC^2}{AC^2}$
(F)	$\frac{1}{CD^2} = \frac{1}{BC^2} + \frac{1}{CA^2}$

- Isosceles triangle:

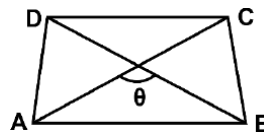


- If Any two sides are equal.
 $AB = AC$, then $\angle C = \angle B$
- If $AD \perp BC$, then
 $BD = CD$ ($AD =$ angle bisector)
- Height (Altitude), $AD = \sqrt{a^2 - \frac{b^2}{4}} = \sqrt{\frac{4a^2 - b^2}{4}}$
- Area $= \frac{1}{2} \times b \times \frac{\sqrt{4a^2 - b^2}}{2} = \frac{b}{4} \sqrt{4a^2 - b^2}$

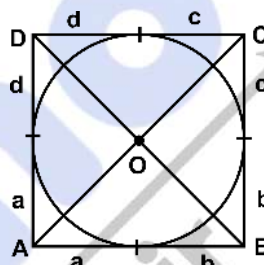
11. QUADRILATERAL



- $\angle A + \angle B + \angle C + \angle D = 360^\circ$
- Area(ABCD) $= \frac{1}{2} AC(h_1 + h_2)$
- Area $= \sqrt{(s-a)(s-b)(s-c)(s-d)}$
Where $s = \frac{a+b+c+d}{2}$
- Area $= \frac{1}{2} \times AC \times BD \times \sin \theta$



- If diagonals intersect at 90° , then
➤ $AB^2 + CD^2 = BC^2 + AD^2$

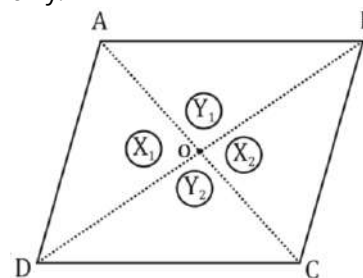


- $AB + CD = BC + AD$

12. PARALLELOGRAM



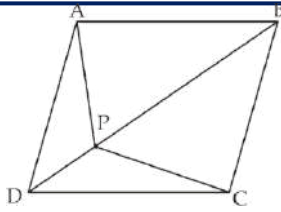
- $AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + AD^2 = 2(AB^2 + BC^2)$
- Area of parallelogram = Base $\times$ height
 $= AD \times AB \sin \angle DAB$
- Two diagonals of a parallelogram divide the parallelogram into triangles of four equal areas but opposite triangles are congruent only.



$$\text{Area} = X_1 = X_2 = Y_1 = Y_2$$

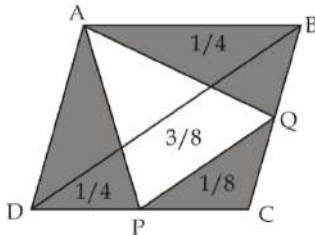
$$\text{Congruent} = \Delta X_1 \cong \Delta X_2 \text{ \& } \Delta Y_1 \cong \Delta Y_2$$

- If 'P' is a point inside a parallelogram then
(Area ΔAPD + Area ΔPBC) = (Area ΔAPB + Area ΔPDC)
 $= \frac{1}{2}$ area of ABCD.

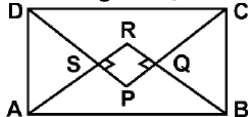


- If P and Q are mid-point of DC & BC respectively then area

$$\Delta APQ = \frac{3}{8} \text{ area of } ABCD$$



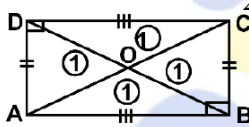
- The angle bisectors of a parallelogram form a rectangle. PQRS is a rectangle.



13. RECTANGLE

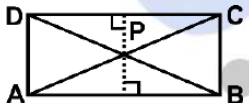
- Area of ABCD = AB x BC .
- Area

$$\begin{aligned} (\Delta ABC) &= \text{area}(\Delta BCD) = \text{area}(\Delta ABD) \\ &= \text{area}(\Delta ACD) = \frac{1}{2}(\Delta ABCD) \end{aligned}$$

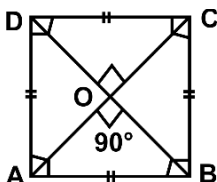


- Diagonals bisect each other
- P is any point in rectangle ABCD, then

$$AP^2 + CP^2 = BP^2 + DP^2$$



14. SQUARE

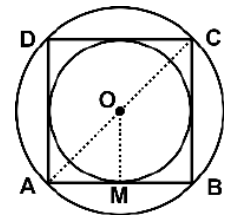


- AREA of ABCD = side²
- AB = BC = CD = DA = a
- AC = BD = a√2

- Diagonals bisect each other O is mid-point of diagonals at 90°

- R = OA = OC & OM = r

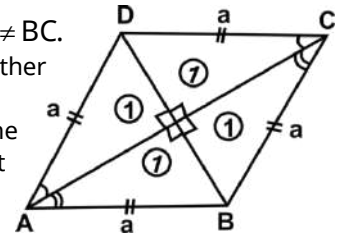
$$R : r = \frac{a\sqrt{2}}{2} : \frac{a}{2} = \sqrt{2} : 1$$



15. RHOMBUS

- Area = $\frac{1}{2} \times AC \times BD$

- All sides are equal but AC ≠ BC.
- Diagonals bisect each other at 90°
- Diagonal divides the rhombus into 4 congruent right triangles.

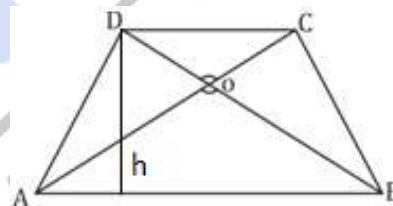


16. TRAPEZIUM

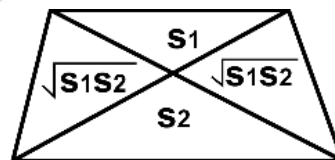
- Area = $\frac{1}{2} \times h(AB + CD)$

$$\frac{OA}{OC} = \frac{OB}{OD} = \frac{AB}{CD}$$

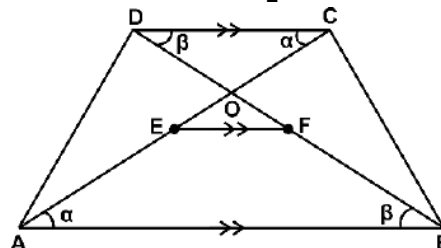
- AC² + BD² = AD² + BC² + 2AB.CD



- Relation between areas formed by the diagonals.



- If E and F are mid-points of AC and BD respectively, then EF = $\frac{1}{2}(AB - CD)$

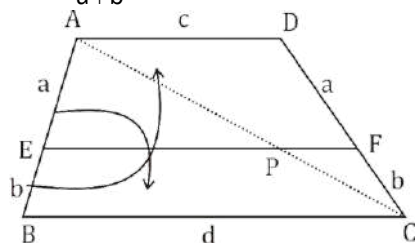


- P and Q are the mid-points of AD and BC respectively, then PQ = $\frac{1}{2}(AB + CD)$



- The length of parallel side EF

$$EF = \frac{ad+bc}{a+b}$$



17. POLYGONS

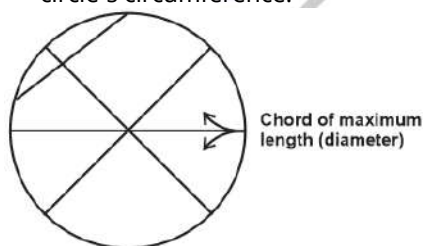
- Sum of all interior angles of a polygon with 'n' sides = $(n-2) \times 180^\circ$
- Interior angle = $\frac{(n-2) \times 180}{n} = 180 - \text{exterior angle}$
- each exterior angle of a regular polygon = $\frac{360^\circ}{n}$
- Number of diagonals = $\frac{n(n-3)}{2}$
- Area of regular polygon = $\frac{na^2}{4} \times \cot\left(\frac{180^\circ}{n}\right)$

18. CIRCLE

It is locus of the points which are equidistance from a fixed point. The fixed point is called the centre of the circle and fixed distance is called the radius.

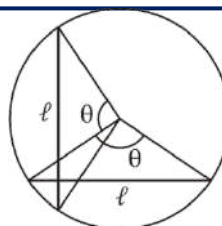
• Chord

The line segment joining the two points on a circle's circumference.

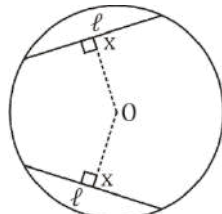


• IMPORTANT PROPERTIES OF CIRCLES-

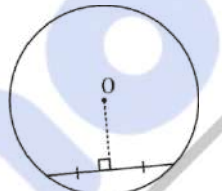
- Equal chords subtend equal angles at the centre and vice versa.



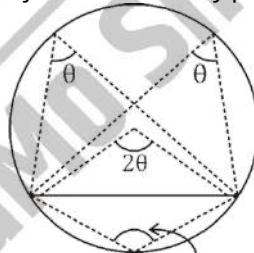
- Equal chords are equidistance from the centre and vice versa.



- The perpendicular from the centre of a circle to a chord bisect the chord.

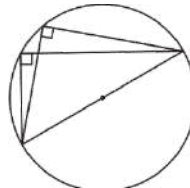


- The angle subtended by an chord at the Centre of a circle is twice the angle subtended by the chord at any point on the major arc.

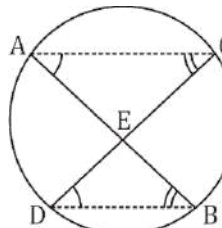


In a Minor Arc it is $180-\theta$

- Angle made by the diameter on circumference is always 90°



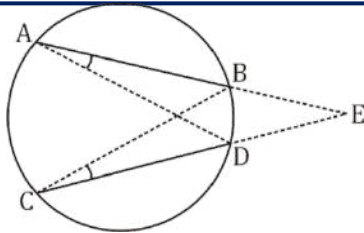
- If two chords intersect inside a circle



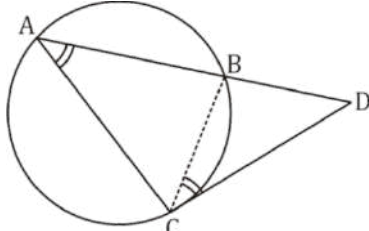
Then, $AE \times EB = CE \times ED$

- If two chords intersect outside a circle or two secant intersect each other.

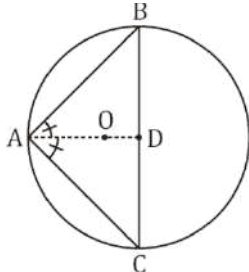
Then $EA \times EB = EC \times ED$



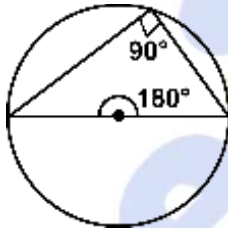
- If a secant and a tangent externally intersect each other. Then $DC^2 = DA \times DB$



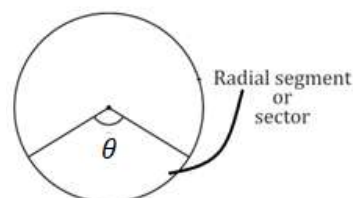
- Two chords AB and AC of a circle are equal. Then the centre of the circle lies on the angle bisector of $\angle BAC$.



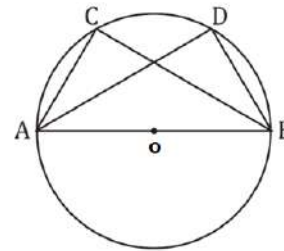
- The angle made by the diameter at circumference is 90°



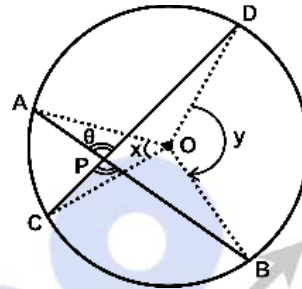
- Length of arc = $\frac{2\pi r}{360} \times \theta$ (θ in degree)
 $l = r\theta$ (θ in radian) ($\because 2\pi = 360^\circ$)
 * area of sector = $\frac{\pi r^2}{360} \times \theta$ (in degree)
 $= \frac{1}{2} r^2 \theta$ (θ is in radian) ($2\pi = 360^\circ$) = $\frac{1}{2} l r$ ($l = r\theta$)



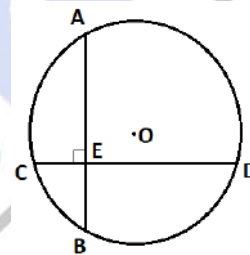
- Angles by the same segment of a circle are equal.
 i.e., $\angle ACB = \angle ADB$



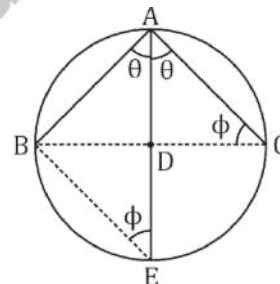
- If two chords AB and CD intersect at P, then
 $\angle APC = \angle BPD = \frac{1}{2}(\angle x + \angle y)$



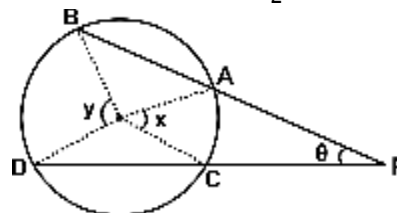
- If two chords AB and CD intersect each other at 90° , then $4r^2 = AE^2 + BE^2 + CE^2 + DE^2$



- In the above figure, AE is angle bisector of $\angle BAC$ then
 $AB \cdot AC + DE \cdot AE = AE^2$

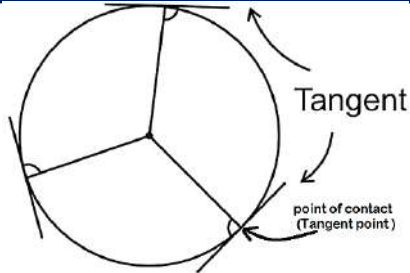


- When two chords AB and CD are extended to meet at P, then $\angle APC = \frac{1}{2}|\angle y - \angle x|$

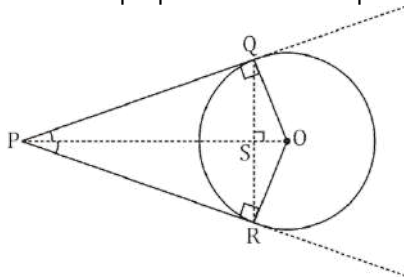


• **Tangents**

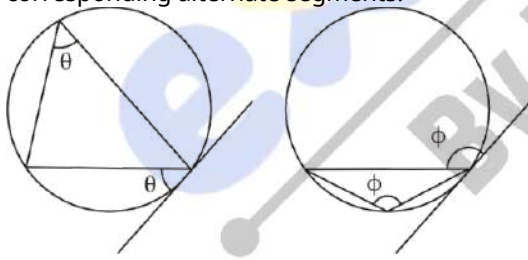
A line segment touches only at one point of a circle is called a tangent of the circle and the point of contact or tangent point.



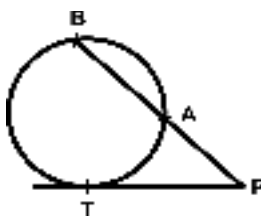
- Important Properties of Tangent
 - The line joining centre of circle and point of contact is perpendicular to the point.



- Tangents drawn from an external point to a circle
 - $PQ = PR$
 - $\angle QPO = \angle RPO = \angle SQP$
 - $\angle QOP = \angle ROP = \angle PQS$
 - $\frac{PQ}{QO} = \frac{PR}{RO}$
 - PQOR is a cyclic quadrilateral
- Alternate segment Theorem
The angle between a chord and a tangent drawn at end point of chord is equal respectively to the angle formed in the corresponding alternate segments.



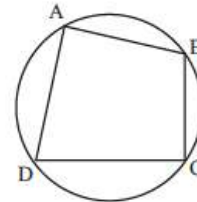
- PT is a tangent and PAB is a secant



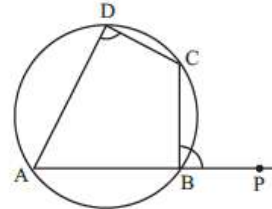
$$PT^2 = PA \times PB$$

19. CYCLIC QUADRILATERAL

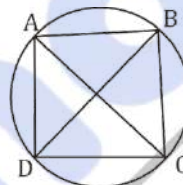
- $\angle A + \angle C = \angle B + \angle D = 180^\circ$



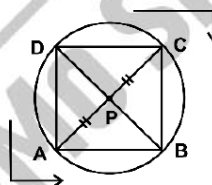
- $\angle ADC = \angle CBP$



- Ptolemy's theorem:
 $AC \cdot BD = AB \cdot DC + AD \cdot BC$

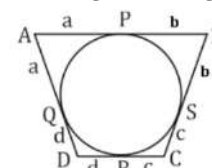


- If one diagonal bisects the other
 $AB \cdot AD = BC \cdot CD$

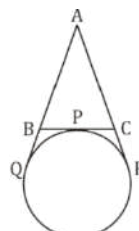


20. SOME IMPORTANT RESULTS

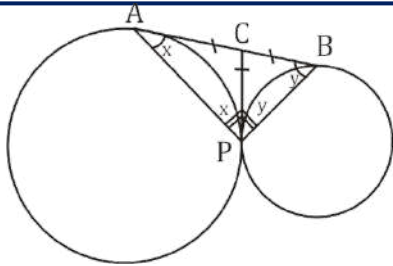
- If a quadrilateral circumscribed a circle.
 $AB + DC = AD + BC$



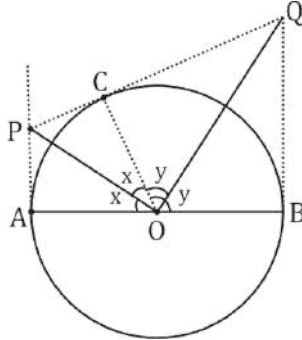
- $AB + BC + AC = 2AQ = 2AR$



- $\angle APB$ is always right angle

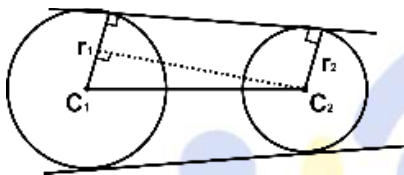


- Remember $\angle POQ = 90^\circ$ and $r^2 = PC \times CQ$



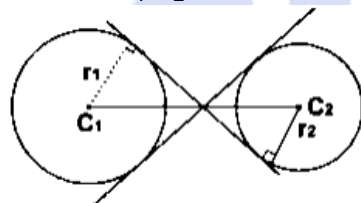
21. DIRECT COMMON TANGENT

- Here, length of $\overline{C_1C_2} = d$
DCT = $\sqrt{d^2 - (r_1 - r_2)^2}$



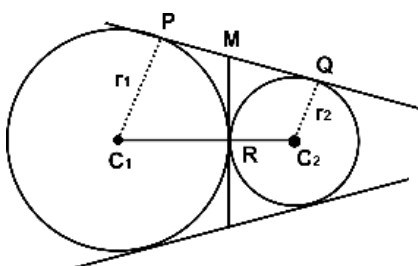
22. TRANSVERSE COMMON TANGENT

- TCT = $\sqrt{d^2 - (r_1 + r_2)^2}$
Here, $(d > (r_1 + r_2))$

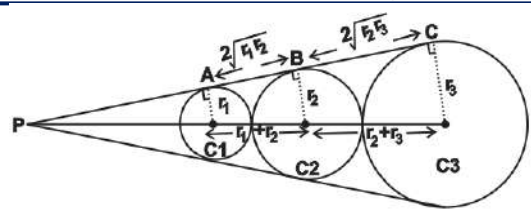


23. FOR EXTERNALLY TOUCHING CIRCLES

- $PQ = 2\sqrt{r_1 r_2}$

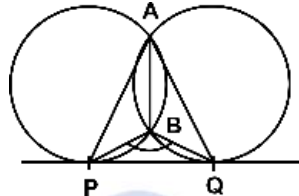


- $r_2 = \sqrt{r_1 r_3}$



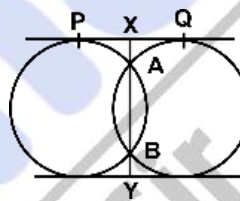
- $\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$, where a, b, c are radii

24. INTERSECTING CIRCLES



- $\angle PAQ + \angle PBQ = 180^\circ$

$$PQ^2 = XY^2 - AB^2$$



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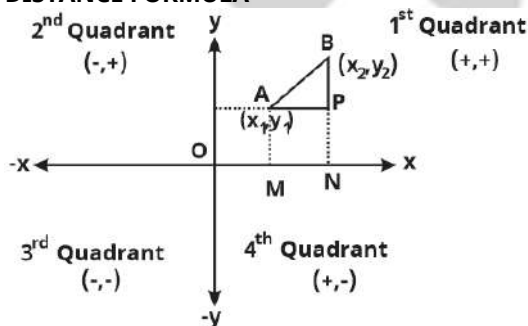
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IMPORTANT CONCEPTS

1. DISTANCE FORMULA



$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2. SECTION FORMULA

Co-ordinates of a point which divides the line segment joining two points A (x_1, y_1) and B (x_2, y_2) in the ratio $m_1 : m_2$ are :



- $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$, for internal division.
- $\left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2} \right)$, for external division
- If $m_1 = m_2$, then the point P is the mid-point of AB whose co-ordinates are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

3. CONDITION OF COLLINEARITY OF THREE POINTS

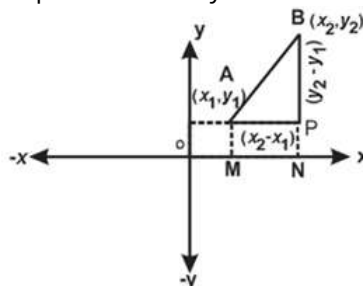
Three points A (x_1, y_1), B (x_2, y_2) and C (x_3, y_3) will be collinear if one of the following four conditions is satisfied.

- If sum of any two distances AB, BC and CA is equal to the third distance.
- $\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_1}{x_3 - x_1}$

- Slope of AB = Slope of BC
- Area of $\triangle ABC = 0$

4. LINE

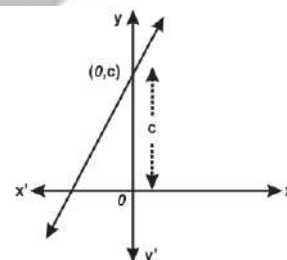
- **Slope of line:** If a line makes an angle θ with positive x-axis, then tangent of that angle is called slope of line. Slope is denoted by m $\Delta m = \tan \theta$



$$m = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}, \quad ax + by + c = 0$$

• Slope Intercept Form

- $y = mx + c$, where m is the slope of the line and c is the y-intercept of the line.
- y-intercept of a line is the y-coordinate of the point where the line intersects the y-axis.



• Slope Point Form

The equation of a line with slope m and passing through a point (x_1, y_1) : $(y - y_1) = m(x - x_1)$

• Two Point Form

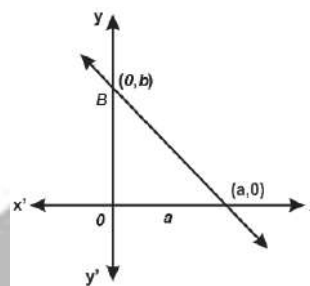
The equation of a line passing through two given

points (x_1, y_1) and (x_2, y_2) is $(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$

• Intercept Form

$$\frac{x}{a} + \frac{y}{b} = 1$$

where a and b are x and y intercept respectively of the line.



5. POSITION OF A POINT RELATIVE TO A LINE

- The point (x_1, y_1) lies on the line $ax + by + c = 0$ if $ax_1 + by_1 + c = 0$
- Length of intercept between both axis = $\sqrt{a^2 + b^2}$
- **Parallel Lines**
 - Two lines are parallel if their slopes m_1 and m_2 are equal i.e., $m_1 = m_2$.



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- Lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ are parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$

• **Perpendicular Lines**

- Two lines are perpendicular if product of their slopes m_1 and m_2 , is -1 i.e., $m_1 \times m_2 = -1$.
- Lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ are perpendicular if $a_1a_2+b_1b_2=0$.

• **Coincident Lines**

Two lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ are coincident

if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

6. DISTANCE OF A LINE FROM A POINT

- The length P of the perpendicular from the point (x_1, y_1) on the line $ax+by+c=0$ is given by

$$P = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

- Length of perpendicular from origin to the line $ax+by+c=0$ is given by $\left| \frac{c}{\sqrt{a^2 + b^2}} \right|$

7. DISTANCE BETWEEN TWO PARALLEL LINES

The distance between two parallel lines $ax + by + c_1 = 0$

and $ax - by + c_2 = 0$ is $\left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$

8. ANGLE BETWEEN TWO STRAIGHT LINES

- If θ be the angle between two straight lines y

$= m_1x + c_1$ and $y = m_2x + c_2$, then $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1m_2} \right|$

- If θ be the angle between two straight lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$, then

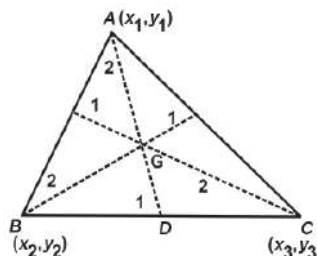
$$\tan \theta = \left| \frac{a_1b_2 - a_2b_1}{a_1b_2 + a_2b_1} \right|$$

9. AREA OF TRIANGLE

Let A (x_1, y_1) , B (x_2, y_2) and C (x_3, y_3) are vertices of a triangle, then area of the triangle ABC

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

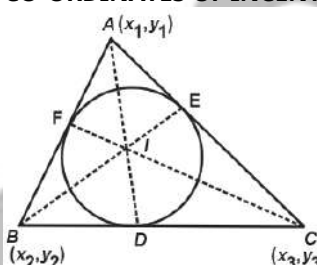
10. CO-ORDINATES OF CENTROID



Co-ordinates of centroid,

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

11. CO-ORDINATES OF INCENTRE



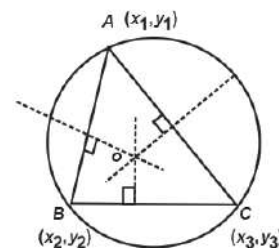
Co-ordinates of incentre,

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right),$$

where a, b, c is length of the sides opposite to vertices A, B, C respectively of triangle ABC.

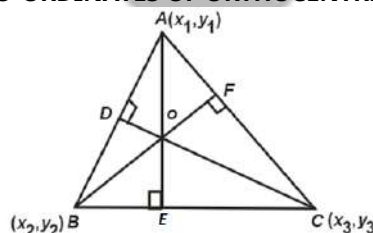
12. CO-ORDINATES OF CIRCUMCENTRE

- If a triangle is right angle, then its circumcentre is the midpoint of hypotenuse.
- If A (x_1, y_1) , B (x_2, y_2) and C (x_3, y_3) are the vertices of a $\triangle ABC$, then coordinates of the circumcentre of the triangle ABC is



$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

13. CO-ORDINATES OF ORTHOCENTRE



Coordinate of the orthocentre of $\triangle ABC$



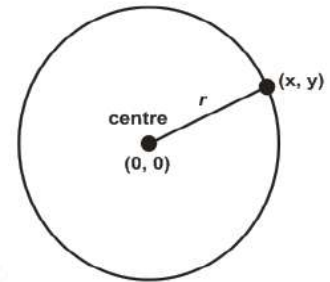
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$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right)$$

14. CIRCLE

- $x^2 + y^2 = r^2$ is called equation of circle.
- **Standard equation of a circle**
 $x^2 + y^2 + 2gx + 2fy + c = 0$ where Centre = $(-g, -f)$
- Radius = $\sqrt{g^2 + f^2 - c}$





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#1 Square :

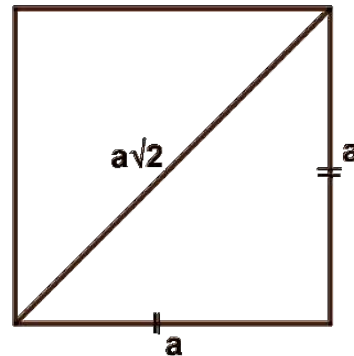
1.1)

Perimeter = $4a$

Area = $(\text{side})^2$

$= \frac{1}{2}(\text{diagonal})^2$

Diagonal = $a\sqrt{2}$



1.2)

ABCD is a square.

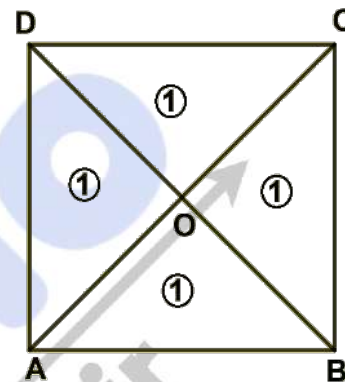
Diagonal AC & BD

intersects at O.

Here,

$\text{ar}(\text{AOB}) = \text{ar}(\text{BOC})$

$= \text{ar}(\text{COD}) = \text{ar}(\text{DOA})$



#2 Circle :

2.1)

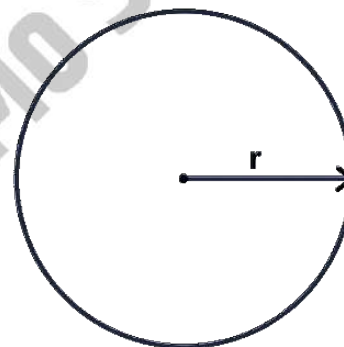
Radius = r

$d = \text{diameter} = 2r$

$C = \text{circumference}$

$= 2\pi r = \pi d$

$\text{area} = \pi r^2$

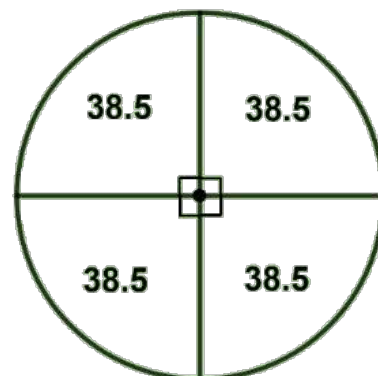


2.2)

Standard values in circle : {RaMo Mantra}

Radius : **Circumference** : **Area**

7	:	44	:	154
14	:	88	:	616
21	:	132	:	1386
3.5	:	22	:	38.5



2.3)

$r = 7$

Area of quarter circle: 38.5

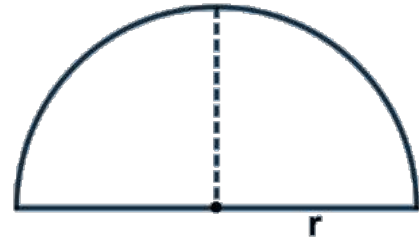
2.4)

Semi Circle :-

$$P = \text{Perimeter} = \pi r + 2r$$

$$\text{area} = \frac{\pi r^2}{2}$$

r	:	P	:	area
7	:	36	:	77
14	:	72	:	308



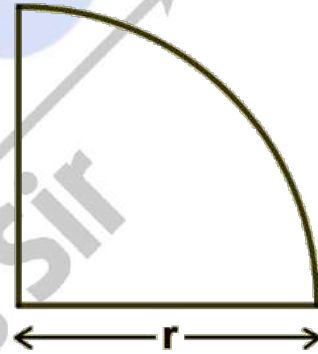
2.5)

Quarter circle :-

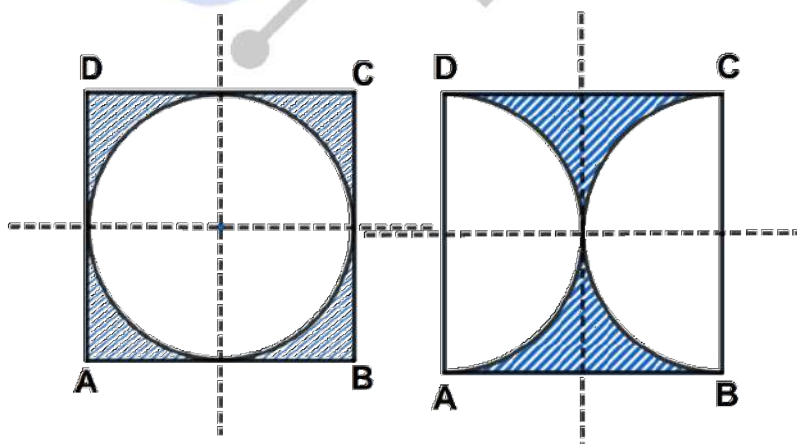
$$P = \frac{\pi r}{2} + 2r, \quad \text{area} = \frac{\pi r^2}{4}$$

Standard Value :

r	:	P	:	A
7	:	25	:	38.5
14	:	50	:	154



2.6)



Area of shaded region is equal in each case.

2.7)

Area of shaded region

$$= a^2 - \frac{\pi a^2}{4} = \frac{a^2}{4} (4 - \pi)$$

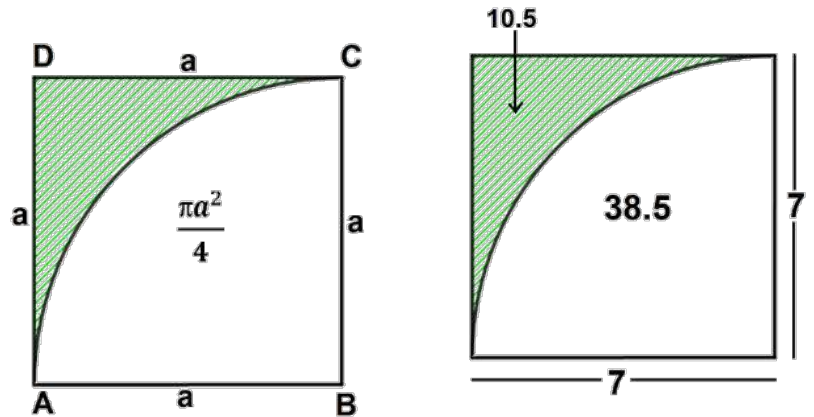
If side of square = 7 unit

Area of Qt. circle = 38.5 unit²

Area of shaded region

= 10.5 unit²

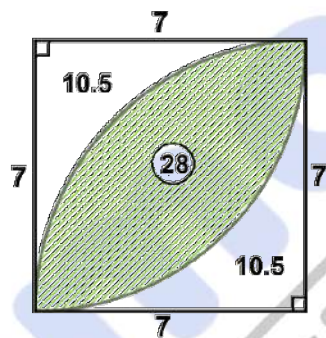
Area if square = 49 unit²



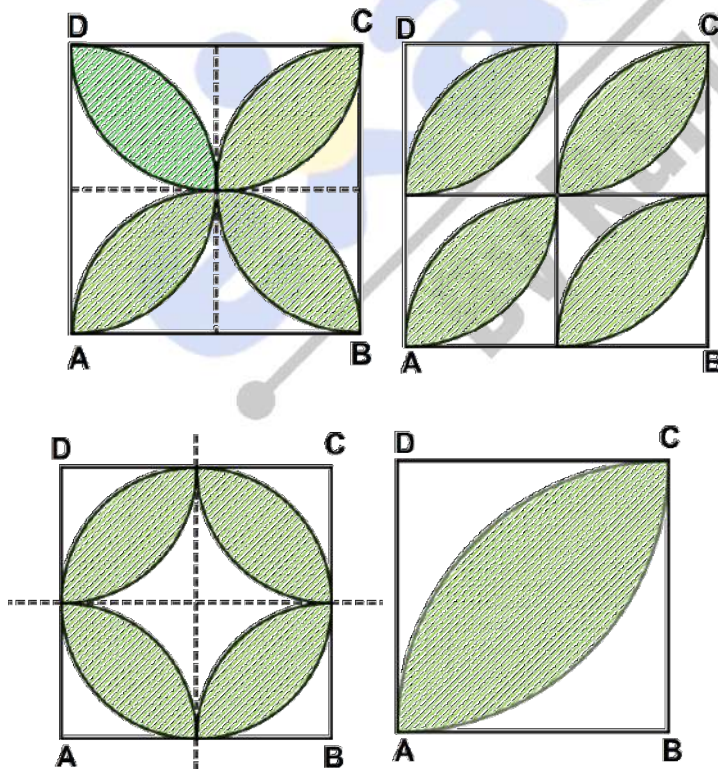
2.8)

If side of square is = 7 unit

Area of leaf = 28 unit²



2.9)



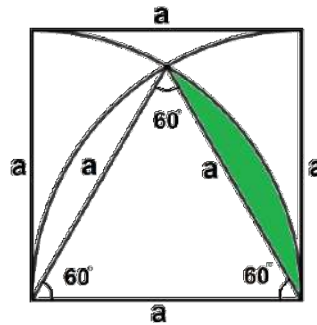
Area of shaded region is equal in each case.

2.10)

Area shaded region

$$= \frac{a^2}{12} \{2\pi - 3\sqrt{3}\}$$

= 9.05% of square area

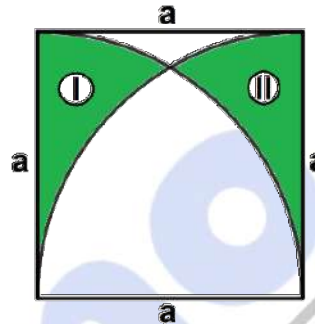


2.11)

area of (I) = area of (II)

$$\text{area of (I) or (II)} = \frac{a^2}{12} \{3\sqrt{3} - \pi\}$$

= 17.1208% of square area

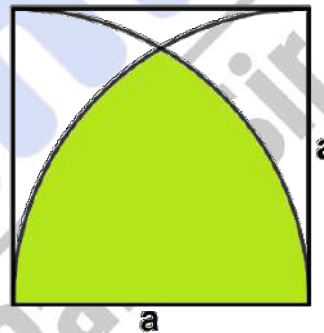


2.12)

Area of shaded region

$$= \frac{a^2}{12} \{4\pi - 3\sqrt{3}\}$$

= 61.4166% of square area

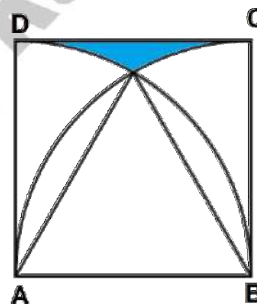


2.13)

Area of shaded region

$$= \frac{a^2}{12} \{12 - 2\pi - 3\sqrt{3}\}$$

= 4.3388% of square area



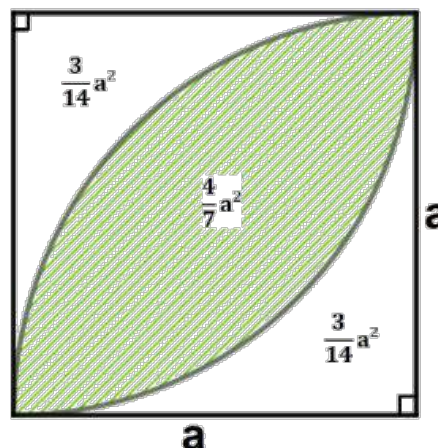
2.14)

Area of shaded region

$$= \frac{a^2}{2} (\pi - 2)$$

$$= \frac{4}{7} a^2$$

= 57.02% of square area

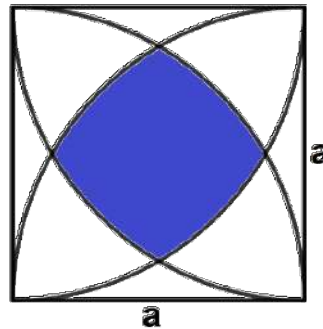


2.15)

Area of shaded region

$$= \frac{a^2}{3} \{3(1 - \sqrt{3}) + \pi\}$$

= 31.5146% of square area



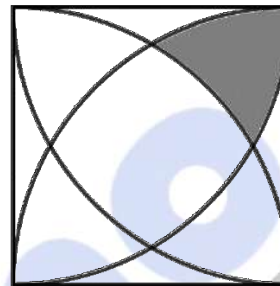
2.16)

Area shaded region

$$= \frac{a^2}{12} [\pi - 12 + 6\sqrt{3}]$$

= 12.78% of square area

= $\frac{1}{8}$ of square area



#3 Square inside a triangle :

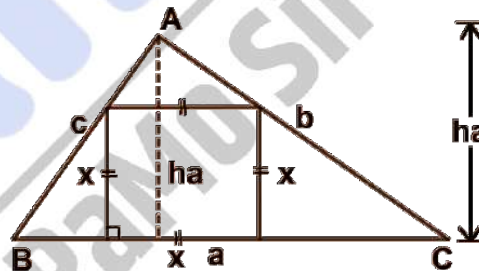
3.1)

Side of square = x

$$\frac{1}{x} = \frac{1}{ha} + \frac{1}{a}$$

$$x = \frac{a \times ha}{a + ha}$$

Where x is on BC.

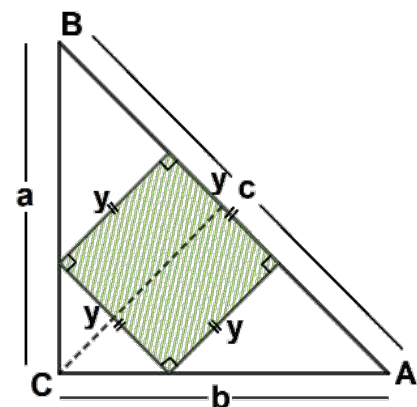
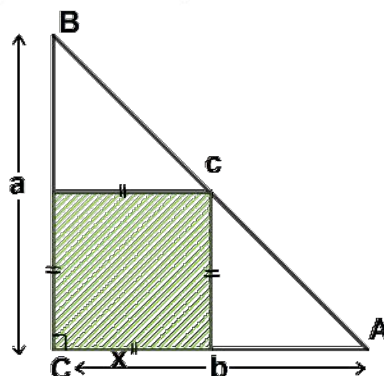


3.2)

$$x = \frac{b \times a}{b + a}$$

$$\frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$y = \frac{abc}{c^2 + ab}$$



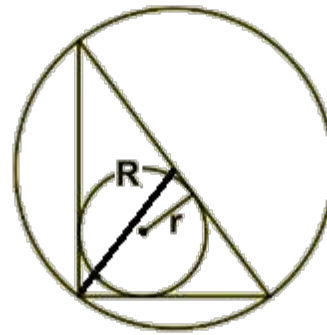
Here $x > y$

#4 Right angled Δ :

4.1)

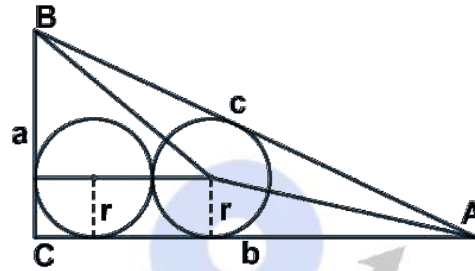
$$r = \frac{p+b-h}{2}; \quad R = \frac{h}{2}$$

$$r = \frac{\Delta}{s}; \quad R = \frac{abc}{4\Delta} = \frac{p \times b \times h}{4\Delta}$$



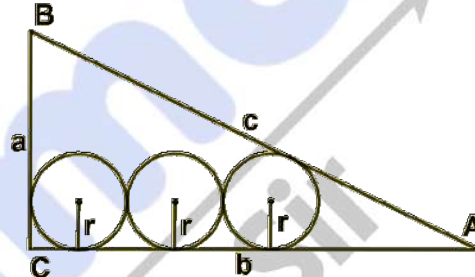
4.2)

$$r = \frac{ab}{3a+b+c}$$



4.3)

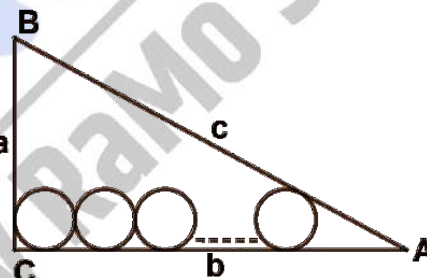
$$r = \frac{ab}{5a+b+c}$$



4.4)

n - identical circles

$$r = \frac{ab}{(2n-1)a+b+c}$$



#5 Pythagoras theorem :

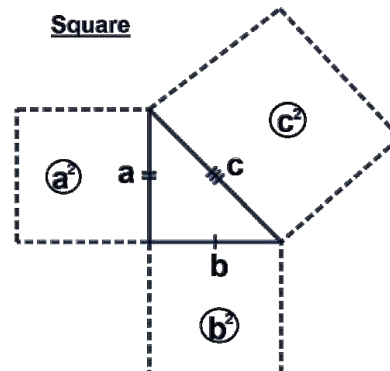
5.1)

Square

$$c^2 = a^2 + b^2$$

$$\text{Area}(\square_c) = \text{ar}(\square_b) + \text{ar}(\square_a)$$

Square

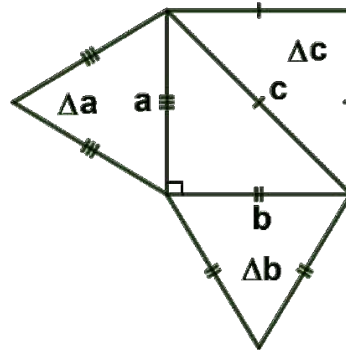


5.2)

Equilateral Δ s

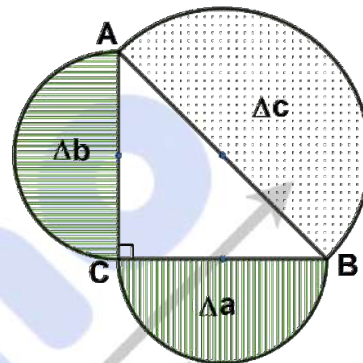
$$\text{ar}(\Delta_c) = \text{ar}(\Delta_a) + \text{ar}(\Delta_b)$$

Equilateral Δ s



5.3)

$$\text{ar}(\Delta_a) = \text{ar}(\Delta_b) + \text{ar}(\Delta_c)$$

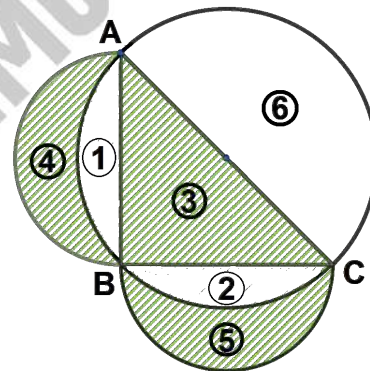


5.4)

Here,

i) $\text{area}(6) = \text{area}(1) + \text{area}(2) + \text{area}(3)$

ii) $\text{area}(4) + \text{area}(5) = \text{area}(3)$



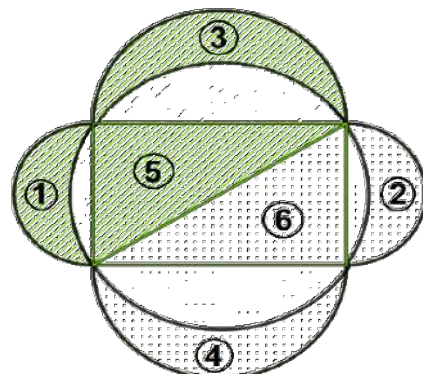
5.6)

Here,

i) $\text{area}(1) = \text{area}(2)$

$\text{area}(3) = \text{area}(4)$

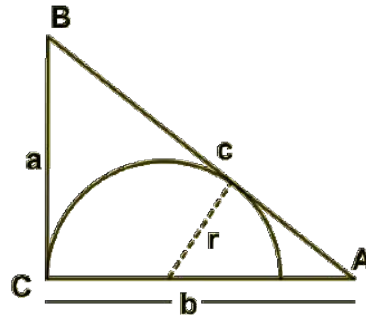
ii) $\text{area of } (1) + (2) + (3) + (4) = \text{area of rectangle}$



#6

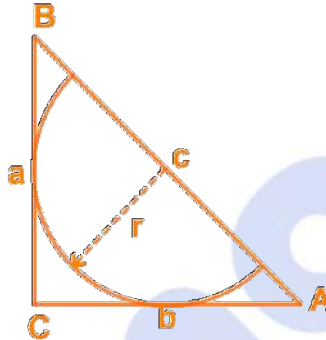
6.1)

$$r = \frac{ab}{a+b}$$



6.2)

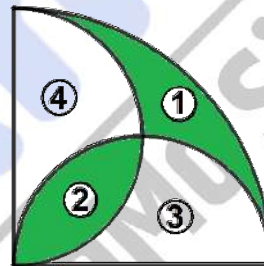
$$r = \frac{ab}{a+b}$$



#7

7.1)

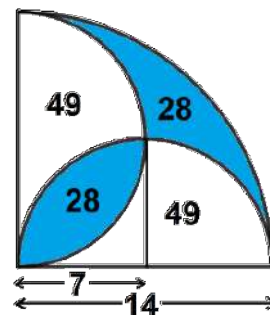
- i) area of region (1) = area of region (2)
- ii) area of region (3) = area of region (4)



Standard value :

If $r = 14$ unit for qt. circle

- Area of region (1) = area of region (2) = 28 unit^2
- Area of region (3) = area of region (4) = 49 unit^2



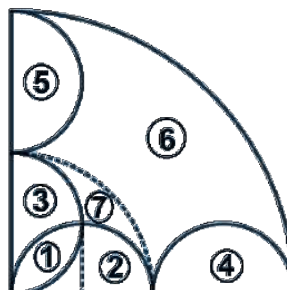
➤ **area ratio :**

ar[(1) : (2) : (3) : (4)]

4 : 4 : 7 : 7

7.2)

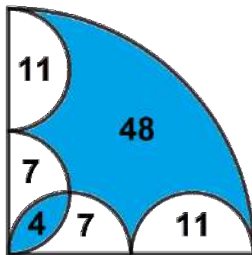
- i) area of (1) = area of (7)



ii) area of (3) = area of (2)

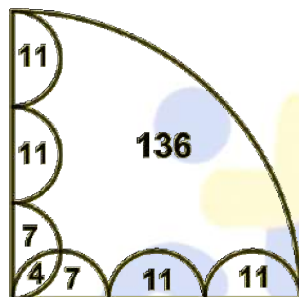
iii) area of (5) = area of (4) = area of [(3) + (1)] = area of [(2) + (1)]

❖

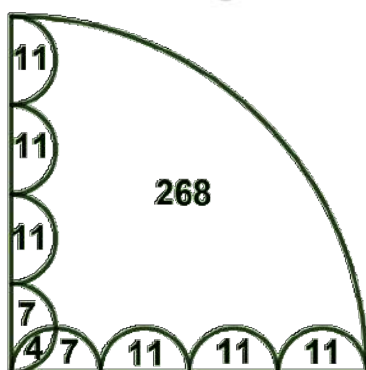


here numbers represent the area values of respective region.

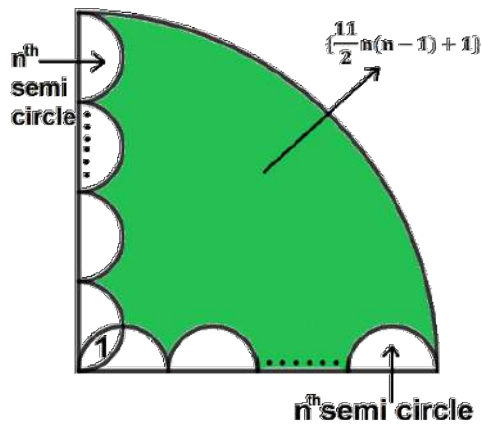
❖



❖



Number represent the area values of respective region.

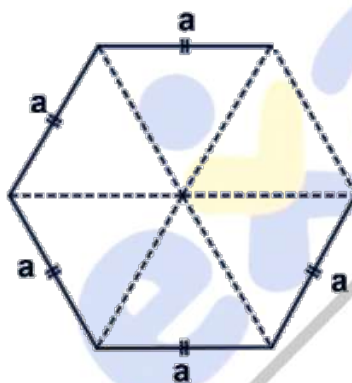


area of leaf : area of shaded region

$$1 : \left\{ \frac{11}{2}n(n-1) + 1 \right\}$$

Regular Polygon :

n sided regular polygon (side length a)



area of n sided regular polygon

i)

$$\Delta = \frac{na^2}{4} \cot \frac{\pi}{n}$$

where a is side of regular polygon

n is no. of sides.

$$\left(\begin{array}{l} \text{if } n = 3 \\ \Delta = \frac{\sqrt{3}a^2}{4} \end{array} \right) \quad \left(\begin{array}{l} \text{if } n = 6 \\ = \frac{3\sqrt{3}}{2}a^2 \quad \{\text{hexagon}\} \end{array} \right)$$

ii) Perimeter = $n \times a$

General formula for Prism (base & top regular polygon) :

(i) CSA : curved surface area :

CSA = Perimeter of base $\times$ height

$$\text{CSA} = na \times h$$

(ii) TSA : Total surface area :

TSA = CSA + 2 $\times$ base area

$$\text{TSA} = nah + 2 \times \frac{na^2}{4} \cot \frac{\pi}{n}$$

(iii) Volume :

= area of base $\times$ height

$$\text{Vol.} = \frac{na^2}{4} \left(\cot \frac{\pi}{n} \right) \times h$$

- here, $n \rightarrow$ no. of side of regular polygon.
- $a \rightarrow$ is side length of reg. polygon.
- $h \rightarrow$ height of prism.

General formulas for Pyramid (base is regular polygon with pointed apex)

i) CSA : curved surface area :

$$\text{CSA} = \frac{1}{2} \text{ base perimeter} \times \text{slant height}$$

ii) TSA : Total surface area :

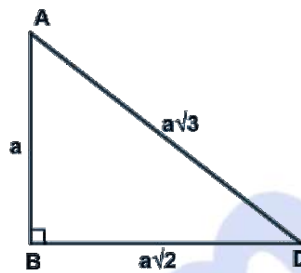
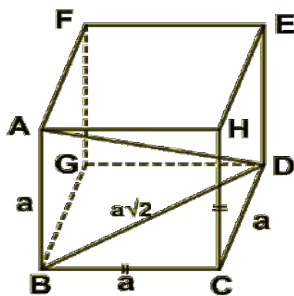
$$\text{TSA} = \text{CSA} + \text{base area}$$

iii) Volume :

$$\text{Vol.} = \frac{1}{3} \times \text{area of base} \times \text{height}$$

Prism :

1) Cube :



Face = 6

Edge = 12

Vertex = 8

(i) $\text{CSA} = 4a^2$ {area of 4 walls}

(ii) $\text{TSA} = 6a^2$

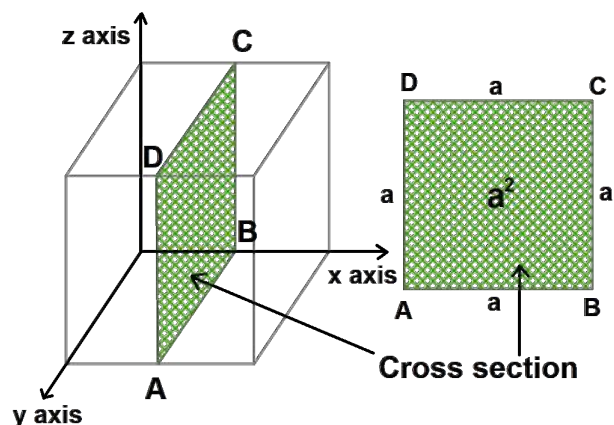
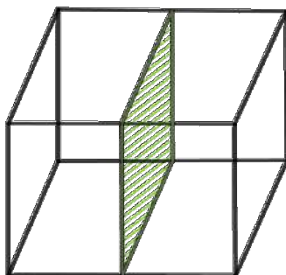
(iii) $\text{Vol.} = a^3$

(iv) diagonal (AD) = $a\sqrt{3}$

(v) cutting of cube in n part parallel to edge in vertical manner:

Cutting of cube in n part, parallel to edge in vertical manner :

➤ By doing 1 cut, we get two cuboids



Now increase in TSA $\rightarrow$ is $[2a^2]$

$$\begin{aligned}\text{TSA of these two parts} &= 6a^2 + 2a^2 \\ &= 8a^2\end{aligned}$$

➤ **By doing n cuts, we get (n+1) cuboids :**

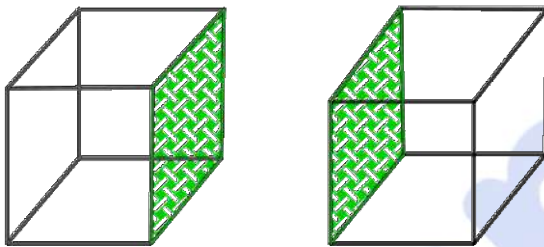
Now increase in TSA $= 2na^2$

$$\text{TSA of these parts } (n+1) = 2a^2(n+3)$$

$$\text{TSA} = 2a^2(n+3)$$

Joining of cubes :

(i) If we join two cubes there will be the loss of two surfaces.



So now,

$$\text{TSA} = 6a^2 + 6a^2 - a^2 - a^2$$

$$\text{TSA} = 10a^2$$

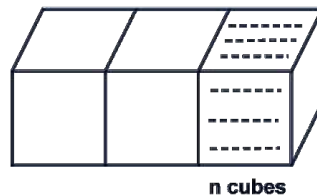
(ii) If we join three identical cubes there will be the loss of $2(3-1) = 4$ surfaces.

Similarly, if we join n - identical cubes, there will be the loss of $2(n-1)$ surfaces.

$$\text{TSA} = n6a^2 - 2(n-1)a^2$$

$$2a^2(2n+1)$$

$$\text{TSA}_2 = 2a^2(2n+1)$$

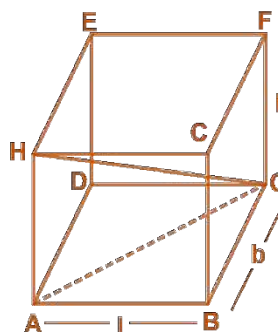


cuboid :

(i) LSA : Lateral surface area

$$= 2bh + 2hl$$

$$= 2(b+l)h$$



(ii) TSA : Total surface area = $2(lb+bh+hl)$

(iii) Volume : = lbh

(iv) diagonal (HC) = $\sqrt{l^2 + b^2 + h^2}$

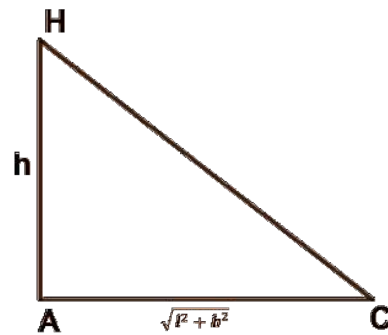
(v) If $S_1 = lb$, $S_2 = bh$, $S_3 = hl$

Volume = $\sqrt{S_1 S_2 S_3}$

(vi) $(l + b + h)^2 = d^2 + TSA$

$d = \sqrt{(l + b + h)^2 - TSA}$

$TSA = (l+b+h)^2 - d^2$



Cylinder :

(i) CSA = Perimeter of base $\times$ height

$$CSA = 2\pi rh$$

(ii) TSA = CSA + 2 $\times$ base area

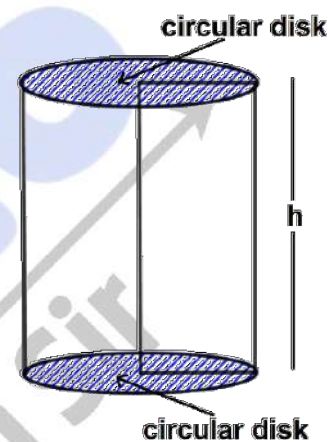
$$= 2\pi rh + 2\pi r^2$$

$$= 2\pi r(h + r)$$

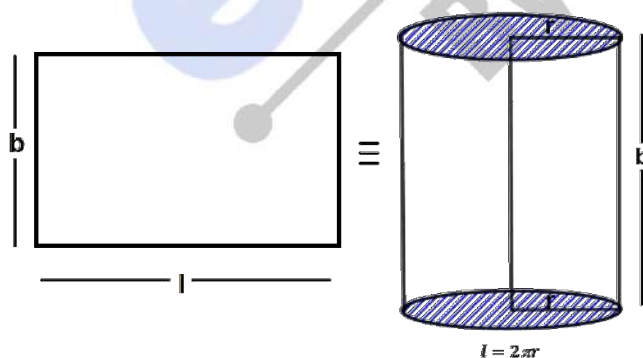
(iii) volume = area of base $\times$ height

$$= \pi r^2 \times h$$

$$\text{Vol.} = \pi r^2 h$$



iv) Cylinder from rectangular sheet :

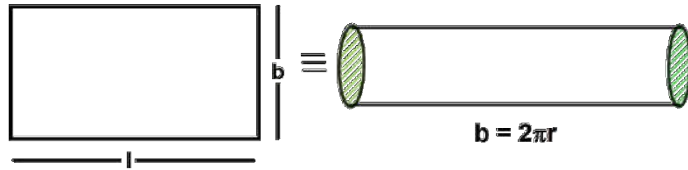


(a) $l = 2\pi r$

$$\text{Vol.}(l) = \frac{l^2 b}{4\pi}$$

(b)

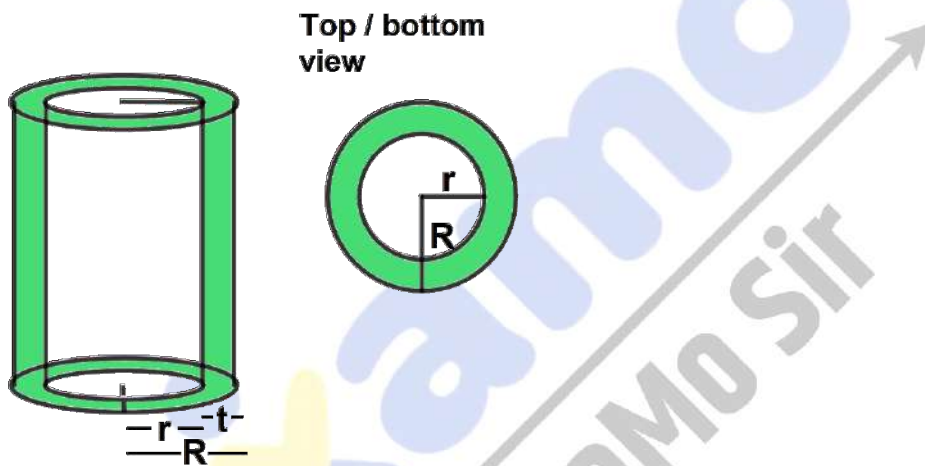
$$\text{Vol}(b) = \frac{b^2 l}{4\pi}$$



(c)

$$\frac{V_L}{V_b} = \frac{l}{b}$$

Hollow cylinder :



r : inner radius
 R : outer radius
 t : thickness = $R - r$

(i) $\text{CSA} = 2\pi(R+r)h$

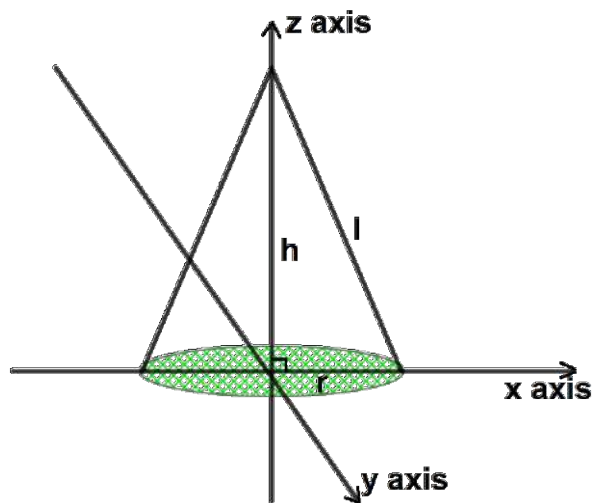
(ii) $\text{TSA} = 2\pi h(R+r) + 2\pi(R^2 - r^2)$

(iii) $\text{Vol.} = \pi(R^2 - r^2)h$

$$= \pi(R+r)(R-r)h$$

$$= \pi(R+r)t \times h$$

Cone :



r : radius of circular base

h : height of cone

l : slant height

$$l^2 = h^2 + r^2$$

(i) $CSA = \pi r l$

(ii) $TSA = \pi r (l + r)$

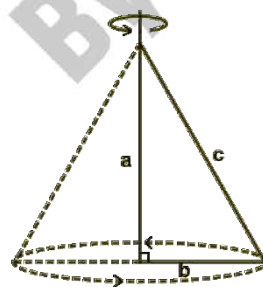
(iii) $vol. = \frac{1}{3} \pi r^2 h$

Cone : by rotating a right angle triangle :

i) rotating about its height :

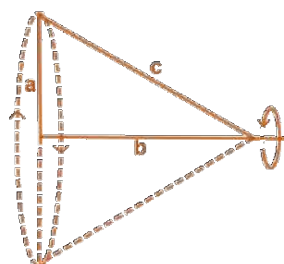
$$c^2 = a^2 + b^2$$

$$V_1 = vol = \frac{1}{3} \pi b^2 a$$



ii) Rotating about its base :

$$V_2 = vol. = \frac{1}{3} \pi a^2 b$$



iii) Rotating about its hypotenuse :

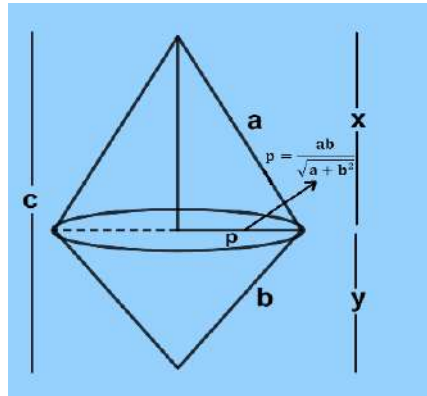
$$p = \frac{ab}{\sqrt{a^2 + b^2}}$$

$$\text{Vol.} = \frac{1}{3}\pi(p^2) \times x + \frac{1}{3}\pi(p^2) \times y$$

$$= \frac{1}{3}\pi p^2 (x + y)$$

$$\text{Vol.} = \frac{1}{3}\pi p^2 \times c$$

$$V_3 = \frac{1}{3}\pi \frac{a^2 b^2}{a^2 + b^2} \times \sqrt{a^2 + b^2} = \frac{1}{3}\pi \frac{a^2 b^2}{\sqrt{a^2 + b^2}}$$



Now, vol. ratios in all three cases :

$$V_1 : V_2 : V_3$$

$$b : a : \frac{ab}{\sqrt{a^2 + b^2}}$$

Cone frustrum :

$$l = \sqrt{h^2 + (R - r)^2}$$

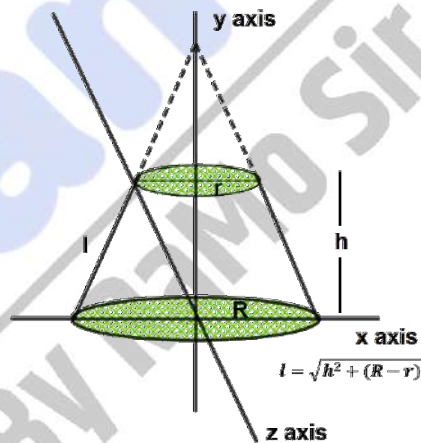
$$\text{CSA} = \pi(R + r) l$$

$$\text{TSA} = \text{CSA} + \text{top area}$$

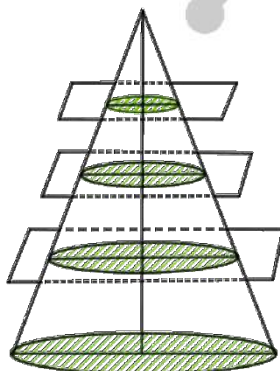
$$+ \text{bottom area}$$

$$= \pi(R + r)l + \pi r^2 + \pi R^2$$

$$\text{Vol.} = \frac{1}{3}\pi(R^2 + r^2 + rR) h$$



Cone Cutting :



C_1, C_2, C_3, C_4 are cones

Where C_1 is smaller one.

i) If a cone is cut into three equal parts of height.

Here,

$$h_1 : h_2 : h_3$$

$$1 : 2 : 3$$

So, volume ratio

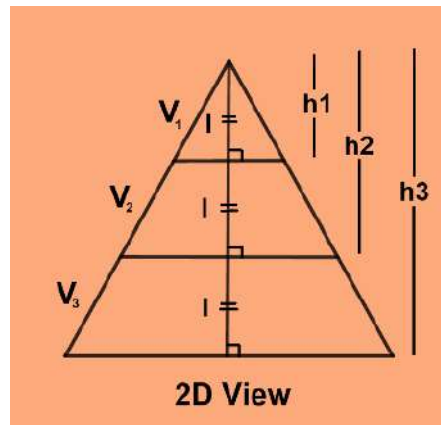
$$V_1 : (V_1 + V_2) : (V_1 + V_2 + V_3)$$

$$1 : 2^3 : 3^3$$

&

$$V_1 : V_2 : V_3$$

$$1 : 7 : 19$$



If cone is cut into n-equal parts of height :

So,

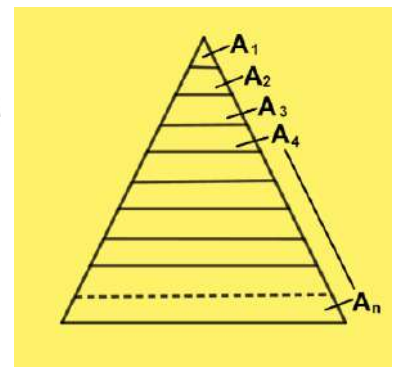
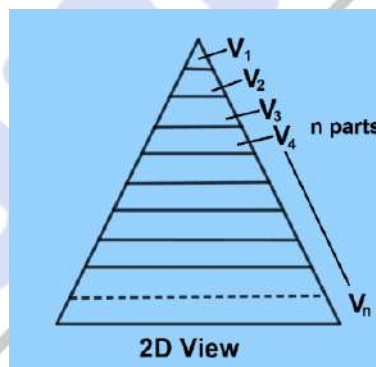
Volume's ratio :

$$V_1 : V_2 : V_3 : \dots : V_n$$

$$1 : 7 : 19 : \dots : n^3 - (n-1)^3$$

$$V_n = 3n^2 - 3n + 1$$

$$V_n = 3n(n-1) + 1$$



CSA ratio :

$$A_1 : (A_1 + A_2) : (A_1 + A_2 + A_3) : \dots : (A_1 + A_2 + \dots + A_n)$$

$$1^2 : 2^2 : 3^2 : \dots : n^2$$

Now,

$$A_1 : A_2 : A_3 : A_4 : \dots : A_n$$

$$1 : 3 : 5 : 7 : \dots : n^2 - (n-1)^2$$

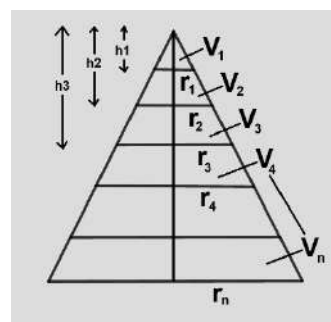
$$A_n = n^2 - (n-1)^2$$

$$A_n = 2n - 1$$

• **If a cone is cut into equal parts of volume :**

$$V_1 : V_2 : V_3 : \dots : V_n$$

$$1 : 1 : 1 : \dots : 1_n$$



So,

$$V_1 : (V_1 + V_2) : (V_1 + V_2 + V_3) : \dots : (V_1 + V_2 + V_3)$$

$$1 : 2 : 3 : \dots : \sqrt[3]{n}$$

Height ratio :

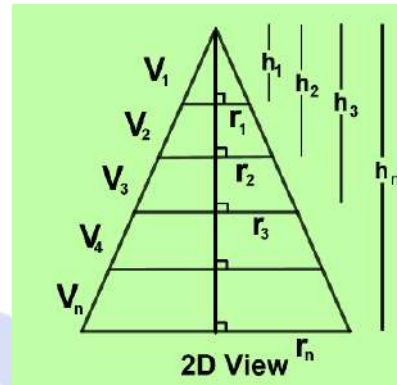
$$h_1 : h_2 : h_3 : \dots : h_n$$

$$1 : \sqrt[3]{2} : \sqrt[3]{3} : \dots : \sqrt[3]{n}$$

Now,

$$h_1 : (h_2 - h_1) : (h_3 - h_2) : \dots : h_n - (h_{n-1})$$

$$1 : \sqrt[3]{2} - 1 : \sqrt[3]{3} - \sqrt[3]{2} : \dots : \sqrt[3]{n} - \sqrt[3]{n-1}$$



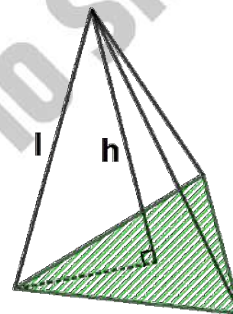
Pyramids :

i) Base is a triangle :

$$CSA = \frac{1}{2} \text{ perimeter of base} \\ \times \text{ slant height}$$

$$TSA = \text{area of base} + CSA$$

$$\text{Vol.} = \frac{1}{3} \text{ area of base} \times \text{height}$$



ii) Base is an equilateral triangle :

ΔABC is an equilateral triangle (side = a)

O is centre of the triangle

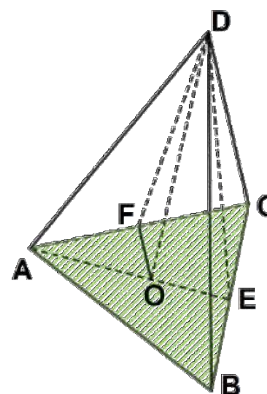
H = OD is the height of the pyramid

L = DE = DF are slant height

ΔAOD is a right triangle

ΔFOD is a right triangle

ΔEOD is a right triangle



$$OF = OE = \text{inradius} = \frac{a}{2\sqrt{3}}$$

In open form :

(Net)

$$l^2 = h^2 + r^2$$

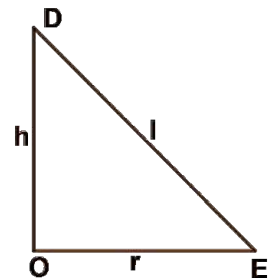
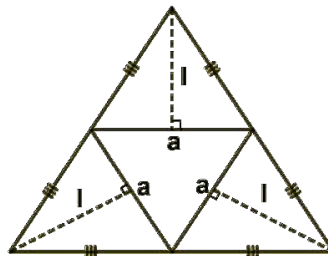
$$l^2 = h^2 + \left(\frac{a}{2\sqrt{3}}\right)^2$$

$$\Rightarrow l = \sqrt{h^2 + \frac{a^2}{12}}$$

$$\Rightarrow CSA = \frac{1}{2}(3a) \times l = \frac{3}{2}al$$

$$\Rightarrow TSA = \left(\frac{\sqrt{3}}{4}a^2\right) + \frac{3}{2}al$$

$$\Rightarrow Vol. = \frac{a^2 h}{4\sqrt{3}}$$



iii) Regular tetrahedron :

$$l = \frac{\sqrt{3}a}{2}$$

$$r = \frac{a}{2\sqrt{3}}$$

$$h = \sqrt{\frac{2}{3}}a$$

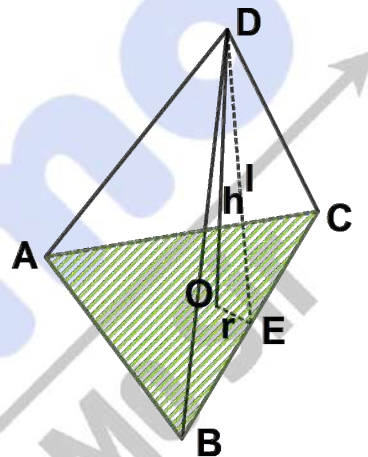
$\triangle ABC, \triangle ADB, \triangle ADC, \triangle BDC,$

are equilateral $\triangle$ s

$\triangle DOE$ in a right $\triangle$.

DO is height, OE is inradius,

ED is slant height.



$$l^2 = h^2 + r^2$$

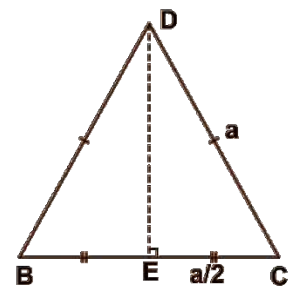
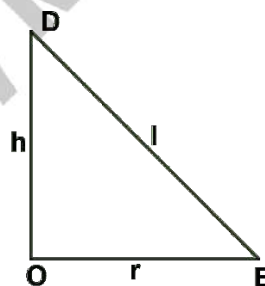
$$l = \sqrt{a^2 - \frac{a^2}{4}}$$

$$r = \frac{a}{2\sqrt{3}}$$

So,

$$\left(a^2 - \frac{a^2}{4}\right) = h^2 + \left(\frac{a}{2\sqrt{3}}\right)^2$$

$$h = \sqrt{\frac{2}{3}}a$$



$$CSA = \frac{1}{2} \text{ Perimeter of base} \times \text{slant height}$$

$$= \frac{1}{2} \times 3a \times \frac{\sqrt{3}}{2}a$$

$$CSA = \frac{3\sqrt{3}}{4}a^2$$

$$TSA = \frac{\sqrt{3}}{4}a^2 + \frac{3\sqrt{3}}{4}a^2$$

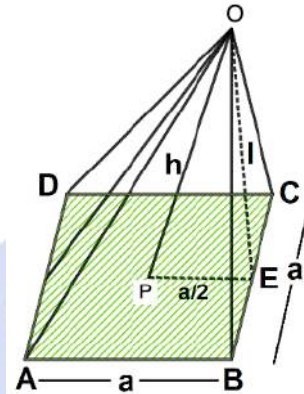
$$TSA = \sqrt{3}a^2$$

$$Vol. = \frac{1}{2} \times \frac{\sqrt{3}}{4} a^2 \times \sqrt{\frac{2}{3}} a$$

$$Vol. = \frac{a^3}{6\sqrt{2}}$$

Base is a square :

- Here ABCD is a square
- OP is height
- OE is slant height
- ΔOPE is a right Δ
- $PE = \frac{1}{2} AB$



$$l^2 = h^2 + \left(\frac{a}{2}\right)^2$$

$$l = \sqrt{h^2 + \frac{a^2}{4}}$$

$$\begin{aligned} \text{CSA} &= \frac{1}{2} \text{perimeter of base} \times \text{slant height} \\ &= \frac{1}{2} \times 4a \times l \\ &= 2al \\ &= a\sqrt{a^2 + 4h^2} \end{aligned}$$

$$\begin{aligned} \text{TSA} &= \text{CSA} + \text{base area} \\ &= a\sqrt{a^2 + 4h^2} + a^2 \end{aligned}$$

$$\text{Vol.} = \frac{1}{3} a^2 h$$

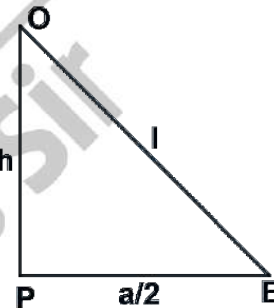
- If $h = a$

$$l = \frac{\sqrt{5}}{2} a$$

$$CSA = \sqrt{5}a^2$$

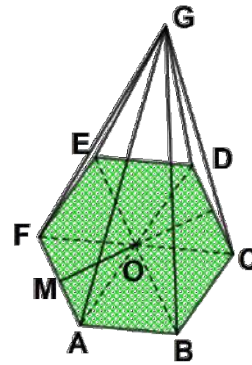
$$TSA = (\sqrt{5}+1)a^2$$

$$Vol. = \frac{1}{3} a^3$$



Base in regular hexagon :

- ABCDEF is a regular hexagon of side a
- GO is the height of pyramid.
- GM is slant height.
- ΔGMO is a right Δ .
- $OM = \frac{\sqrt{3}}{2} a$
- $CSA = \frac{1}{2} \text{perimeter of base} \times \text{height}$

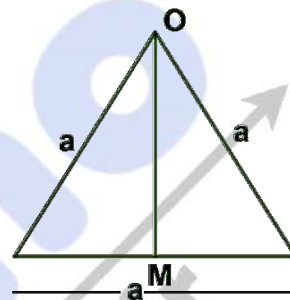


$$= \frac{1}{2} \times 6a \times \sqrt{h^2 + \frac{3}{4}a^2}$$

$$= 3a \sqrt{h^2 + \frac{3}{4}a^2}$$

- $TSA = CSA + \text{base area}$

$$= 3a \sqrt{h^2 + \frac{3}{4}a^2} + \frac{3\sqrt{3}}{2} a^2$$



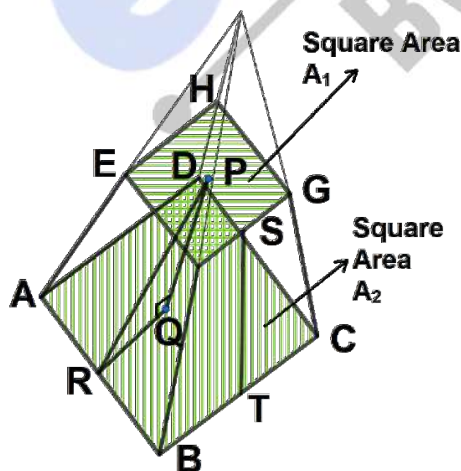
- $\text{Vol.} = \frac{1}{3} \text{area of base} \times \text{height}$

$$= \frac{1}{3} \left(\frac{3\sqrt{3}}{2} a^2 \right) \times h$$

Vol.

$$= \frac{\sqrt{3}}{2} a^2 h$$

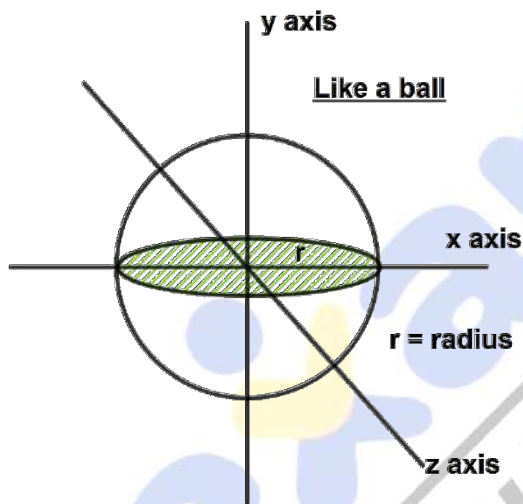
frustrum of square pyramid :



- Here, ABCD is a square with area A_2 & EFGH is a square with area A_1
- PQ is the height of frustrum.
- $RQ = \frac{1}{2} AB$

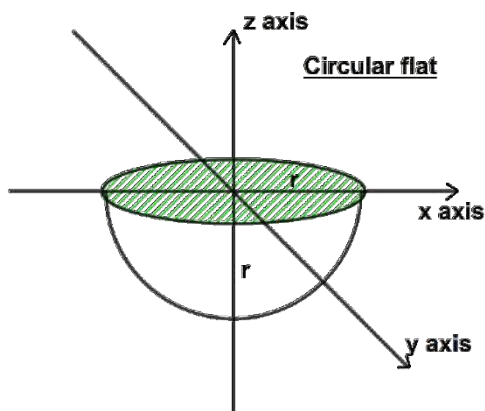
- ST is slant height.
- $CSA = \frac{1}{2}(P1 + P2) \times \text{slant height}$
Where P1 is the perimeter of EFGH.
P2 is the perimeter of ABCD.
- $TSA = CSA + A1 + A2$
Where A1 is the area of EFGH.
& A2 is the area of ABCD.
- $Vol. = \frac{1}{3}(A1 + A2 + \sqrt{A1 \times A2}) \times h$

Sphere :

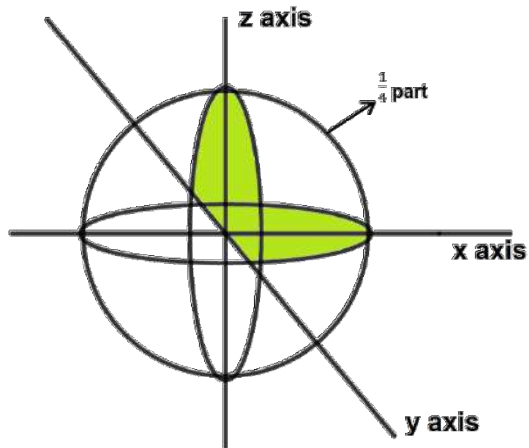


- $CSA / TSA = 4\pi r^2$
- $Vol. = \frac{4}{3} \pi r^3$

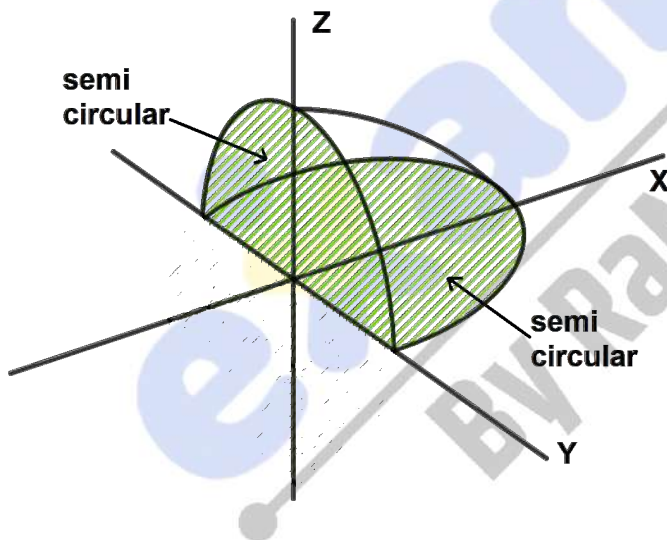
Hemi sphere :



- $CSA = 2\pi r^2$
- $TSA = 3\pi r^2$
- $Vol. = \frac{2}{3}\pi r^3$



- $\frac{1}{4}$ th part of sphere

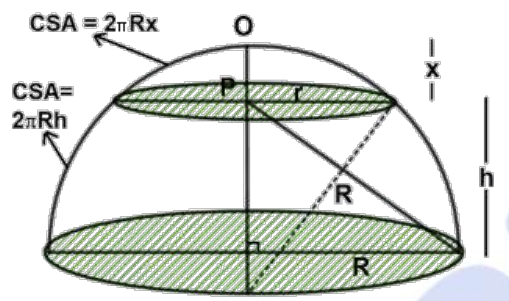
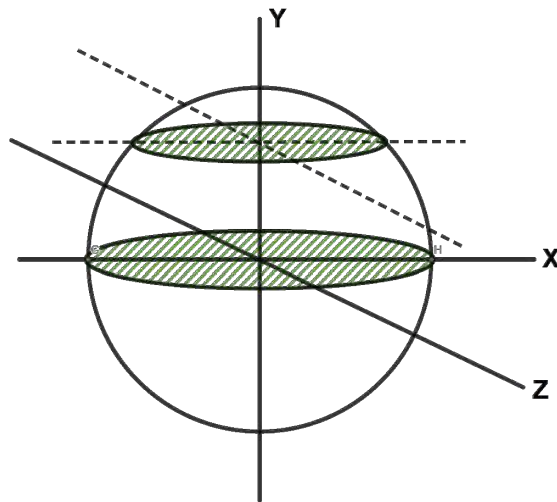


$$CSA = \pi r^2$$

$$TSA = \pi r^2 + \frac{1}{2}\pi r^2 + \frac{1}{2}\pi r^2 = 2\pi r^2$$

$$Vol. = \frac{1}{3}\pi r^3$$

Special case :



$$\begin{aligned}\text{CSA of hemi} &= 2\pi Rh + 2\pi Rx \\ &= 2\pi R(x + h) \\ &= 2\pi R(R) \\ &= 2\pi R^2\end{aligned}$$